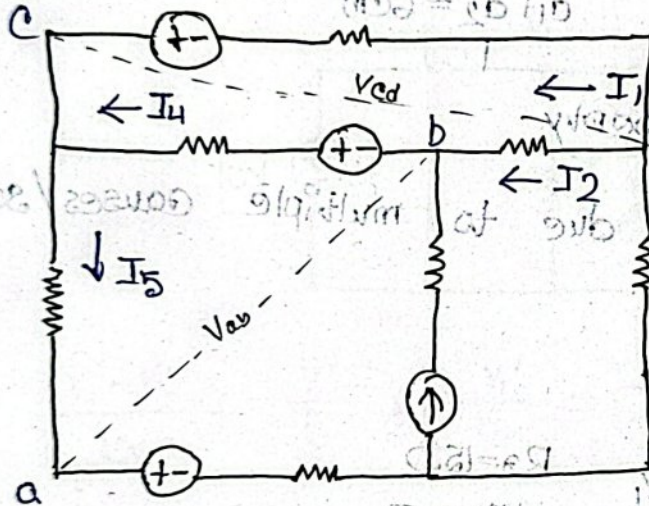


Lec-174

03/08/26

Topic: Superposition Theorem

Node: Universal Method

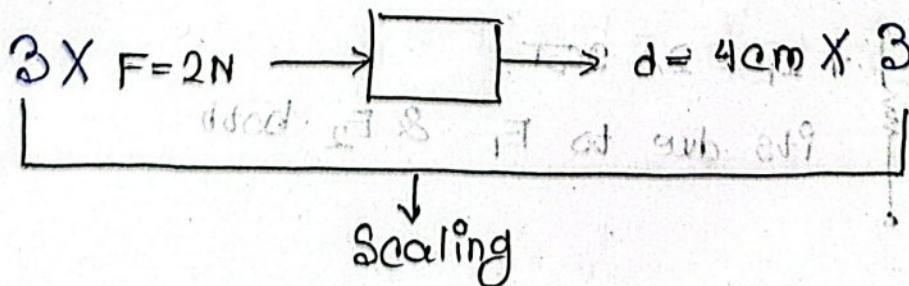


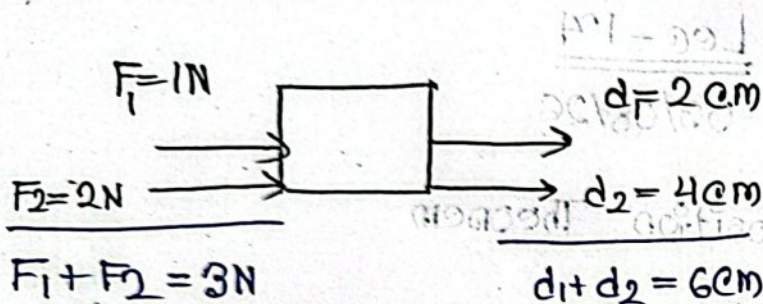
System (Linear)

$$\left. \begin{array}{l} F = 1N \\ F = 2N \end{array} \right\} \begin{array}{l} d = 2cm \\ d = 4cm \end{array} \quad \text{dd } F \text{ (linear Reln)}$$

$$y = mx + c \text{ [Linear eqn]}$$

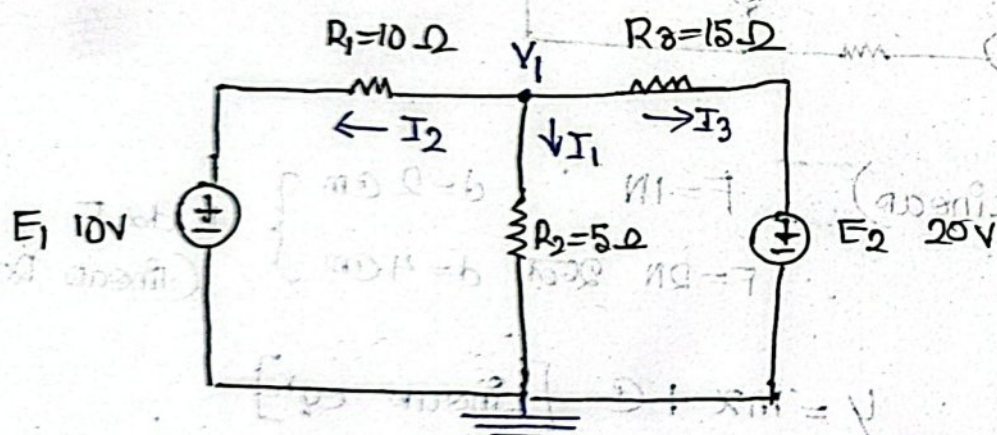
→ Homogeneity Property :-





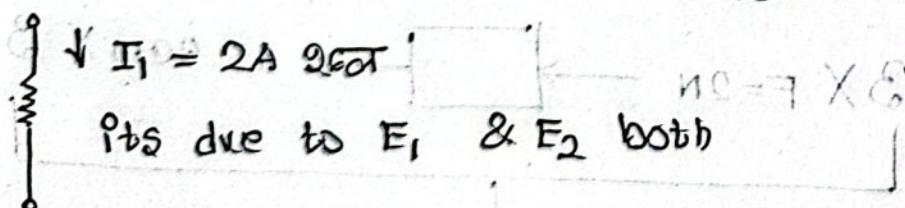
Additive Property

Effect on the system due to multiple causes/sources.



$$\frac{V_1 - 10}{10} + \frac{V_1}{5} + \frac{V_1 - 20}{15} = 0 \Rightarrow I_1 = \frac{V_1}{5} \quad I_2 = \frac{V_1 - 10}{10}$$

$$I_3 = \frac{V_1 - 20}{15}$$



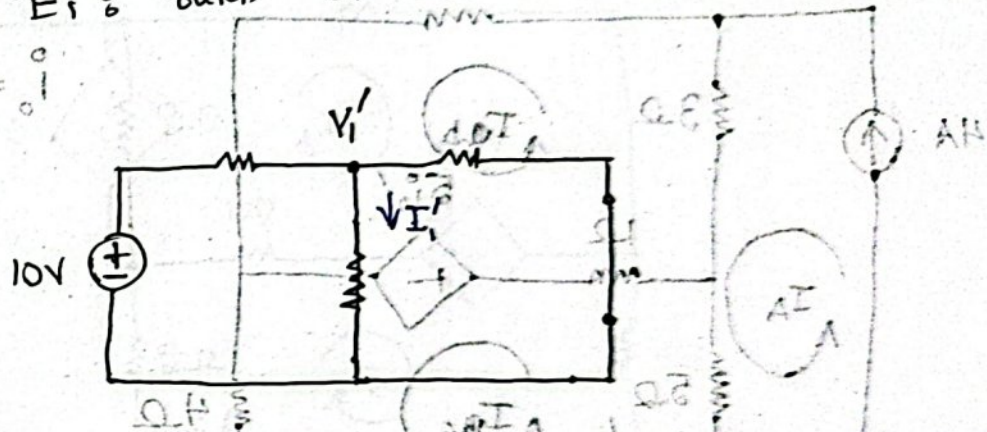
Step 18 Turn of All sources Except 1.

⇒ For E_1 : turn of all the other sources.

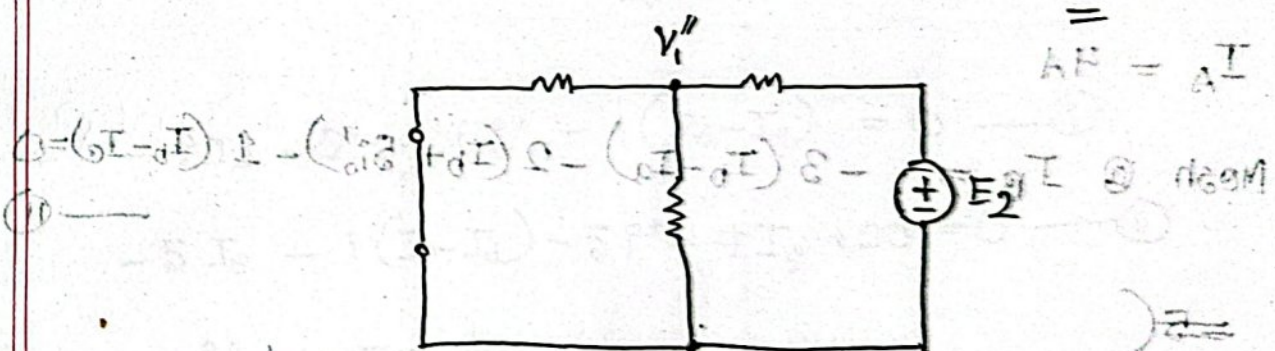
$$V_{oc} + I_{oc} = 0$$

↓
VOC

↓
AP



$$\frac{V_1' - 10}{10} + \frac{V_1'}{5} + \frac{V_1'}{15} = 0 \quad I_1' = \frac{V_1'}{R_2} = \frac{2.7272}{5}$$



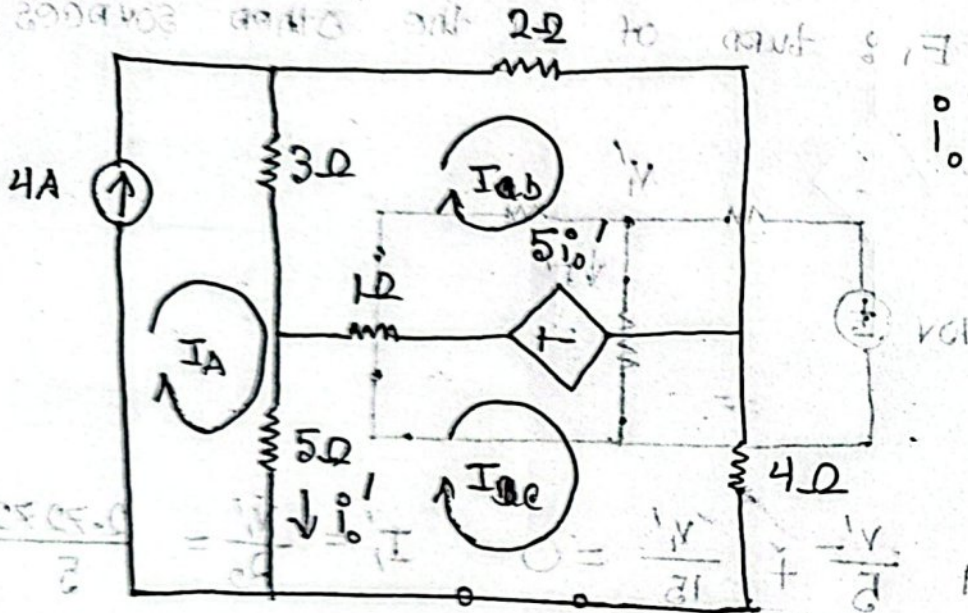
$$\frac{V_1'' - 10}{10} + \frac{V_1''}{5} + \frac{V_1'' - 20}{15} = 0 \quad I_1'' = \frac{V_1''}{R_2} = \frac{3.6363}{5}$$

$$\therefore I_1 = I_1'' + I_1' = \frac{V_1''}{R_2} + \frac{V_1'}{R_2}$$

$$= 1.2727 A$$

$$V_1 = V_1' + V_1''$$

For 4A



$$I_o = I_o' + I_o''$$

$\downarrow$ $\downarrow$
 4A 20V

$$I_A = 4A$$

Mesh @ $I_B \Rightarrow -3(I_B - I_A) - 2(I_B + 5I_o') - 1(I_B - I_C) = 0$ — (i)

5

Mesh @ $I_C \Rightarrow -5(I_C - I_A) - 1(I_C - I_B) - 5I_o' - 4I_C = 0$ — (ii)

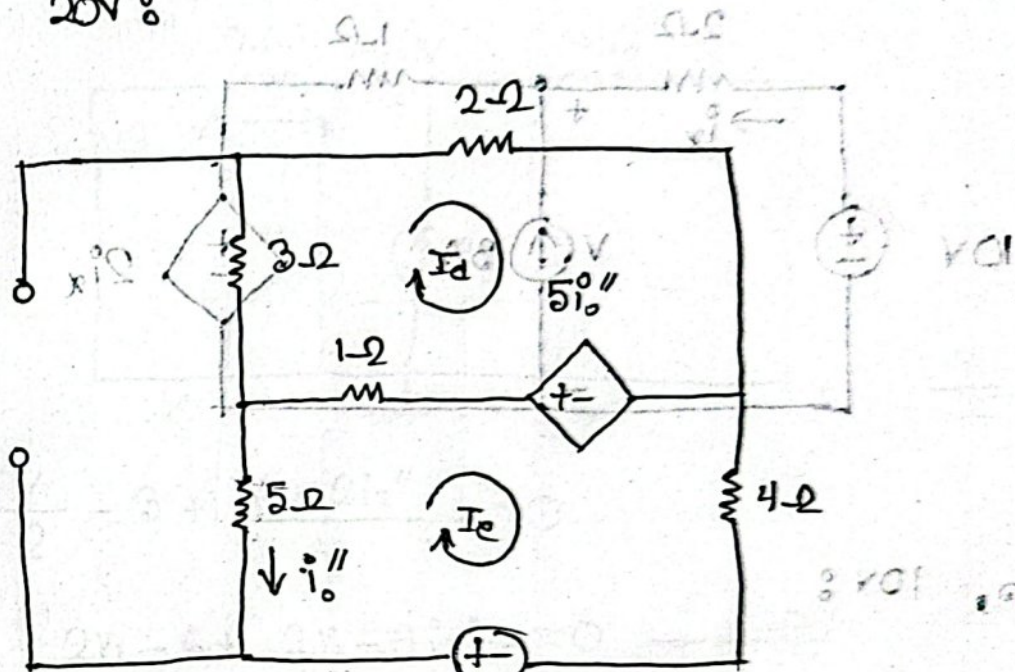
$$I_o' = I_A - I_C$$

$$\frac{1}{2} + \frac{1}{4} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$A \times B = B \times A$$

$$1 + 1 = 2$$

For 20V :

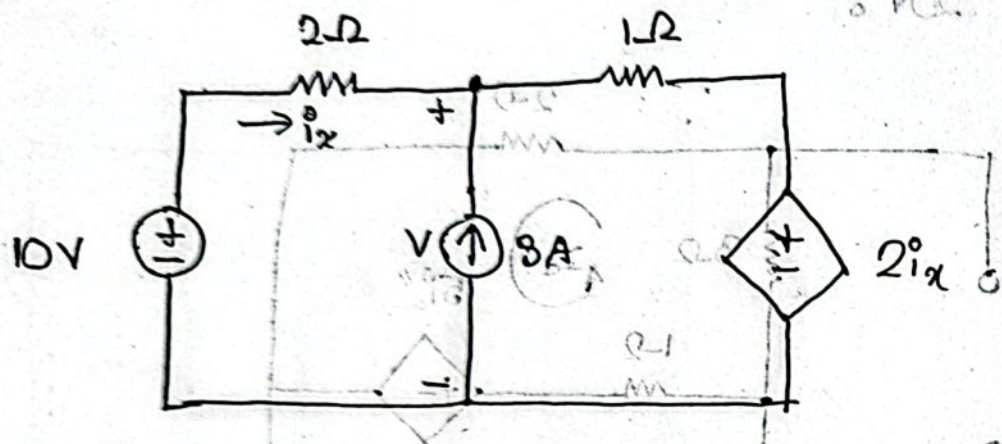


$$\begin{aligned}
 -3I_d + 2I_d + 5i'' - 1(I_d - I_e) &= 0 \quad \text{--- (I)} \\
 -5I_e - 1(I_e - I_d) - 5i'' - 4I_e + 20 &= 0 \quad \text{--- (II)} \\
 P'' &= -I_e
 \end{aligned}$$

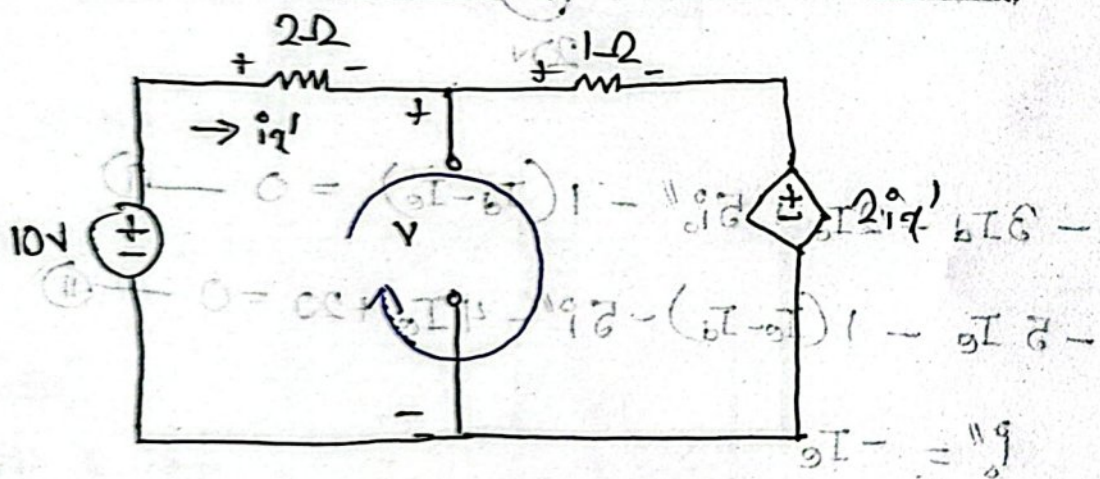
$$0 = i'' - i'' - i'' - 0$$

$$A\Omega = \frac{1}{x''}$$

$$AB = i'' : .$$



For, 10V :

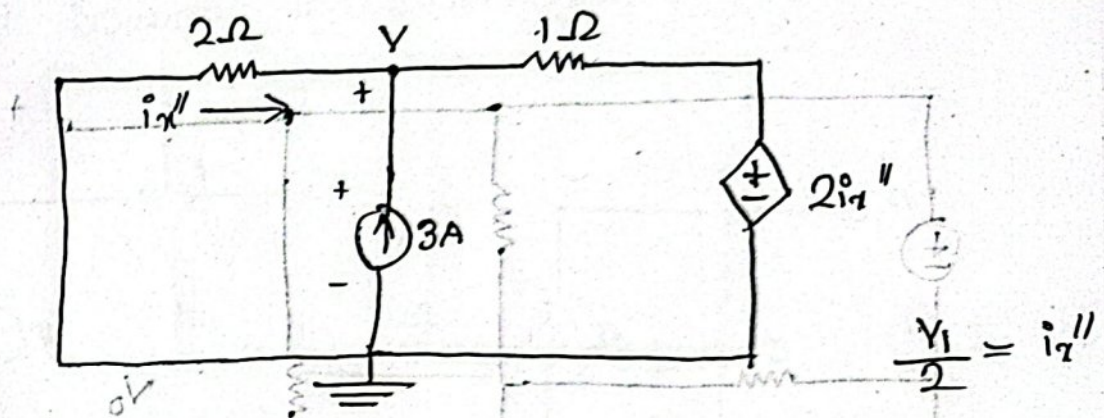


$$10 - 2i_x - 1i_x - 2i_x = 0$$

$$i_x' = 2A$$

$$\therefore 2i_x = 4A$$

Next Week Wed: Quiz 3 (Mesh + Superposition)



$$\frac{v_1}{2} - 3 + \frac{v_1 - 2i_x''}{4} = 0$$

$$\Rightarrow 2v_1 - 6 + 2v_1 - 4i_x'' = 0$$

$$\Rightarrow 4v_1 - 4i_x'' = 6$$

$$4(v_1 - i_x'') = 6$$

$$\Rightarrow 4\left(v_1 - \frac{v_1}{2}\right) = 6$$

$$\therefore v_1 = 3 \text{ V} \quad i_x'' = 1.5 \text{ A}$$

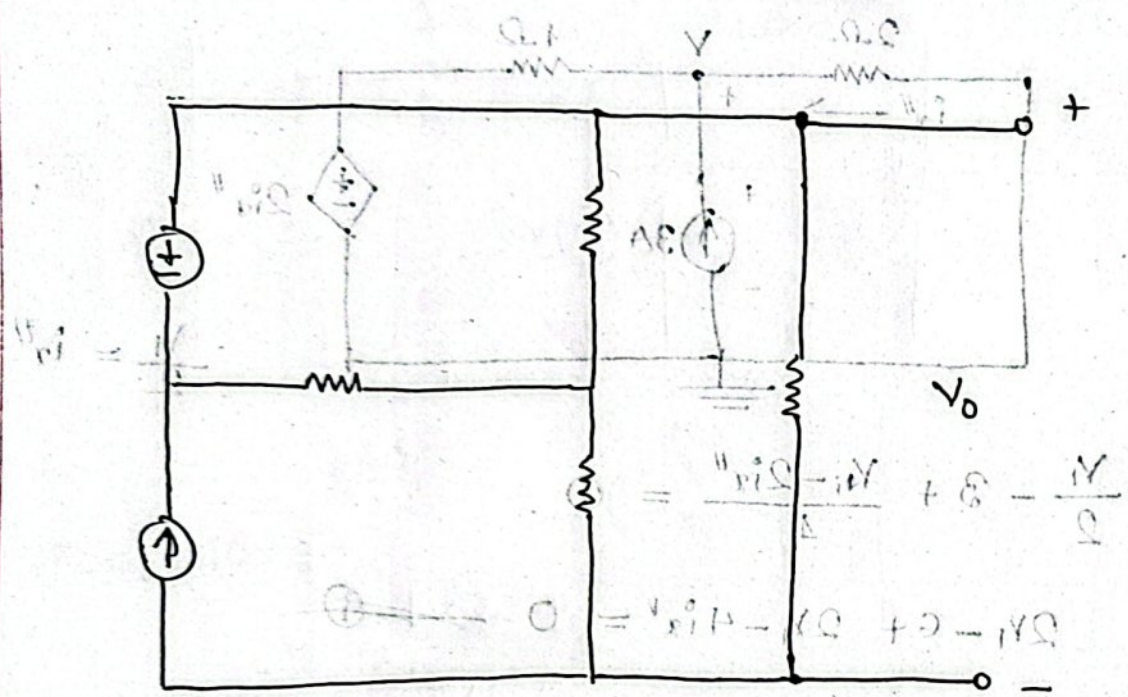
$$i_x = i_x' + i_x''$$

$$= 2 + 1.5$$

$$= 3.5 \text{ A}$$



Next Week Web 8 Quiz 8 (Week 1 Substitution)

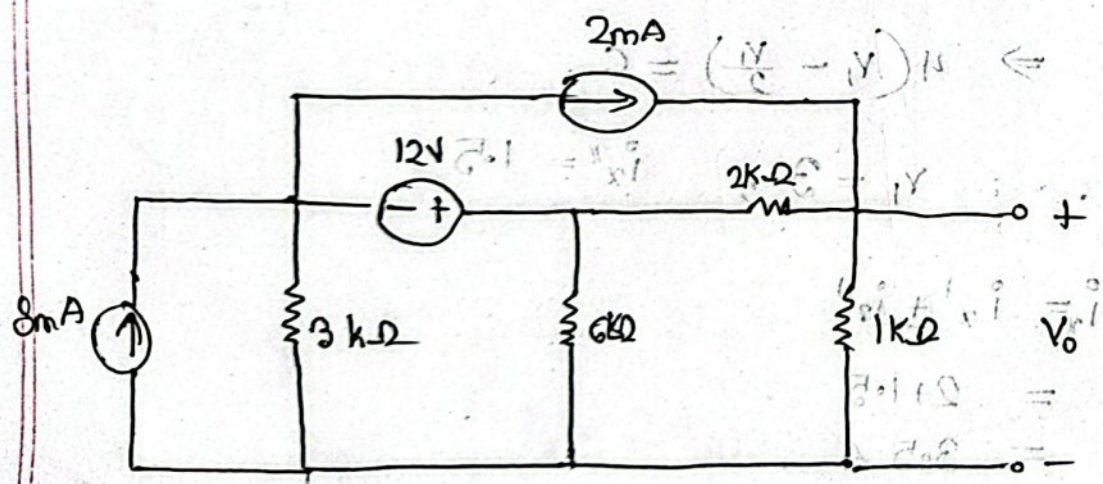


$$0 = \frac{12V - V}{10k} - \frac{V}{10k} + \frac{V}{10k} - \frac{V}{10k}$$

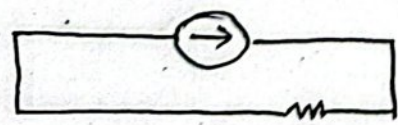
$$0 = \frac{12V - V}{10k} - \frac{V}{10k} + \frac{V}{10k} - \frac{V}{10k}$$

$$0 = \frac{12V - V}{10k} - \frac{V}{10k} + \frac{V}{10k} - \frac{V}{10k}$$

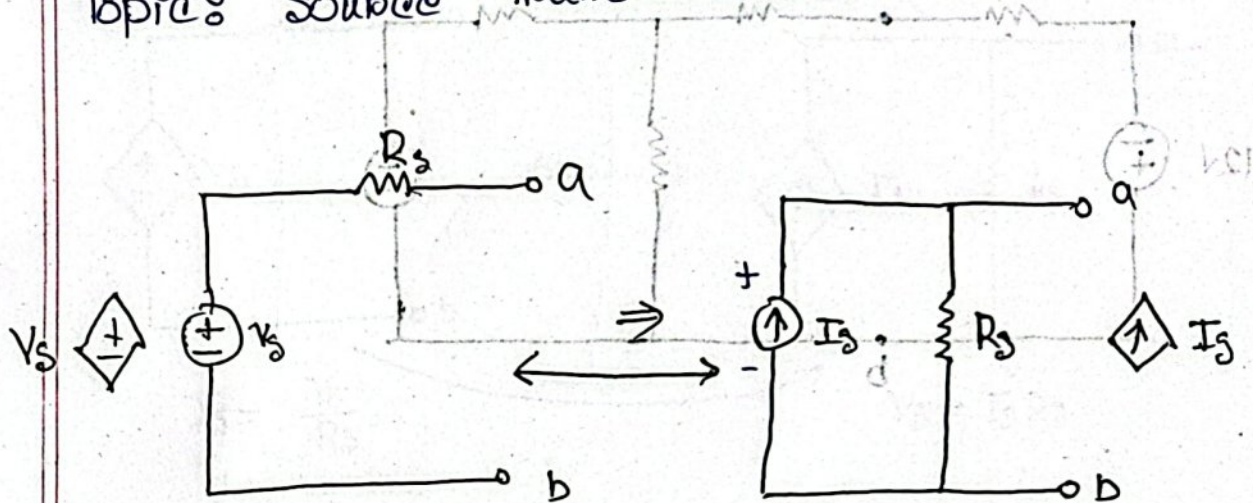
$$0 = \left(\frac{12V}{10k} - \frac{V}{10k} - \frac{V}{10k} + \frac{V}{10k} \right)$$



For 2mA:



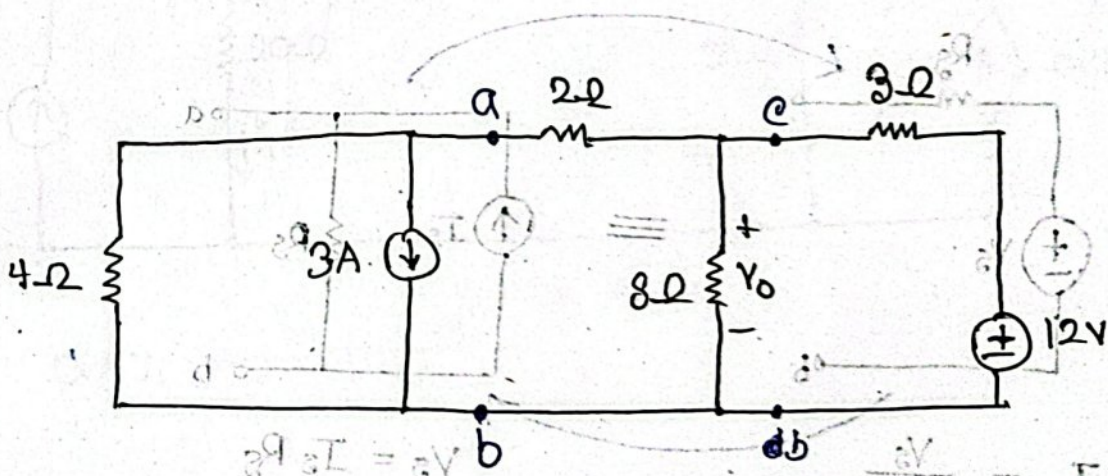
Topic: Source Transformation



$$I_s = \frac{V_s}{R_s}$$

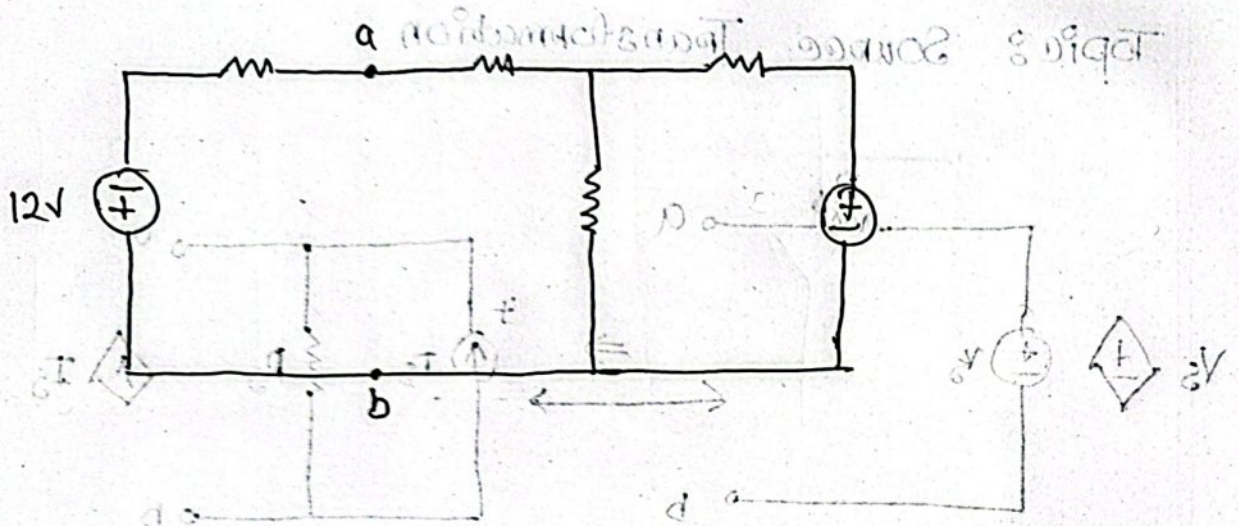
$$V_s = I_s R_s$$

* Same Node



$$V_1 = I_1 \times R_1 = 3 \times 4 = 12V$$

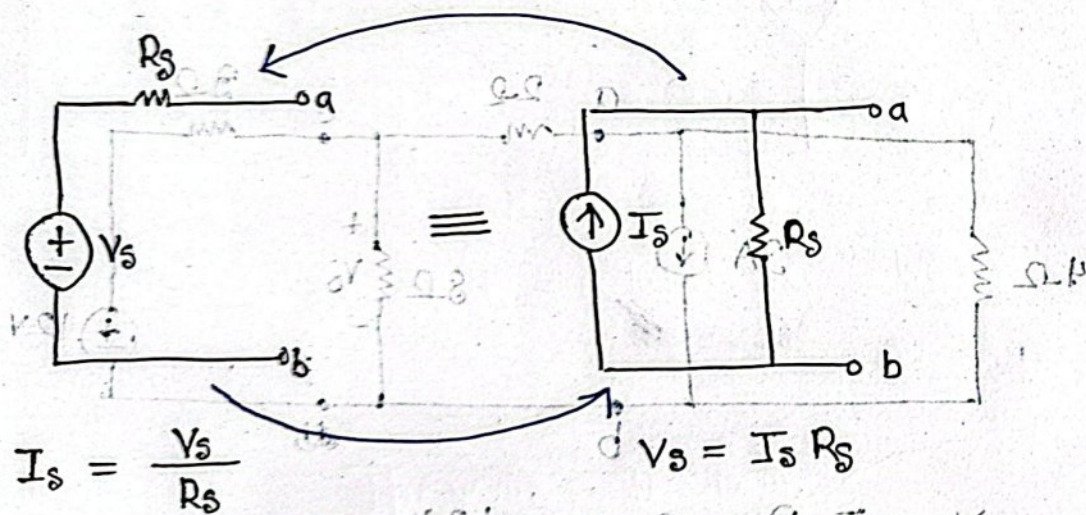
$$I_2 = \frac{V_2}{R_2} = \frac{12}{3} = 4A$$



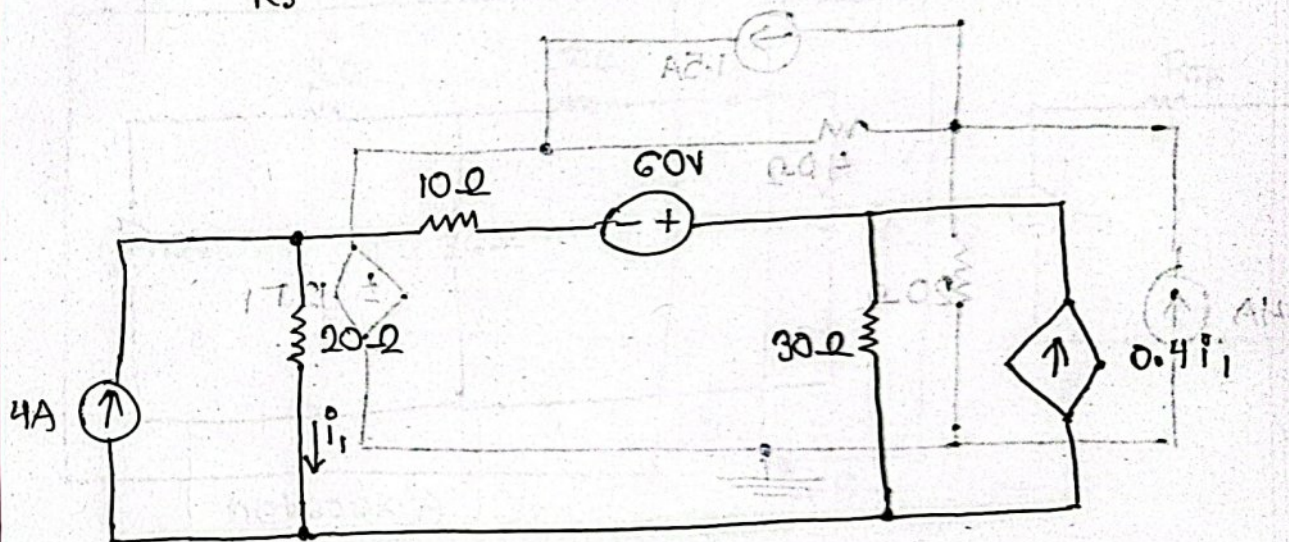
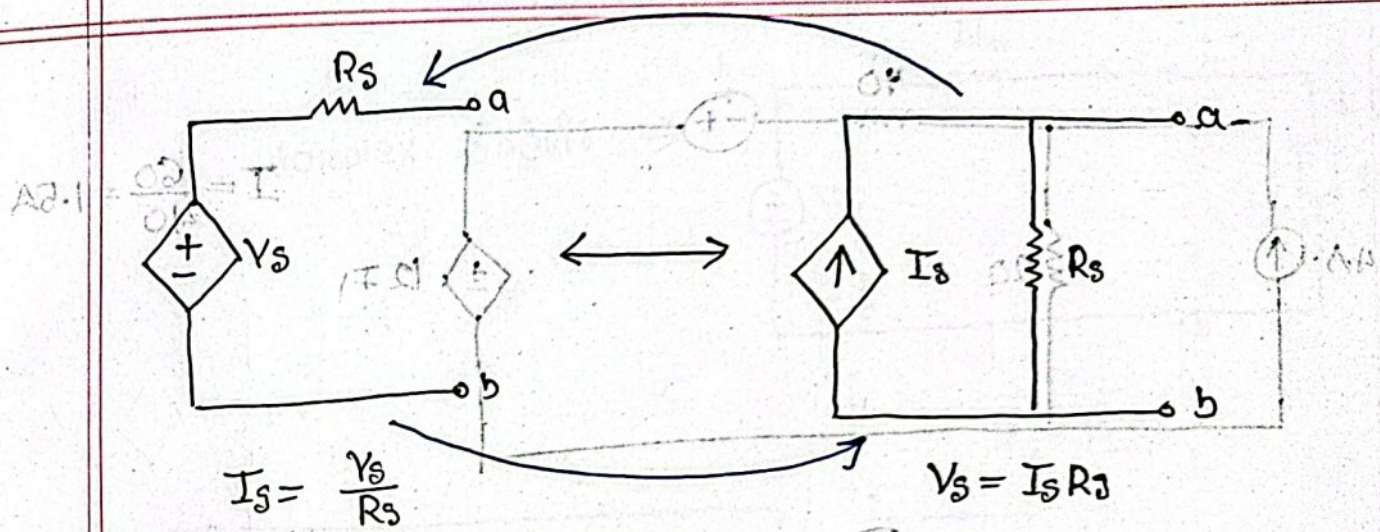
Lec-15

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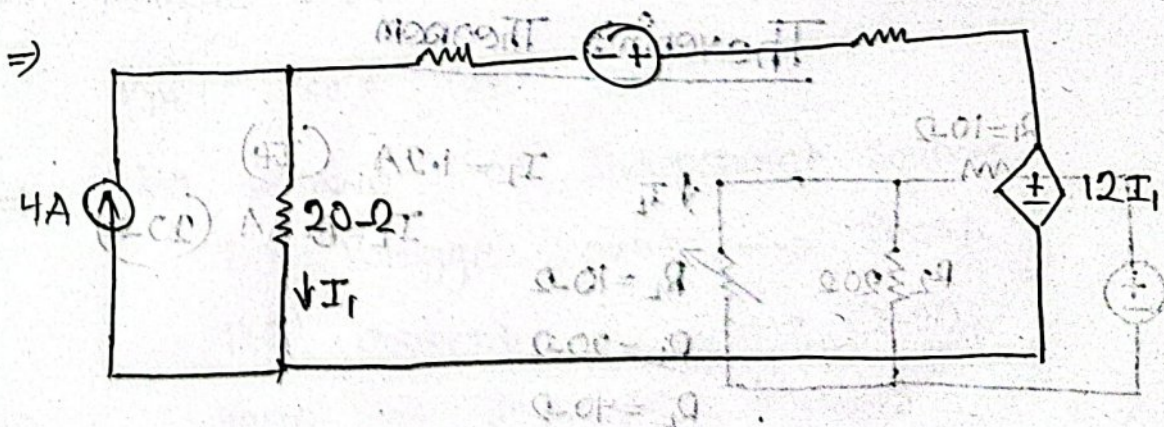
$$\frac{2V}{2} = 2I$$



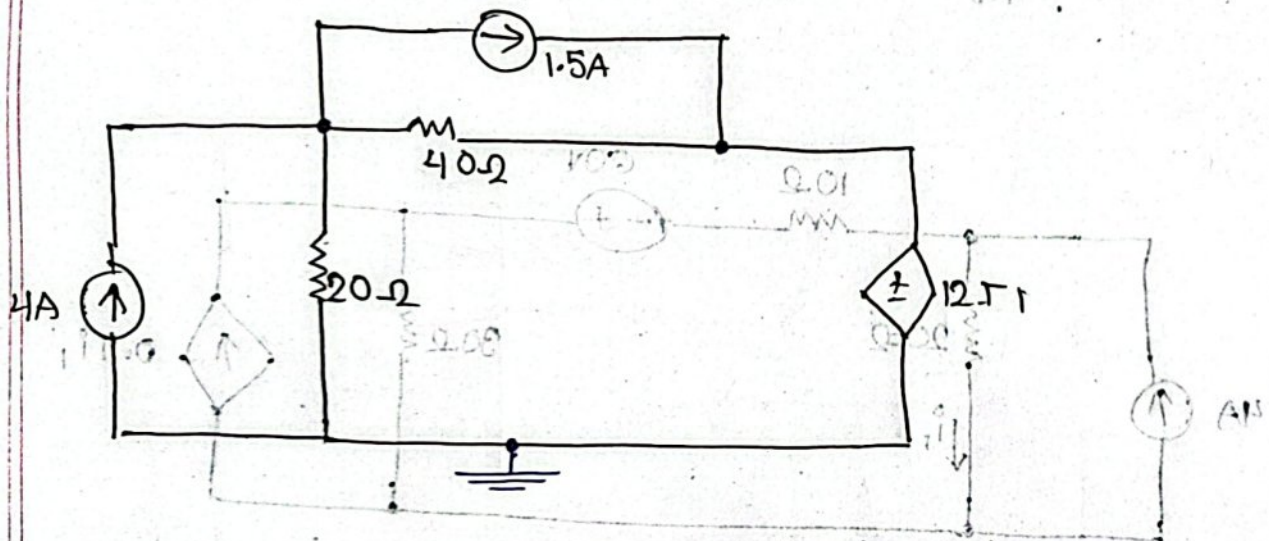
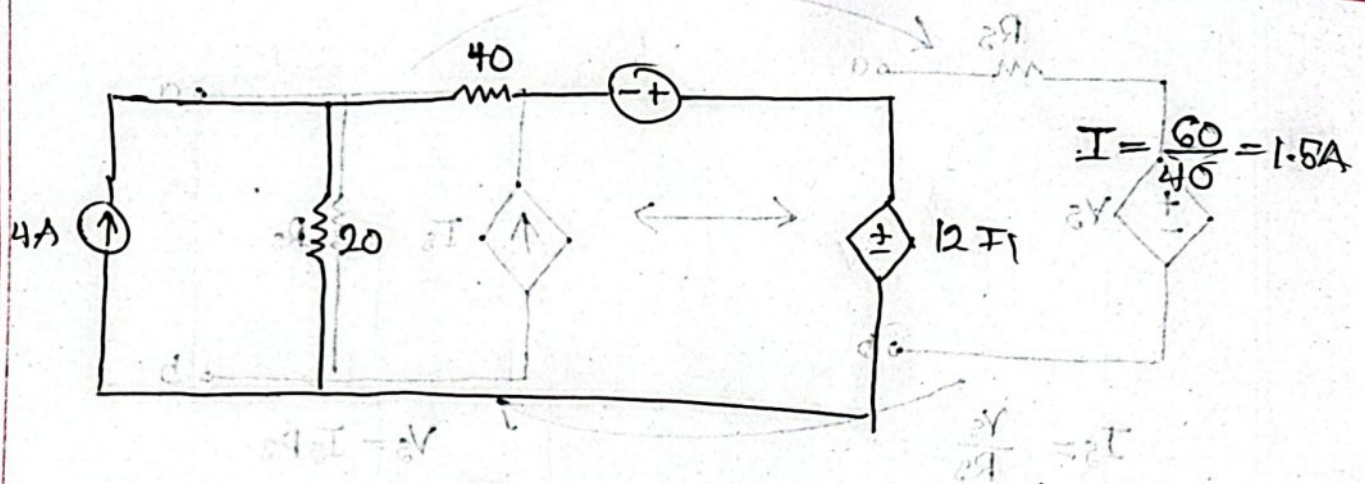
* Current Direction: to the +ve terminal. $\uparrow \oplus$



$0.4 I_1 \times 30 = 12 I_1$



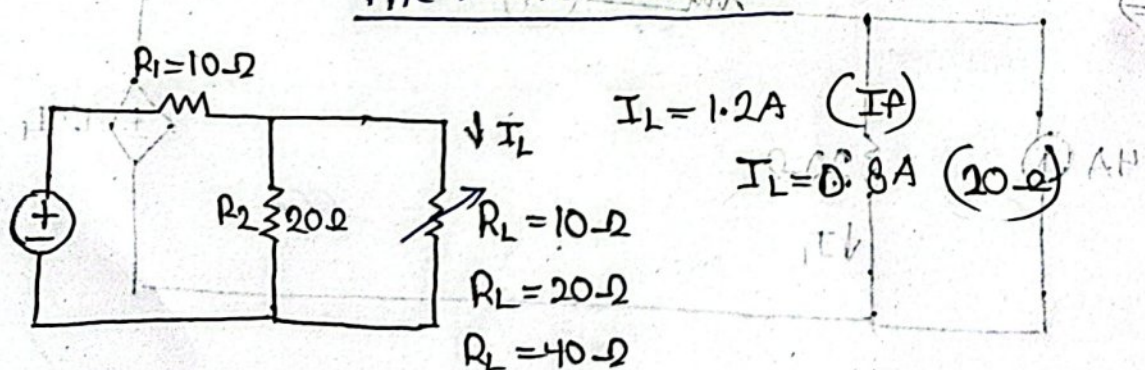
* Grading is only in A



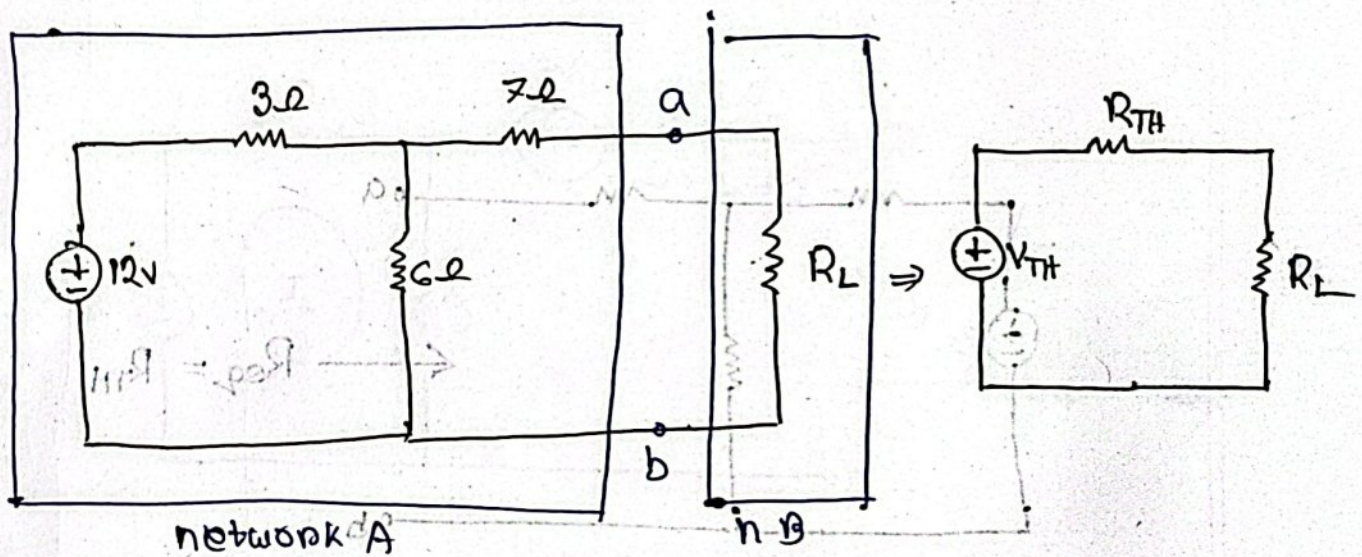
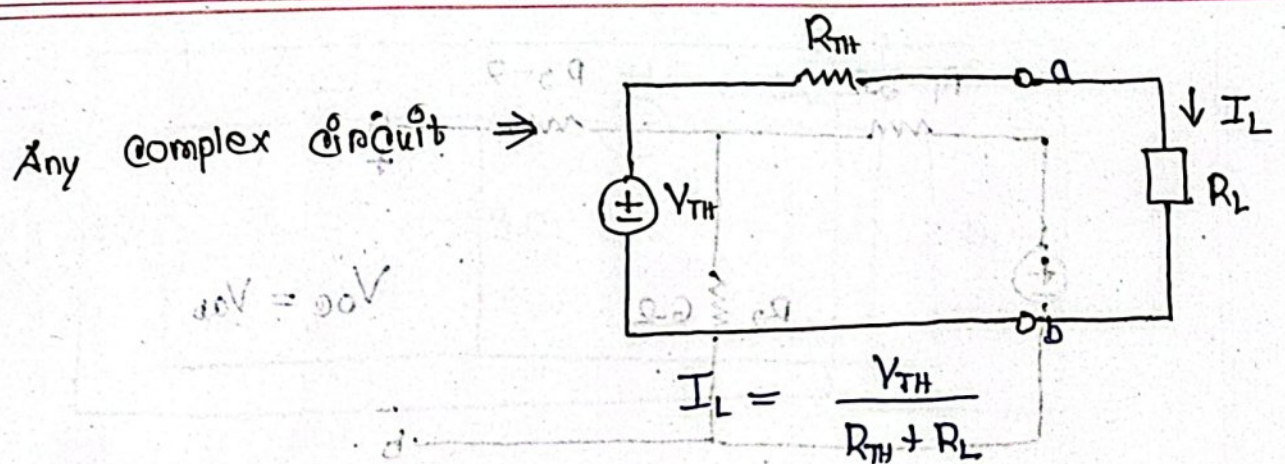
⇒ Apply Nodal or Mesh analysis to solve.

$$I_{L(1)} = 0.8 \times I_{L(2)}$$

Thevenin's Theorem



* Change is only in R_L



$\Rightarrow$ Divide into 2 Parts: 1 with complex circuits, 2 with R_L

$\Rightarrow$ Remove Network B by opening the circuit

$$V_{TH} = V_{oc} = V_{ab}$$

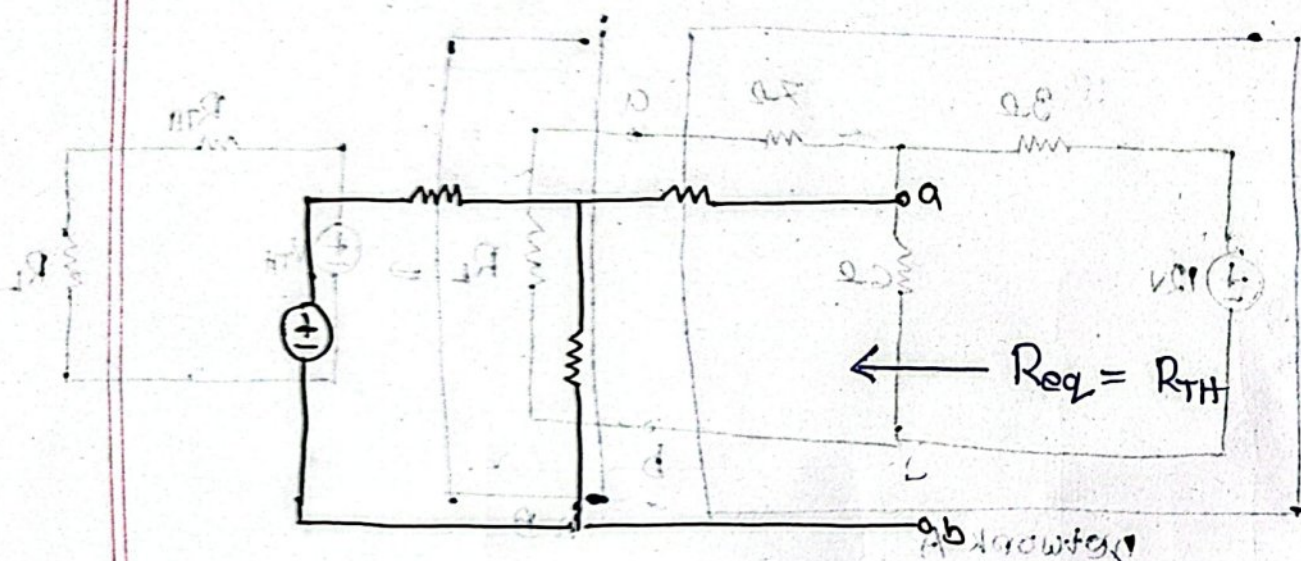
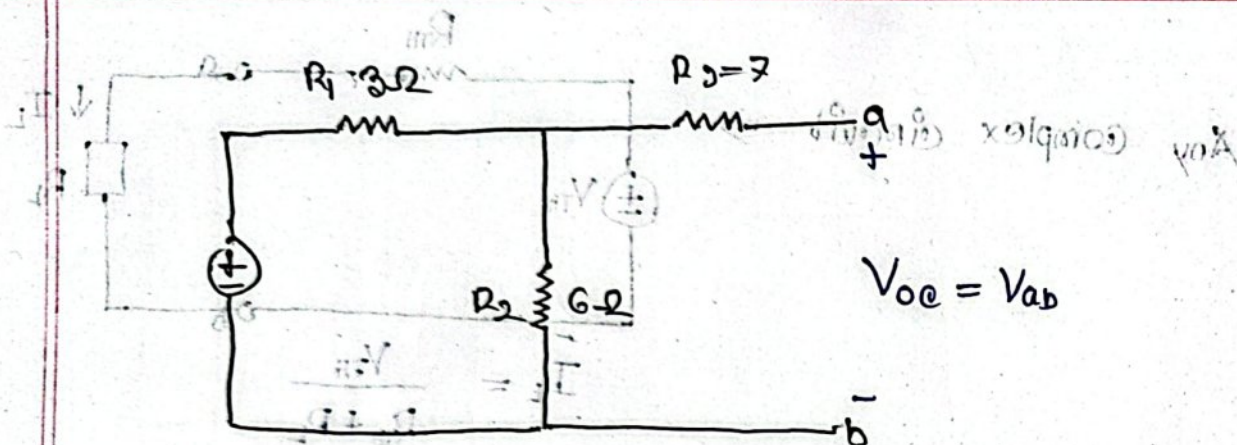
$\Rightarrow$ ~~Be~~ Remove all Independent Sources

DC Voltage source $\rightarrow$ Short

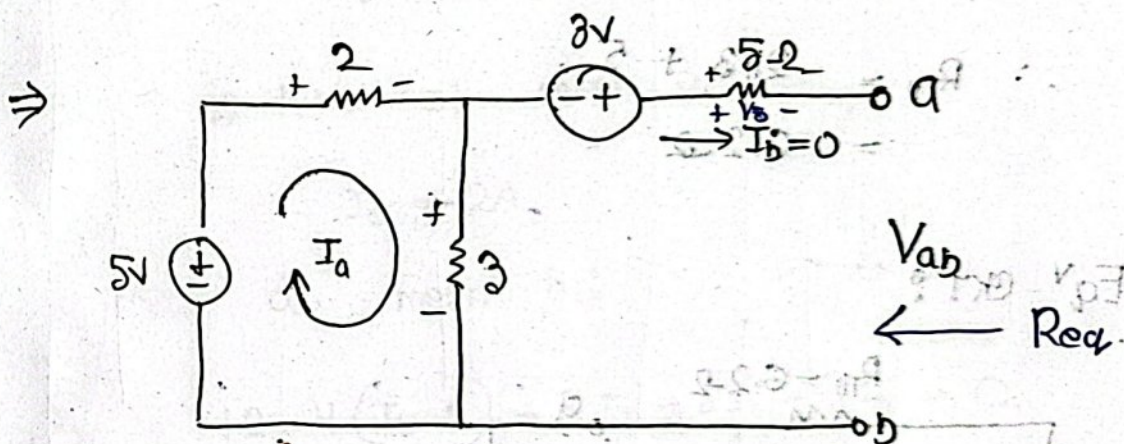
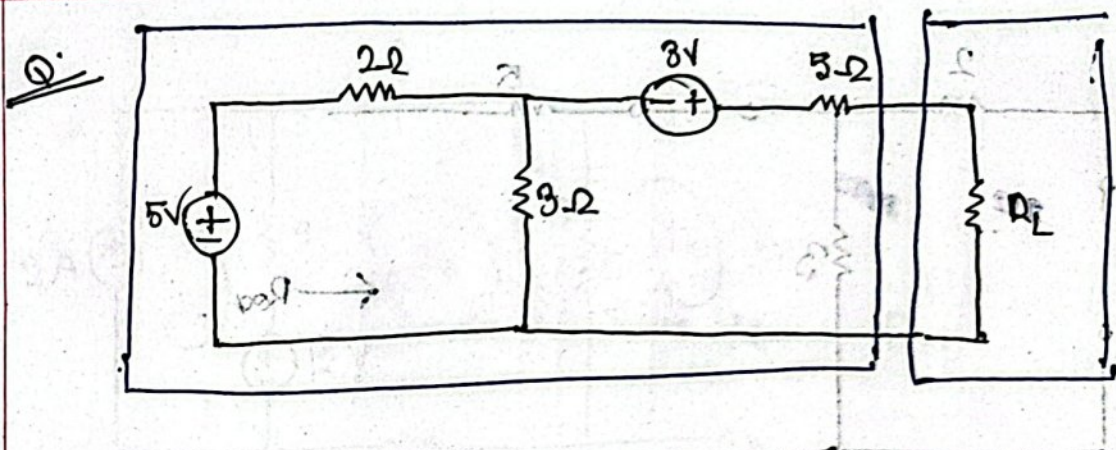
DC current " $\rightarrow$ Open

$\Rightarrow$ Keep all dependent sources

$\rightarrow$ Test Voltage



- Test Voltage
- Keep all dependent sources
- DC current $I_0 \rightarrow$ open
- DC Voltage sources $\rightarrow$ short
- DC Remove all independent sources
- $V_{TH} = V_{oc} = V_{op}$
- ⇒ Remove Network B by opening the circuit
- ⇒ Divide int 2 parts. A with complex circuit. B with



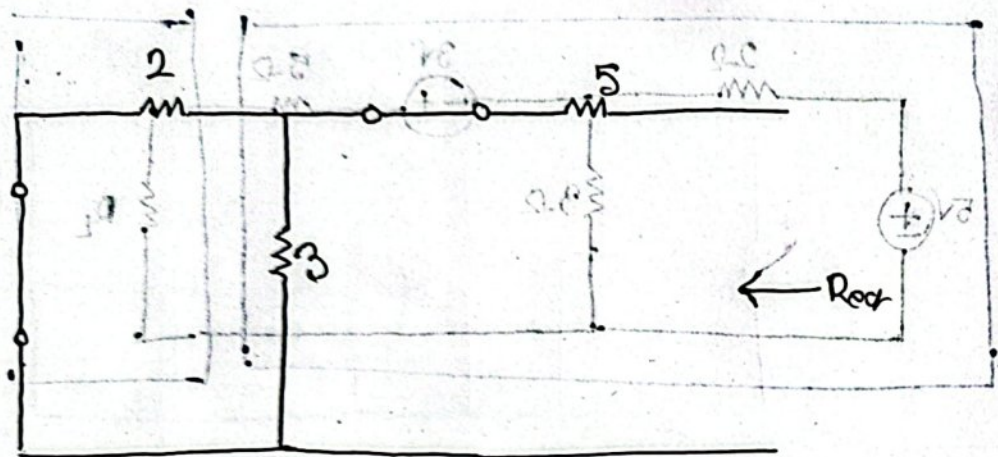
$$5 - 5I_a = 0 \quad \therefore I_a = 1A$$

$$+ 3I_a + 3 - V_s - V_{ab} = 0$$

$$\Rightarrow 3I_a + 3 - 5I_b - V_{ab} = 0$$

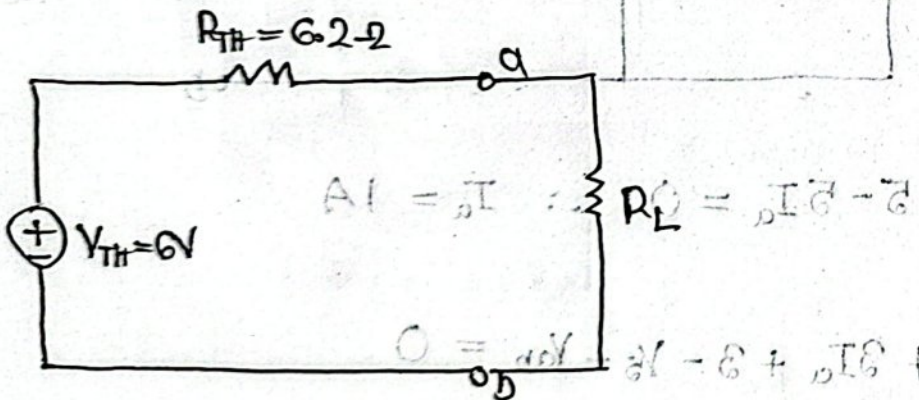
$$\Rightarrow 3 + 3 - 0 = V_{ab} = 6V$$

$$V_{oc} = V_{TH} = 6V$$



$$\therefore R_{TH} = \frac{2 \parallel 3 + 5}{1} = 6.2 \Omega$$

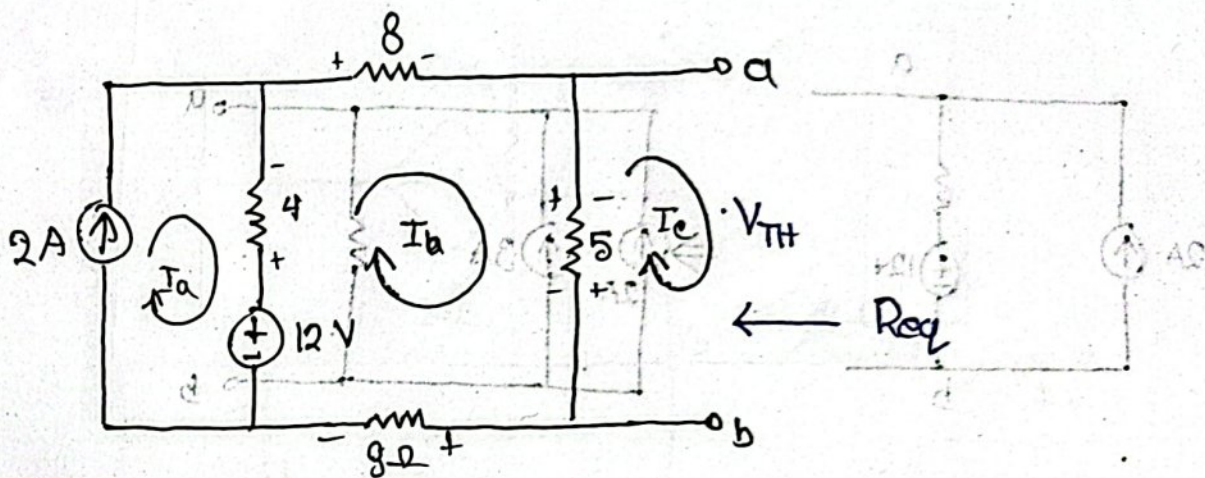
$\therefore E_{qV}$ K.T.?



$$0 = 6V - 6.2I_L - 6 + I_L 6$$

$$V_L = 6V - 6 + 6 = 6V$$

$$V_L = V_{TH} = 6V$$



⇒ #2 KVL at mesh a:

$$I_a = 2A$$

KVL at mesh b:

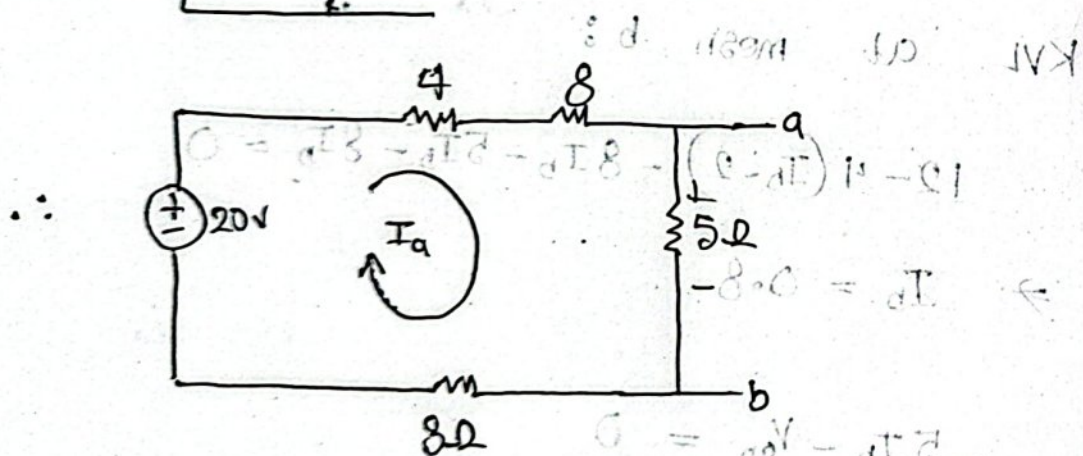
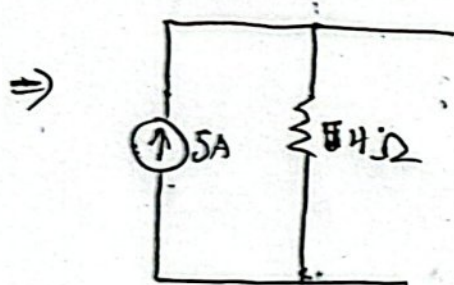
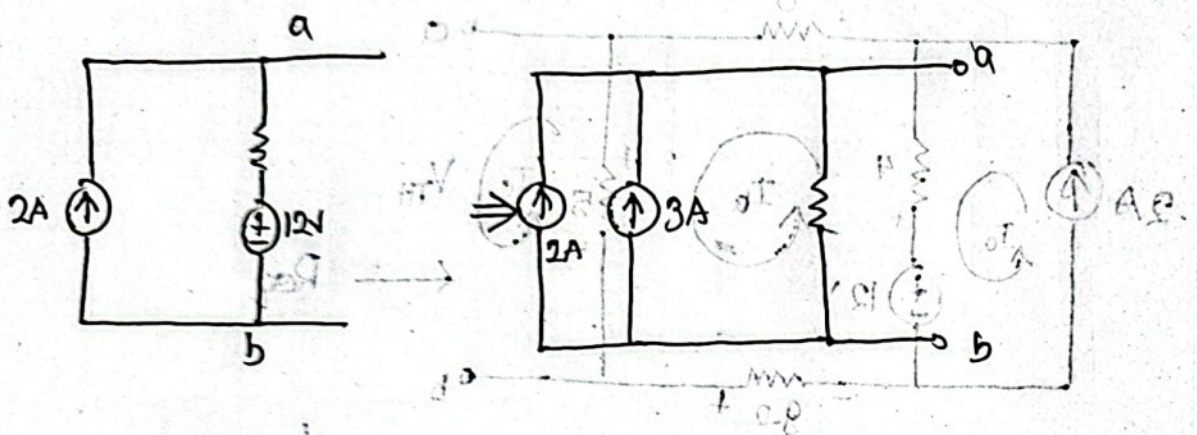
$$12 - 4(I_b - 2) - 8I_b - 5I_b - 8I_b = 0$$

$$\Rightarrow I_b = 0.8$$

$$\therefore -5I_b - V_{ab} = 0$$

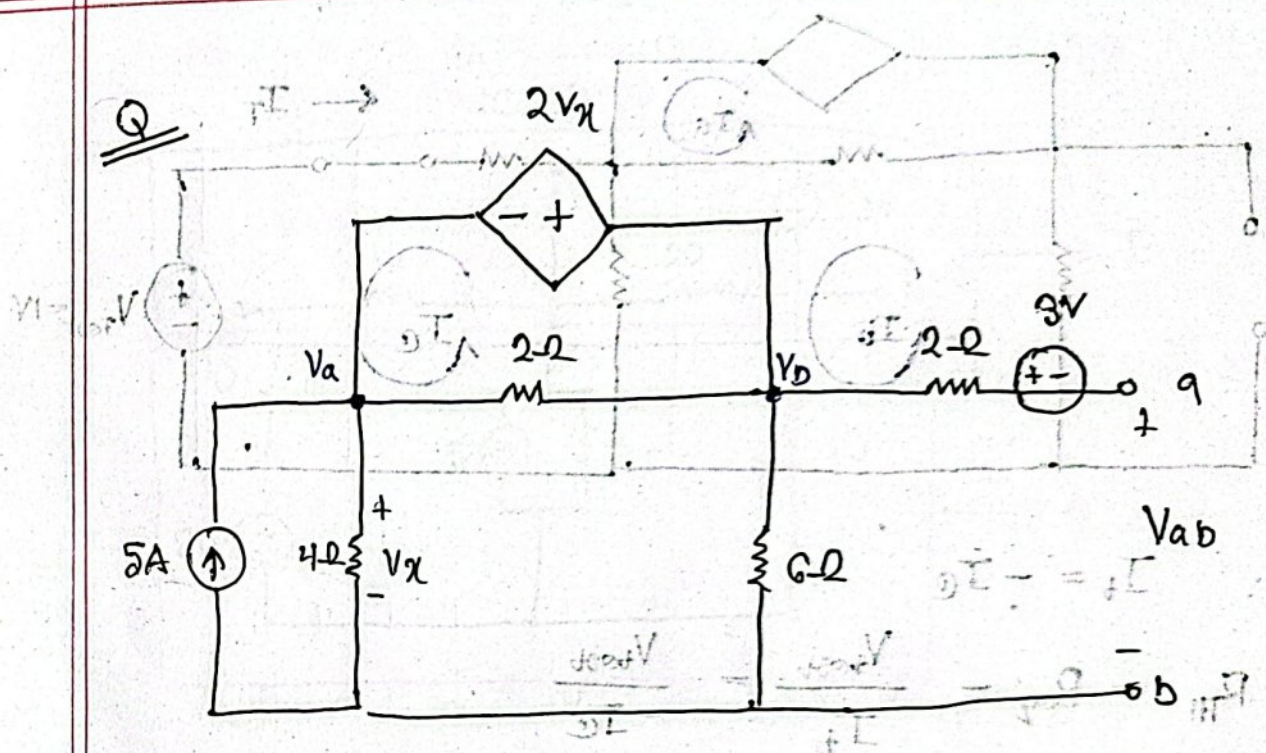
$$\therefore -5 \times 0.8 = V_{ab} \Rightarrow -4V = V_{TH}$$

2 hr 8



Now,

$R_{TH} :$



$$\therefore V_b - V_a = 2V_x$$

$$\text{KCL @ Node a: } \underbrace{-5 + \frac{V_a}{4} + \frac{V_a - V_b}{2}}_{\text{Node a}} + \underbrace{\frac{V_b - V_a}{2} + \frac{V_b}{6}}_{\text{Node b}} = 0$$

$$V_a - 0 = V_x \quad \therefore \text{yes}$$

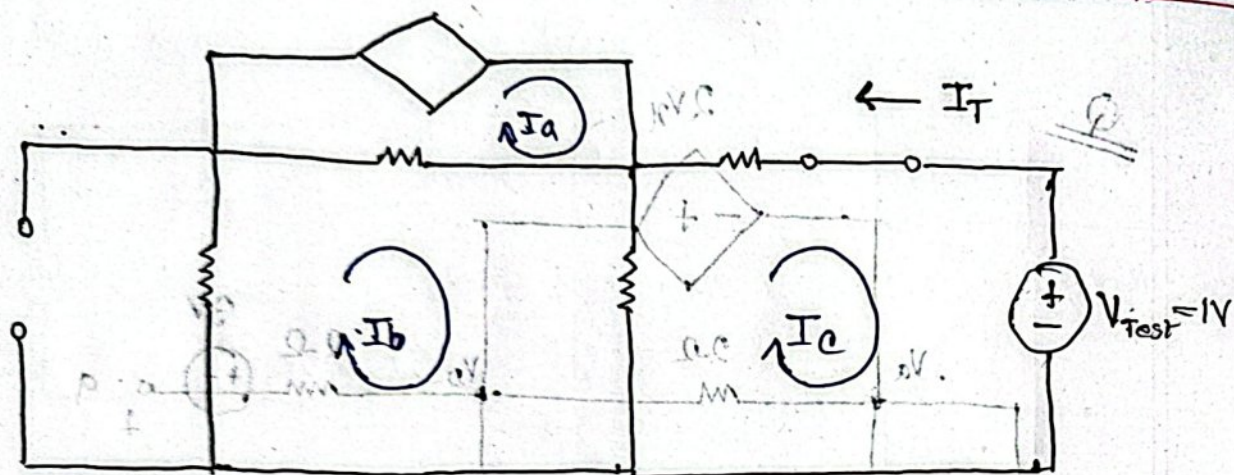
$$\therefore V_b = 3V_{x_a} \quad \therefore -5 + \frac{V_a}{4} + \frac{3V_a}{6} = 0$$

$$\therefore V_a = 6.67 \quad V_b = 20.01$$

$$\therefore V_b - 0 - V_2 - 3 - V_{ab} = 0$$

$$\therefore 20.01 - 2I_1 - 3 = V_{ab}$$

$$\therefore 17.01 = V_{ab}$$



$$\therefore I_T = -I_c$$

$$R_{TH} = R_{eq} = \frac{V_{test}}{I_T} = \frac{V_{test}}{-I_c}$$

$$V_0 = 0V = V_1 - V_2$$

$$KCL @ Node a: \frac{V_0}{5} + \frac{V_0 - V_1}{2} + \frac{V_0 - V_2}{10} = 0$$

$$V_0 - V_1 = V_2$$

$$0 = \frac{V_0}{5} + \frac{V_0 - V_2}{2} + \frac{V_0 - V_2}{10} = 0$$

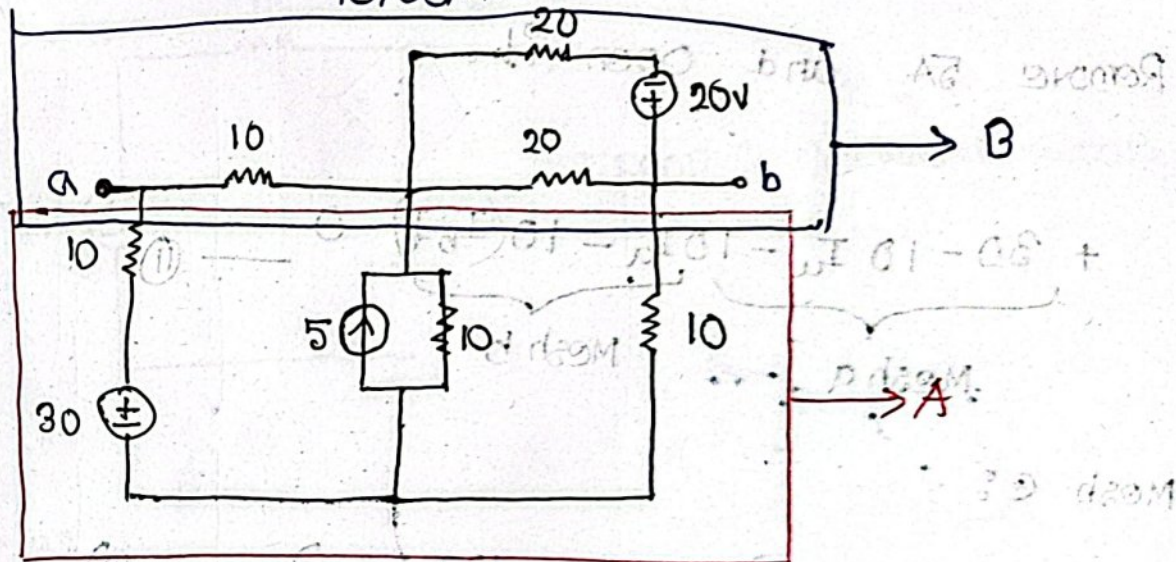
$$V_0 = 0.5V_2$$

$$V_0 - V_1 = V_2 - V_0 = 0$$

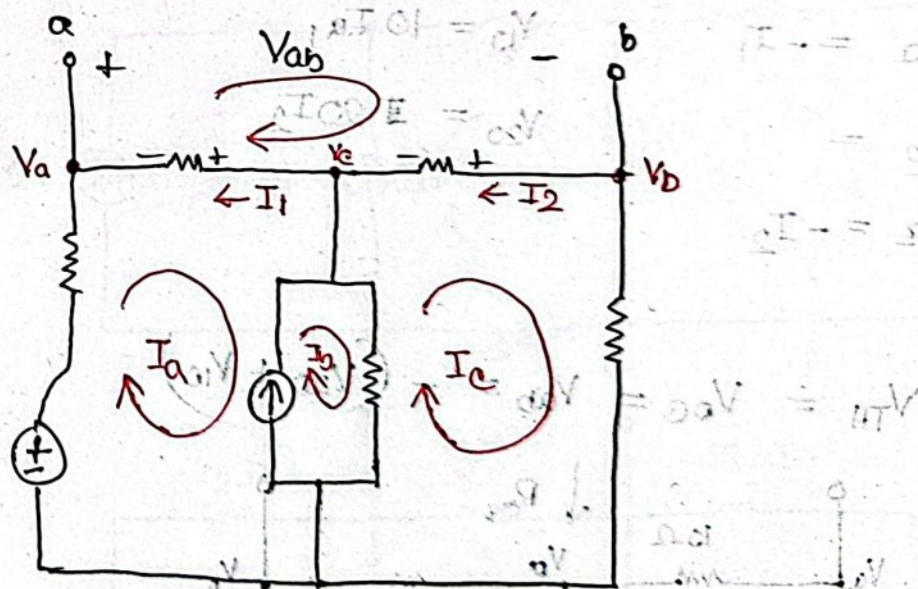
$$V_0 - V_1 = V_2 - V_0 = 0$$

$$V_0 = 0.5V_2$$

Loc-16
10/08/26



ii) Remove Network B



BYW-Δ

$$-V_{ab} - V_{20} - V_{10} = 0$$

$$-V_{ab} = V_{20} + V_{10}$$

SuperMesh eqn : $I_b - I_a = 5A$ (i)

Remove 5A and Open it.

$$+ 30 - 10 I_a - 10 I_a - 10(I_b - I_a) = 0 \quad \text{--- (ii)}$$

Mesh a Mesh b

Mesh c:

$$-10(I_c - I_b) - 20 I_c - 10 I_c = 0 \quad \text{--- (iii)}$$

$\therefore I_a = -I_1$

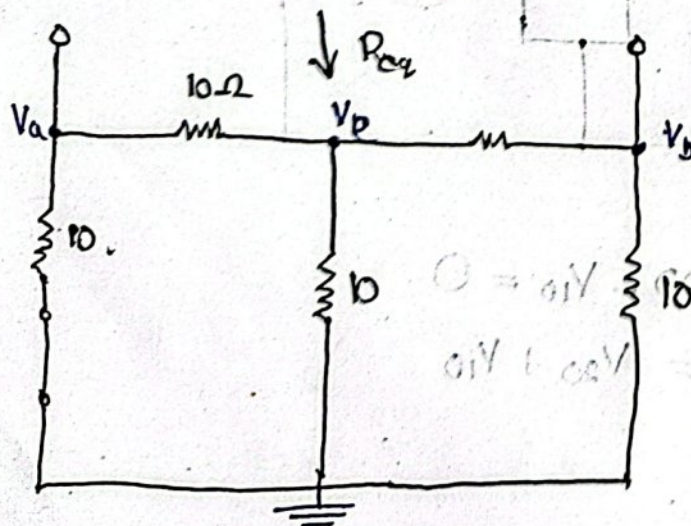
$I_b =$

$I_c = -I_2$

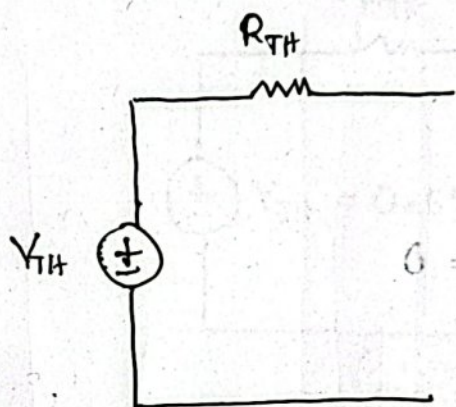
$V_b = 10 I_a$

$V_{20} = 20 I_2$

$V_{TH} = V_{OC} = V_{ab} = -(V_{20} + V_{10})$



Δ -WYE
नाजदर



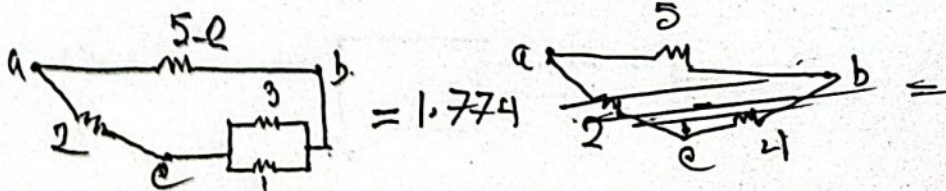
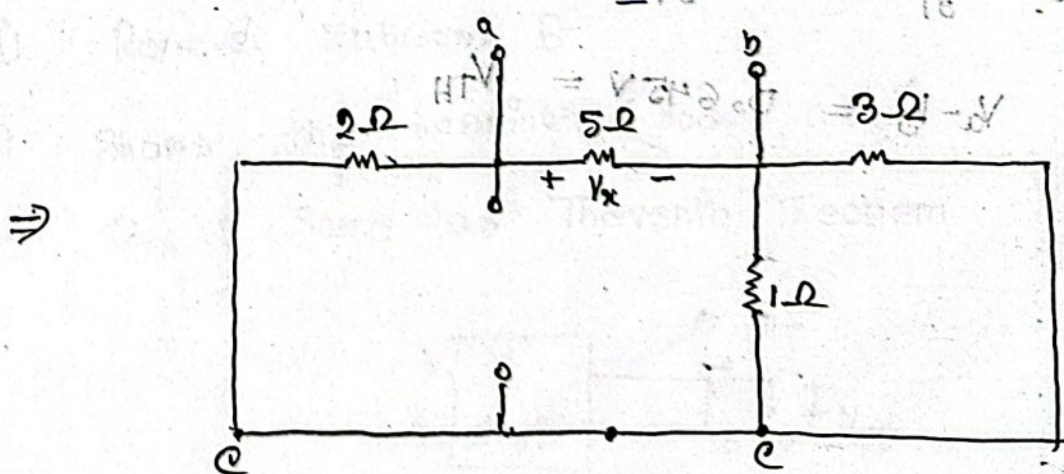
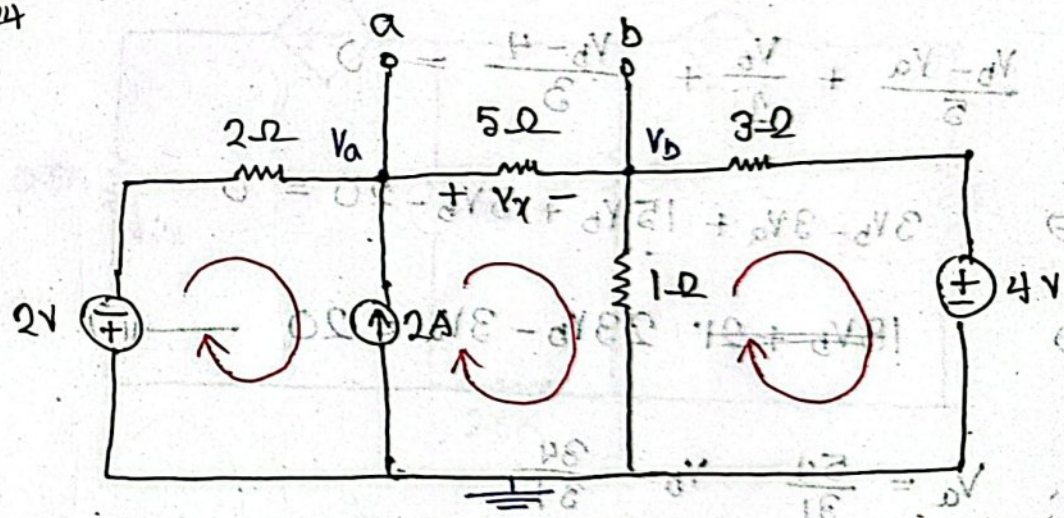
$$C = \frac{V - V'}{R} + R = \frac{V + V'}{R}$$

ERROR in Network choosing

$$C = \frac{V - V'}{R} + R = \frac{V + V'}{R}$$

$$C = \frac{V - V'}{R} + R = \frac{V + V'}{R}$$

Q. 32
Fig 5.74



$$= 1.774$$

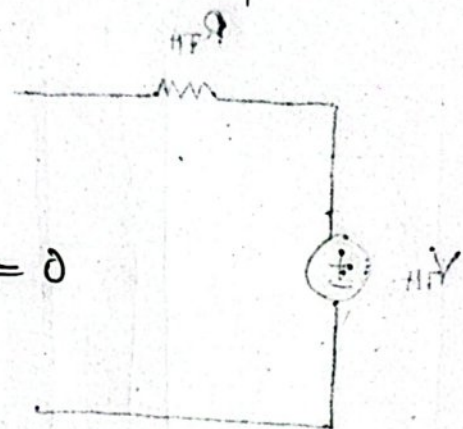
KCL @ node a:

$$\frac{V_a + 2}{2} - 2 + \frac{V_a - V_b}{5} = 0$$

ERROR in Network (Circuit)

$$\Rightarrow 5 + 5V_a + 10 - 20 + 2V_a - 2V_b = 0$$

$$\Rightarrow 7V_a - 2V_b = 10 \quad \text{--- (i)}$$



KCL @ Node b:

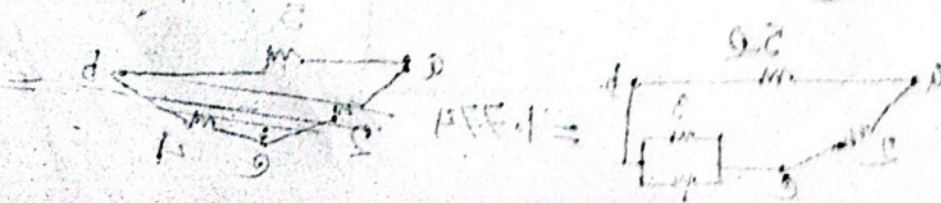
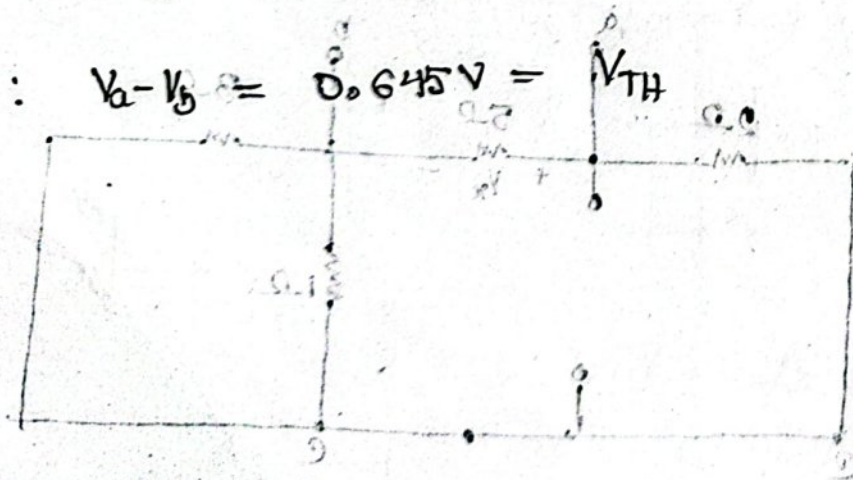
$$\frac{V_b - V_a}{5} + \frac{V_b}{1} + \frac{V_b - 4}{3} = 0$$

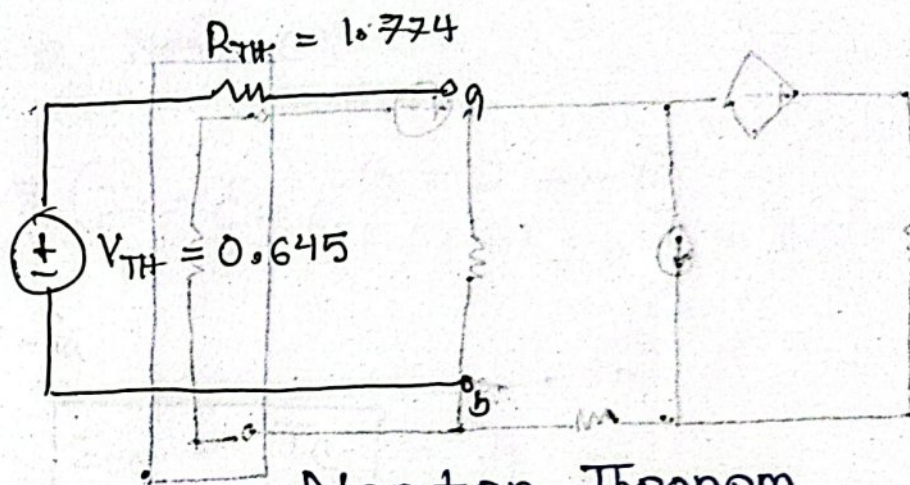
$$\Rightarrow 3V_b - 3V_a + 15V_b + 5V_b - 20 = 0$$

$$\Rightarrow 18V_b - 3V_a = 20 \quad \text{--- (ii)}$$

$$\therefore V_a = \frac{54}{31} \quad V_b = \frac{34}{31}$$

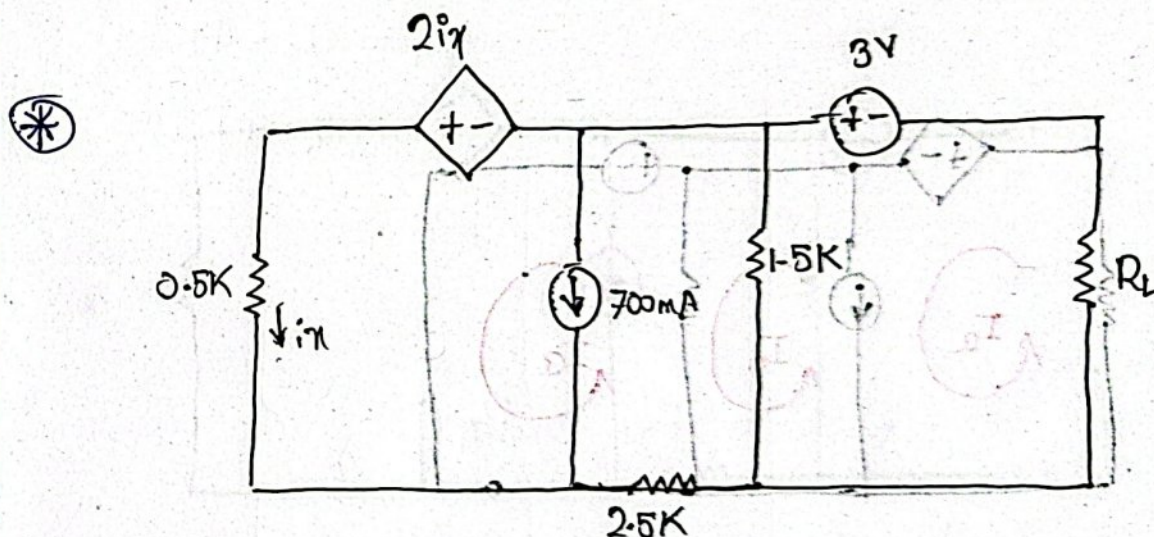
$$\therefore V_a - V_b = 0.645V = V_{TH}$$





Norton Theorem

Remove load and short-circuit terminals a-b

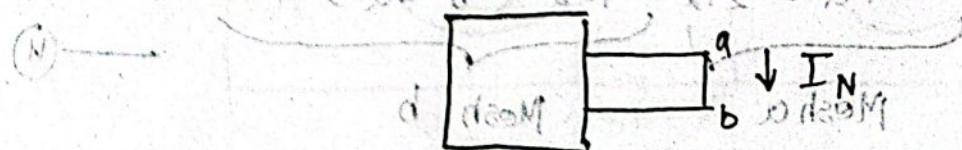


i) Remove Network B. $A_{NOC} = I_a - I_b$

ii) Short-circuit the terminals (nodes a, b)

iii) R_{TH} : Same as Thevenin Theorem

$$0 = I_a - I_b - (I_a - I_b) \cdot R_{TH} - I_b \cdot R_{TH}$$



$$0 = I_a - I_b - (I_a - I_b) \cdot R_{TH} - I_b \cdot R_{TH}$$

KVL @ Mesh ③

$$0 = 0.2I_c - (1 + 0.2I_c)2.1 - 0.2I_c - 1.5(I_c - I_b) - 3 = 0 \quad \text{--- (iii)}$$

① $i_x = -i_a$

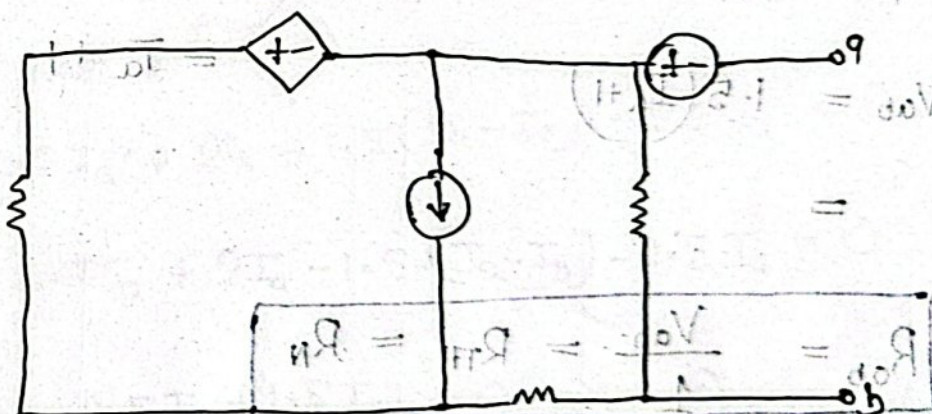
$\therefore I_{ac} = I_c = I_N$

$2.1 - 0.2I_c = 1.5$

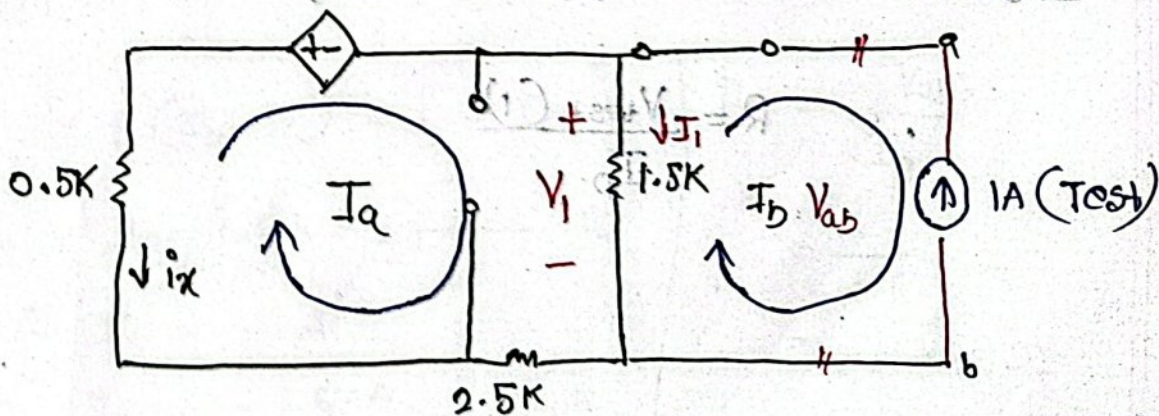
$0 = 0.2I_c - 1.5 - 3$

$0 = 0.2I_c - 4.5$

$\Rightarrow V_{oc} = 1.5 - 1.5 = 0$



Instead of 1A test source $\Rightarrow$ In test source $\Rightarrow$



$I_b = 1A$

$$-0.5I_a - 2i_x - 1.5(I_a + 1) - 2.5I_a = 0$$

$$\Rightarrow -0.5I_a + 2I_a - 1.5(I_a + 1) - 2.5I_a = 0$$

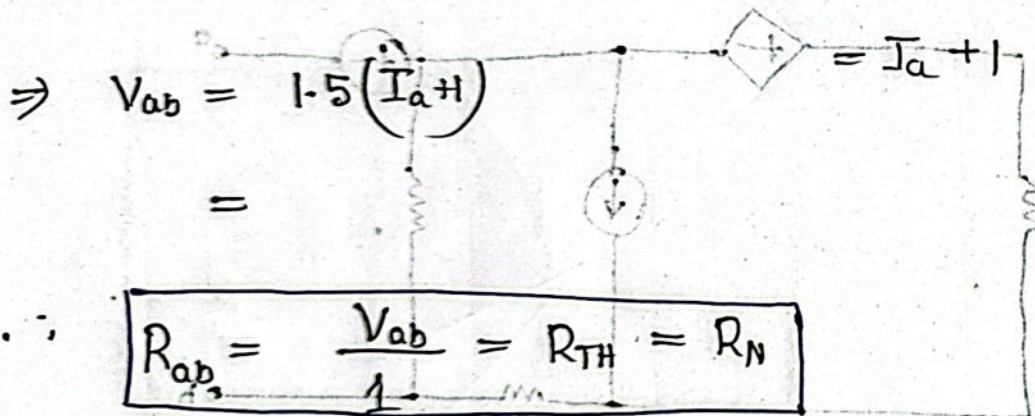
1A Test Source

$$+V_1 - V_{ab} = 0$$

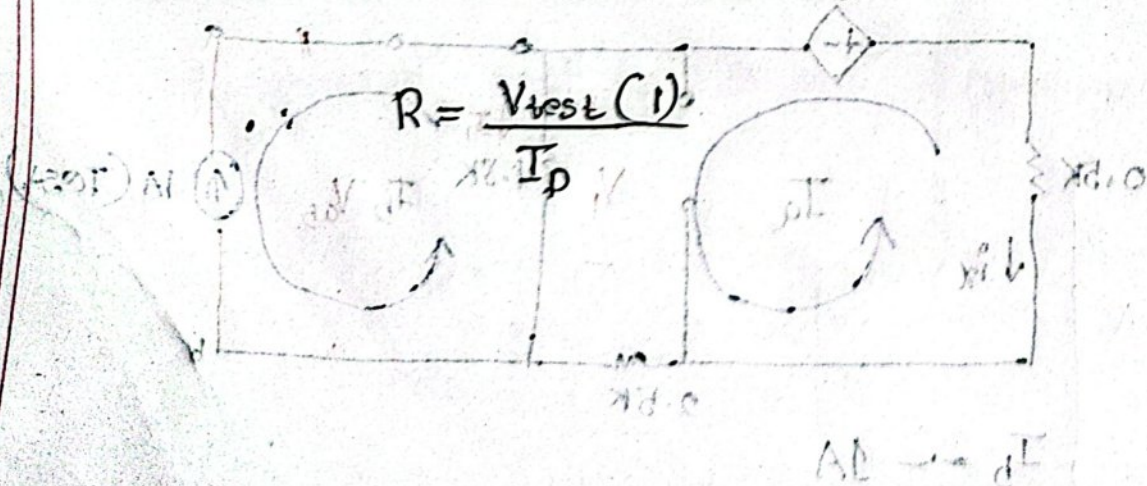
$$I_1 = I_a - I_b$$

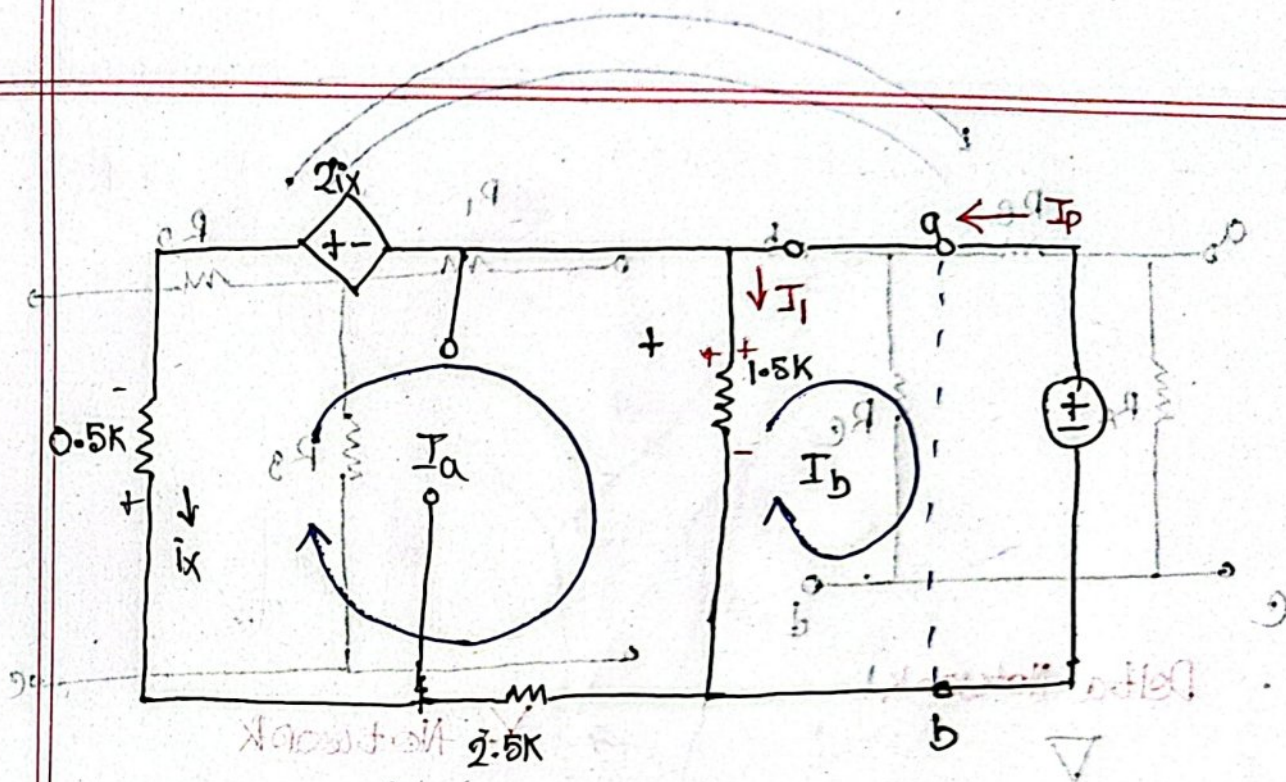
$$1.5I_1 - V_{ab} = 0$$

$$I_1 = I_a - I_b$$



Instead of 1A Test Source, Use 1V Test Source





KVL @ Mesh a:

$$-0.5I_a + 2i_x - 1.5(I_a - I_b) - 2.5I_a = 0$$

$$-0.5I_a + 2I_a - 1.5(I_a - I_b) - 2.5I_a = 0$$

$$\Rightarrow -2.5I_a + 1.5I_b = 0 \quad \text{--- (i)}$$

KVL @ Mesh b:

$$-1.5(I_b - I_a) - 1 = 0 \quad \Downarrow$$

$$1.5I_a - 1.5I_b = 1 \quad \text{--- (ii)}$$

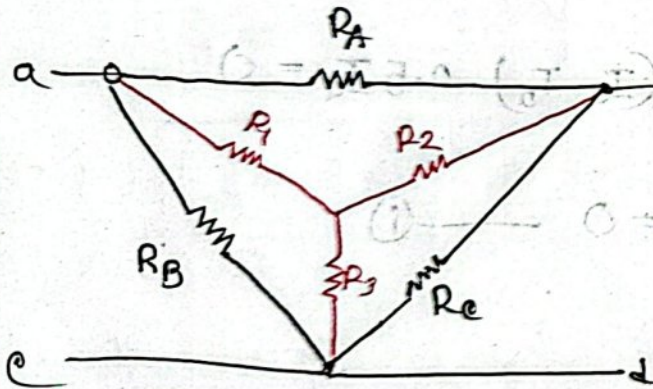
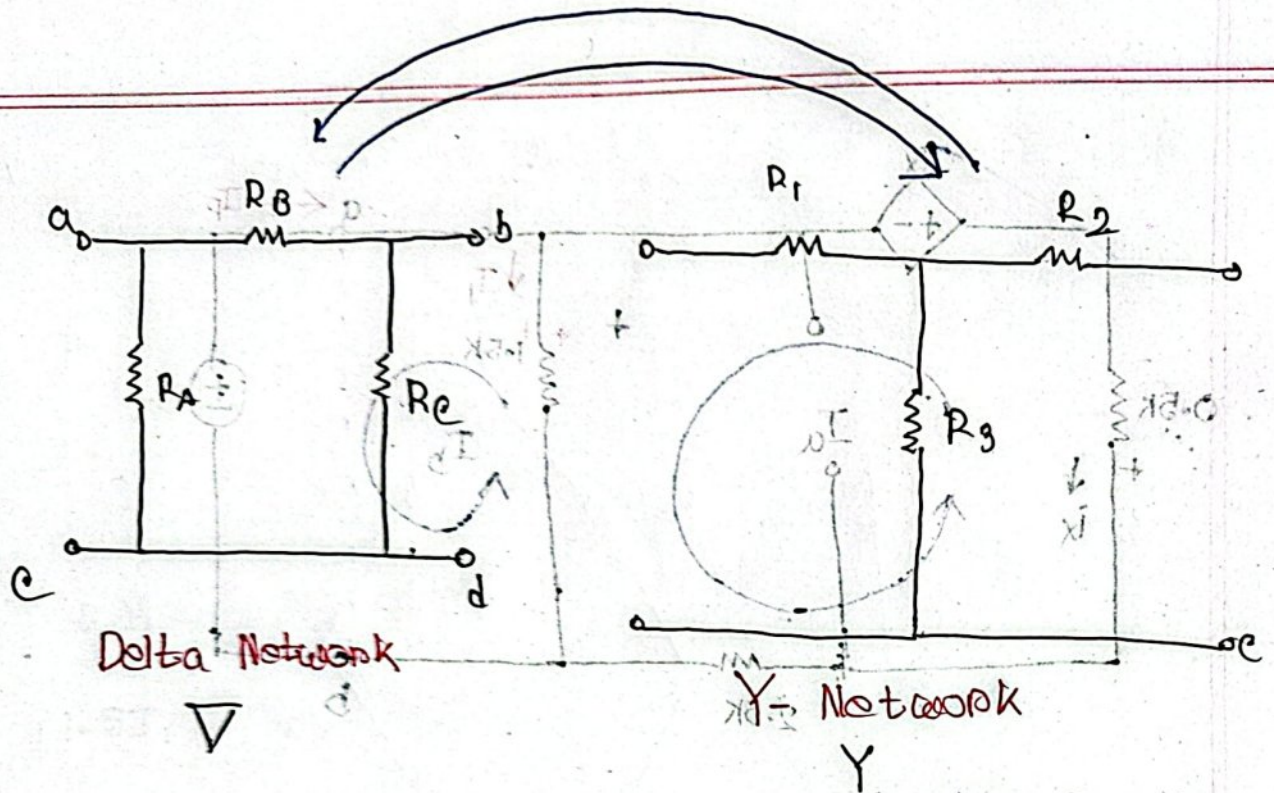
$$I_a = -1A$$

$$I_b = -1.67A$$

$$\therefore I_p = -I_b = 1.67A$$

$$R_{eq} = \frac{V_{ab}}{I_p}$$

$$= \frac{1}{0.5988} = 1.67A$$



$$R_1 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_B R_C}{R_A + R_B + R_C}$$

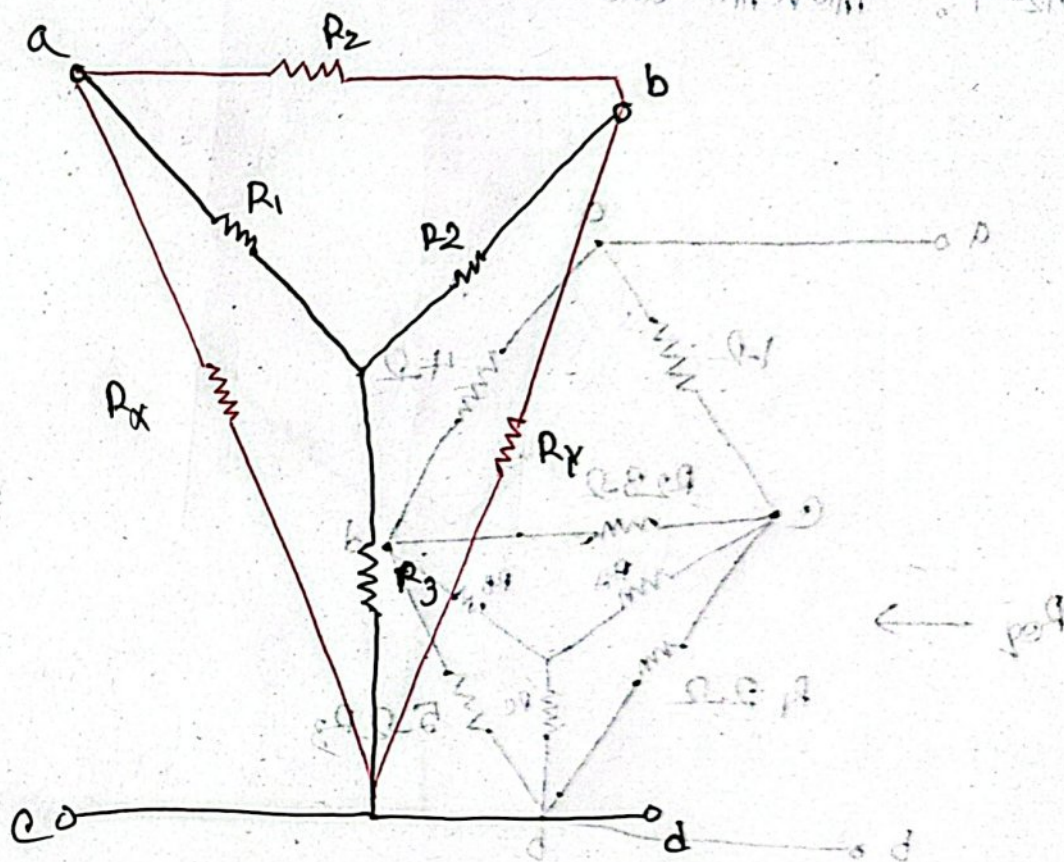
$$\Downarrow \quad C = 1 - (I_A - I_B) R_C -$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_1 = \frac{R_A R_B}{R_A + R_B + R_C}$$

Adjacent



$$R_x = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_y = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_z = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

[Opposite]

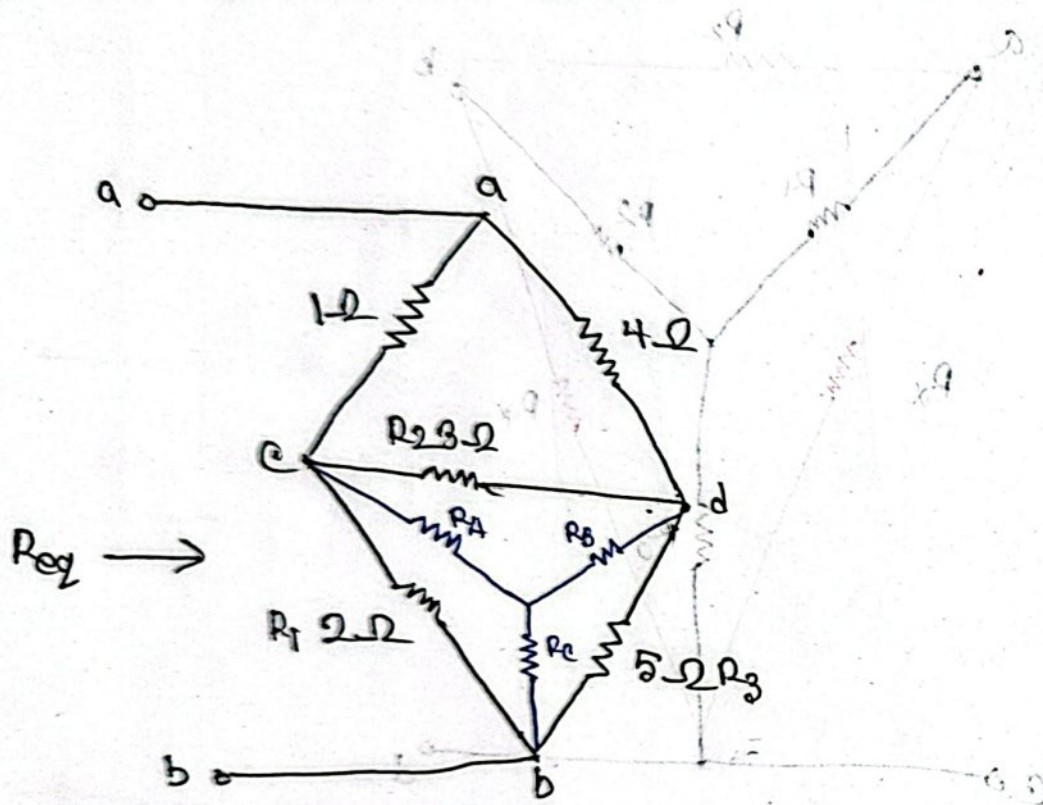
[a, b, c]

$$\frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = R_z$$

[a, b, c]

LEC-17

Quiz-4: Thevenin and Norton



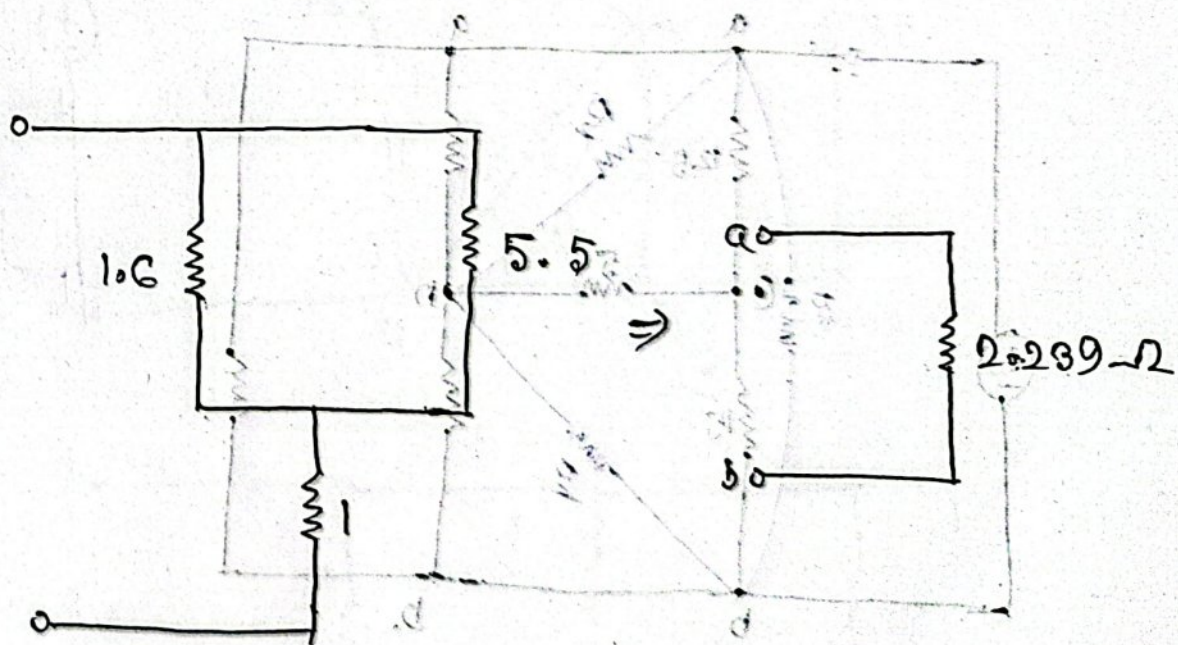
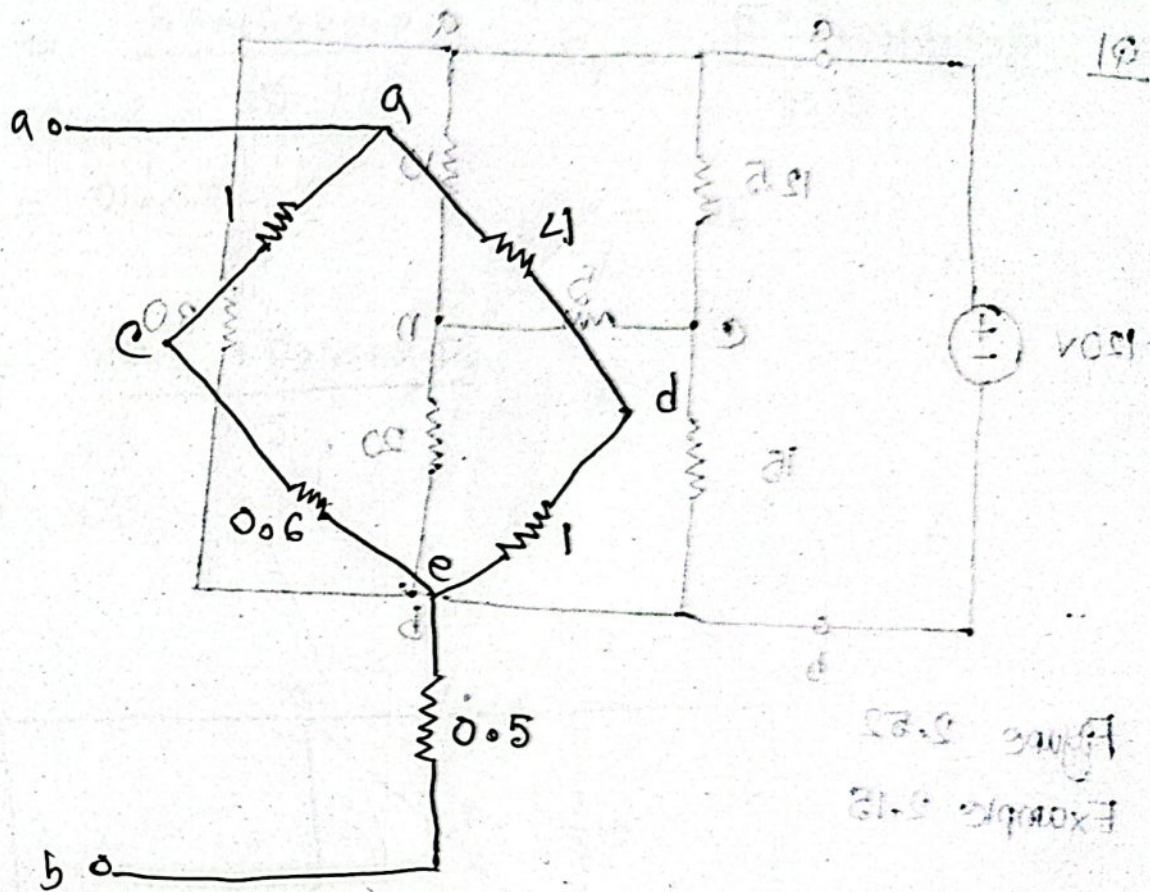
① Label the Nodes

② Ask $\rightarrow$

③ No series, No parallel

$$R_A = \frac{R_1 R_2}{R_1 + R_2 + R_3} \quad R_B = \frac{R_2 R_3}{R_1 + R_2 + R_3} \quad \boxed{1}$$

$$R_C = \frac{R_1 R_3}{R_1 + R_2 + R_3} \quad \boxed{0.5 \Omega}$$



Q1

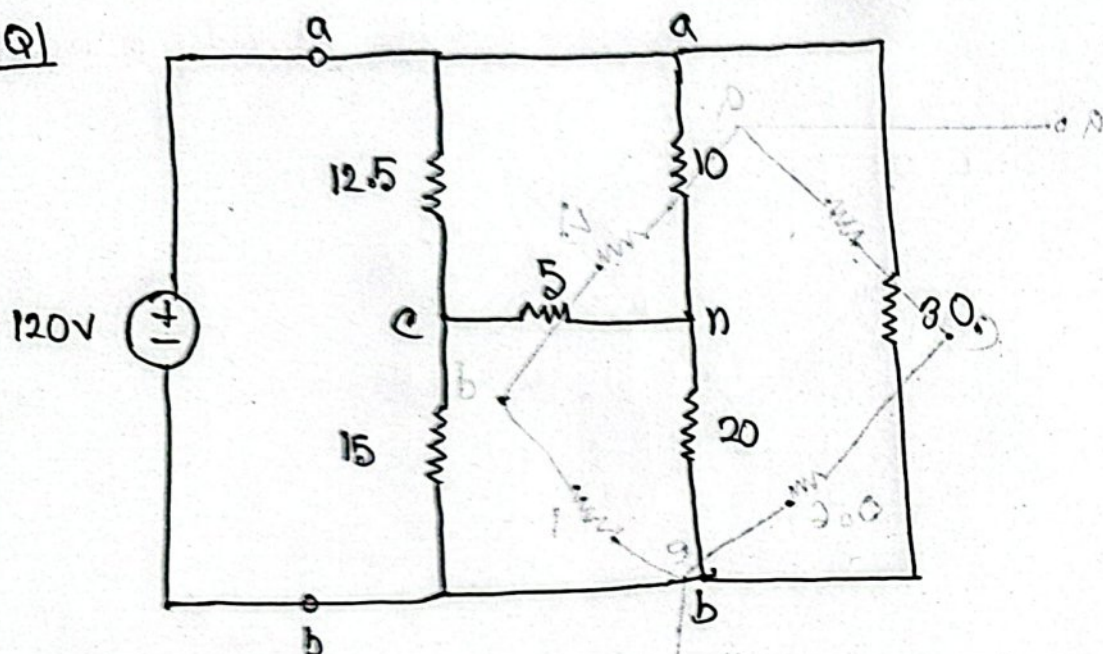
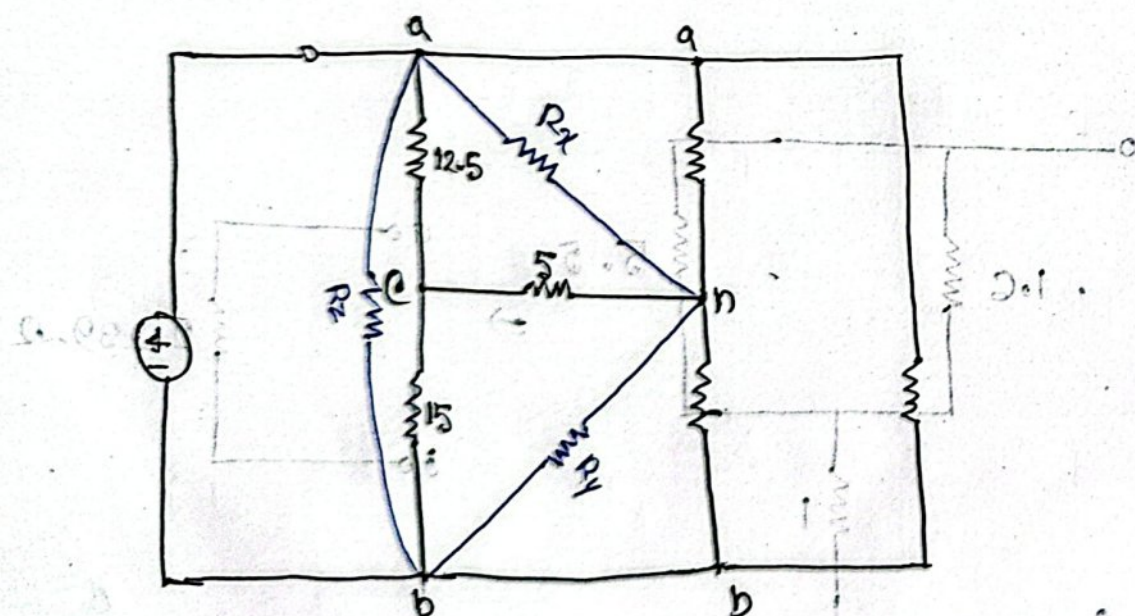


Figure 2.52

Example 2.15



$$R_x = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{15}$$

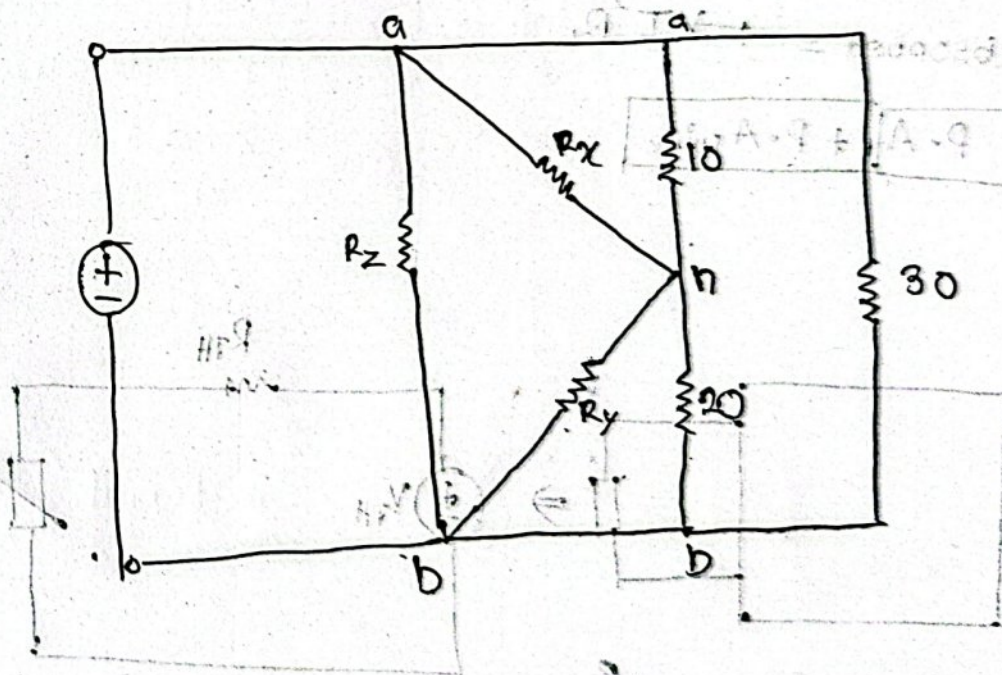
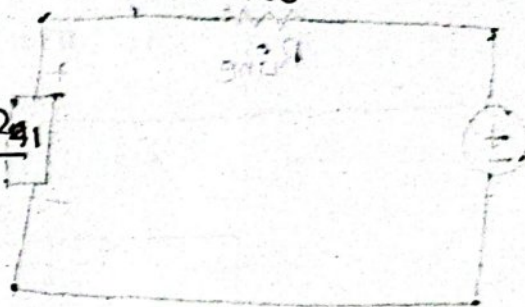
$$R_y = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{12.5}$$

$$= 21.67$$

$$= 26$$

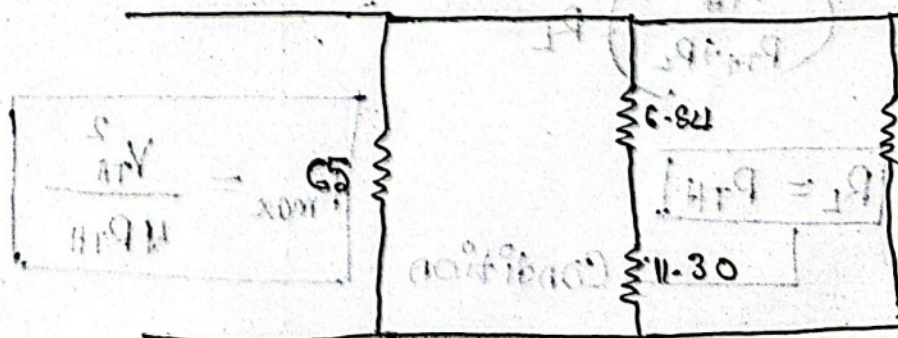
$$R_z = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{5}$$

$$= 65$$



$$6.842$$

$$\frac{260}{28}$$

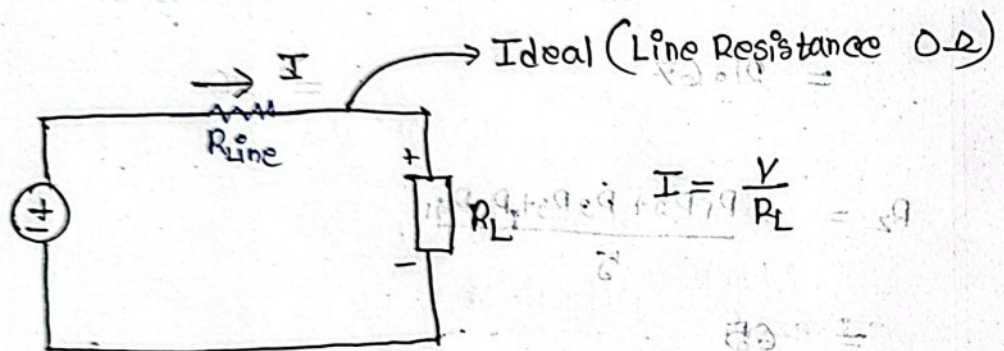


$$\frac{120}{2.63}$$

$$\Rightarrow 120 \text{ V} \Rightarrow 9.63 \text{ A} \Rightarrow I = \frac{120}{2.63} = 45.6 \text{ A}$$

L15/09-08
L17/12-08

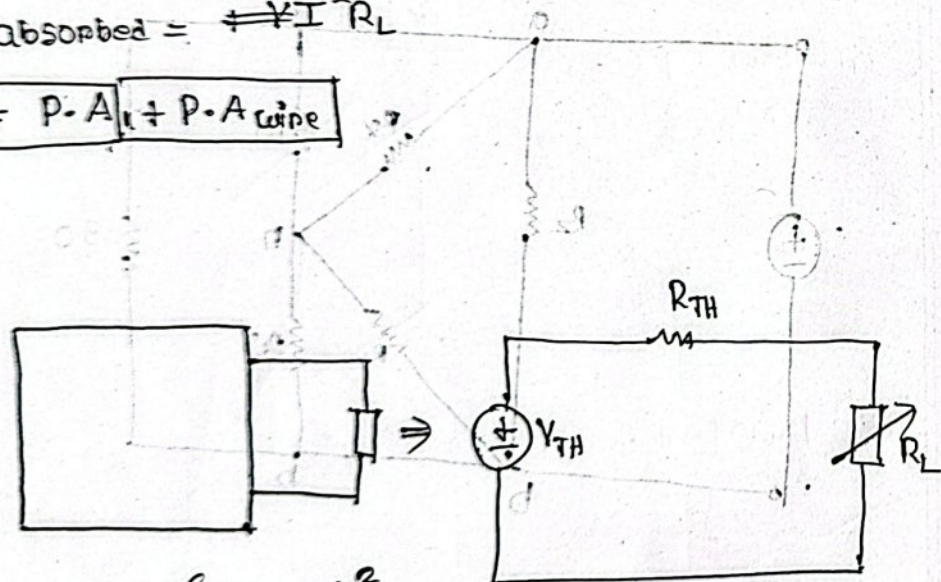
Maximum Power Transfer Theorem:



Power delivered = $-VI$

absorbed = $+I^2 R_L$

$P.D = P.A + P.A_{wire}$



$P = i_1^2 R_L = \left(\frac{V_{TH}}{R_{TH} + R_L} \right)^2 R_L$

$\frac{dP}{dR_L}$

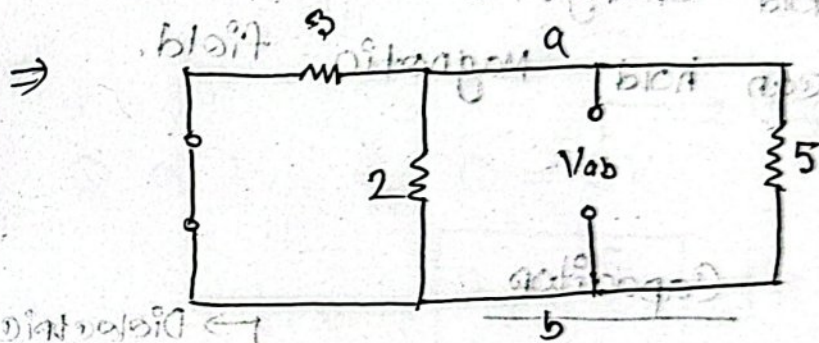
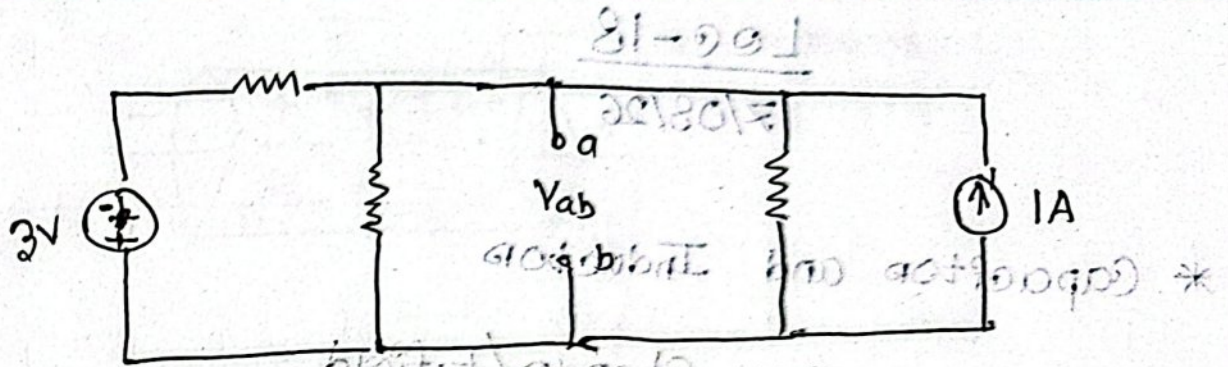
$\Rightarrow$

$R_L = R_{TH}$

Condition

$P_{max} = \frac{V_{TH}^2}{4 R_{TH}}$

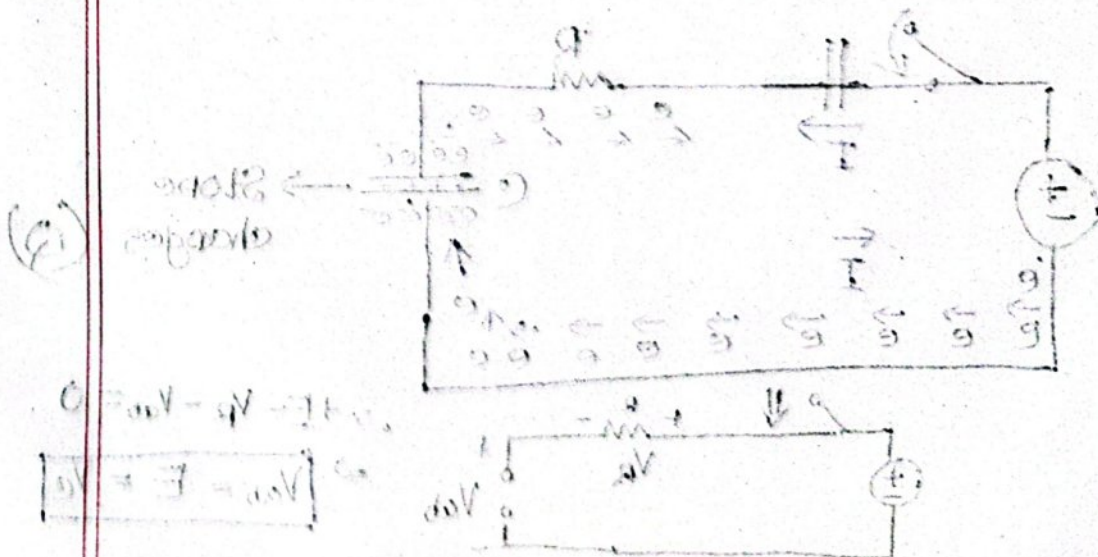
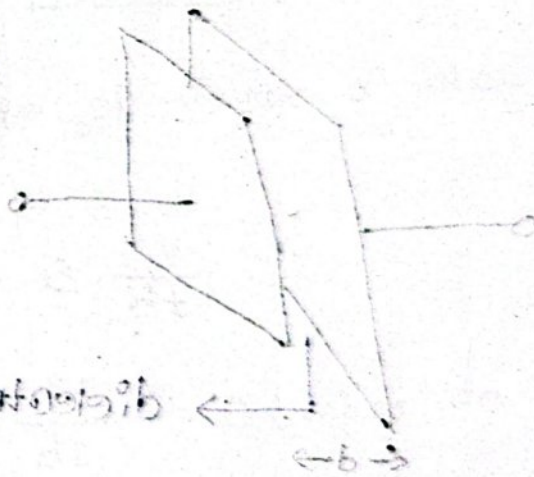
$\frac{dP}{dR_L} = \frac{0.01}{0.01} = 1$



$$\therefore P_{\text{avg}} = \frac{30}{31} \text{ W}$$

$$V_{\text{TH}} = \text{TBD}$$

$$\frac{A \cdot b \cdot 3}{b} = 0$$



L18/17-08

Capacitor
+
Inductor

L15/09-08
L17/12-08

Lee-18

17/08/26

* Capacitor and Inductor

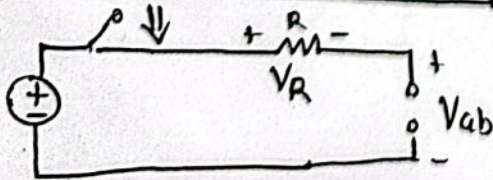
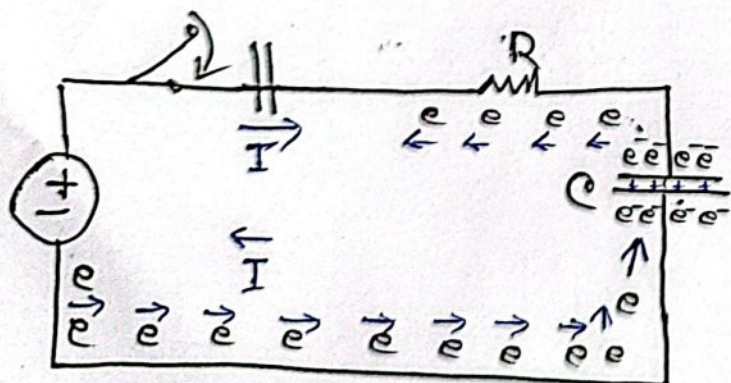
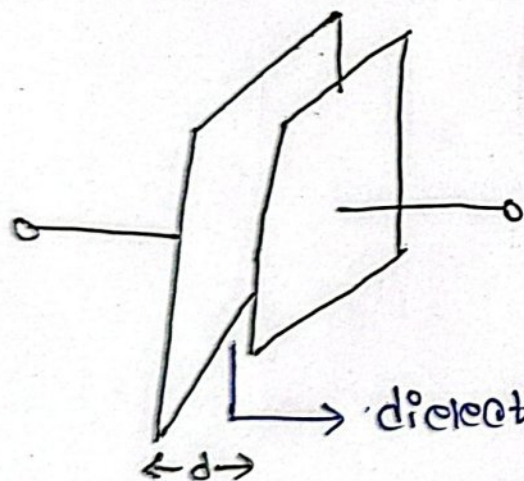
⇒ Capacitors Hold Charge/E-field

⇒ Inductors can hold Magnetic field.

Capacitor

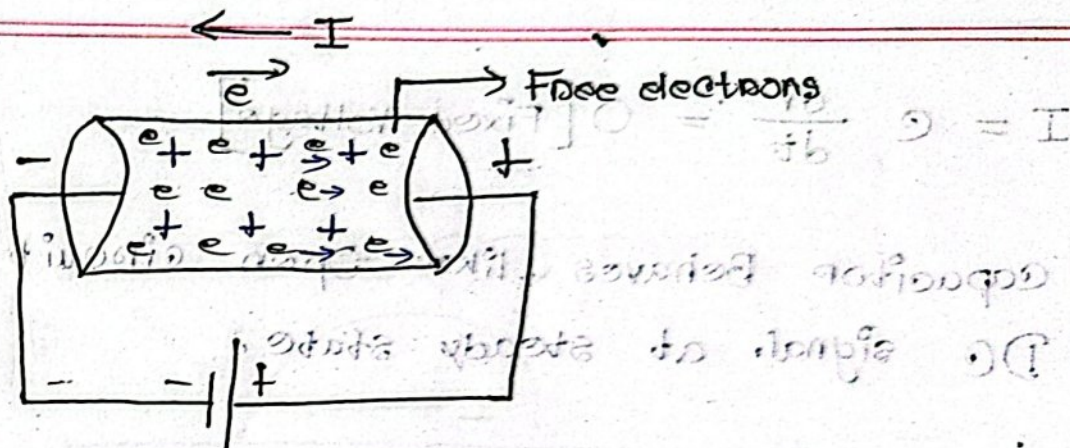
→ Dielectric constant

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$



$$\therefore +E - V_R - V_{ab} = 0$$

$$\Rightarrow \boxed{V_{ab} = E = V_c}$$



Capacitance (Farad)

$$Q \propto V \Rightarrow Q = CV$$

$$C = \frac{Q}{V}$$

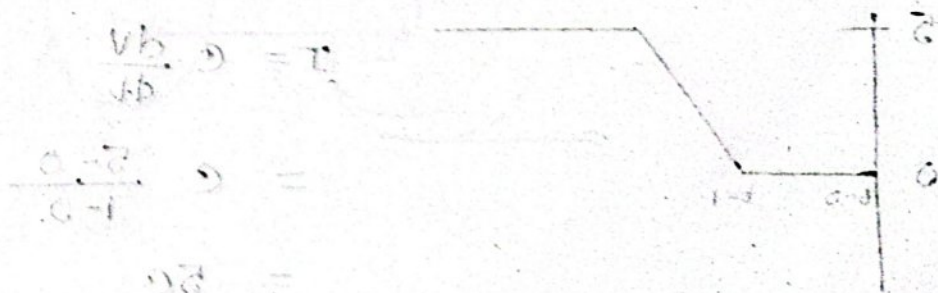
$$1F = \frac{1 \text{ Coulomb}}{1V}$$

rate of change of charge

$$I = \frac{dQ}{dt}$$

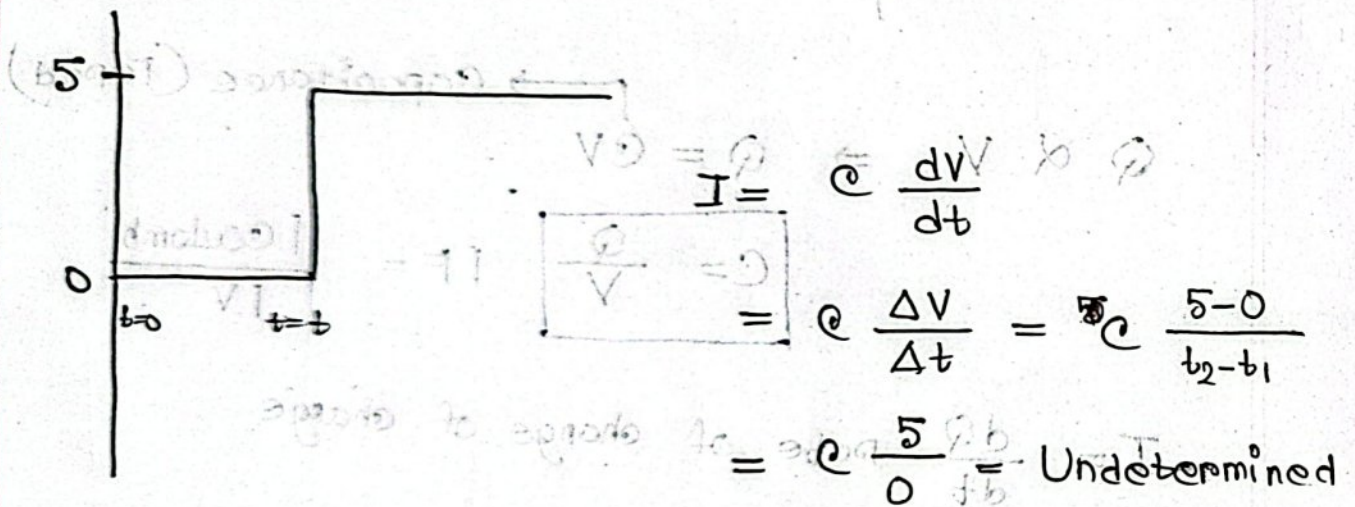
$$\Rightarrow \frac{dQ}{dt} = C \frac{dV}{dt}$$

$$\therefore I = C \frac{dV}{dt} \quad (\text{when } \frac{dV}{dt} \neq 0)$$

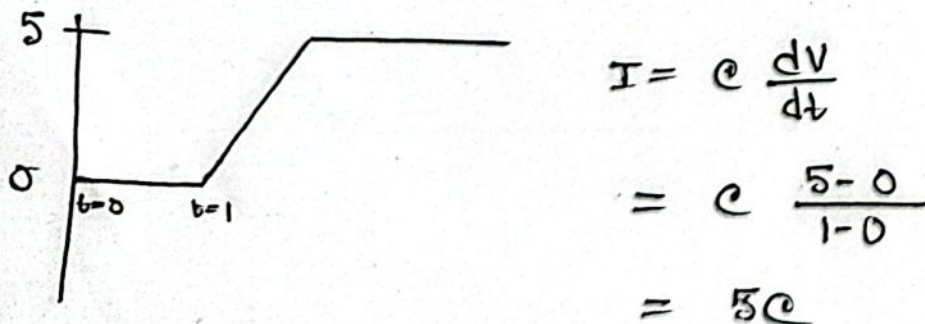


$$I = C \frac{dV}{dt} = 0 \text{ [Fixed Voltage]}$$

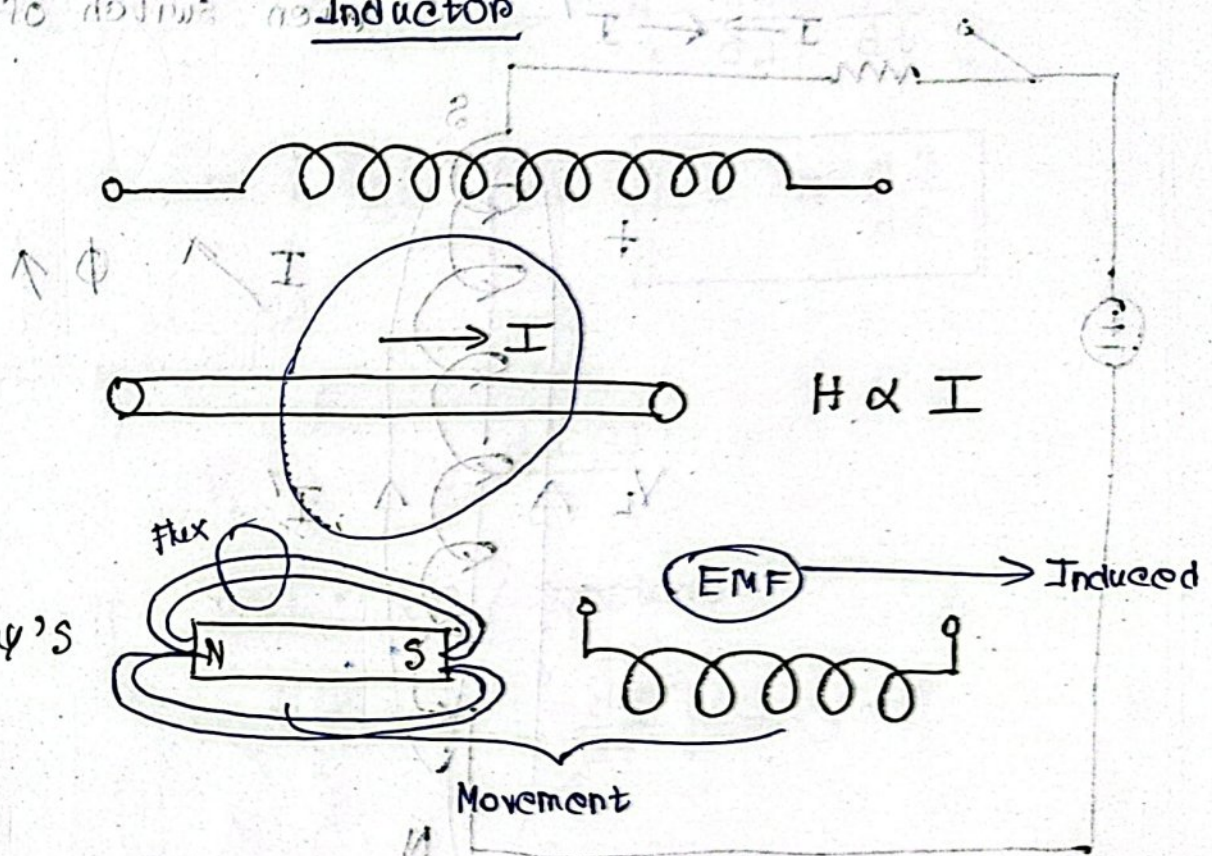
* Capacitor Behaves like Open circuit to DC signal at steady state.



* Voltage across the capacitor can't change instantaneously.



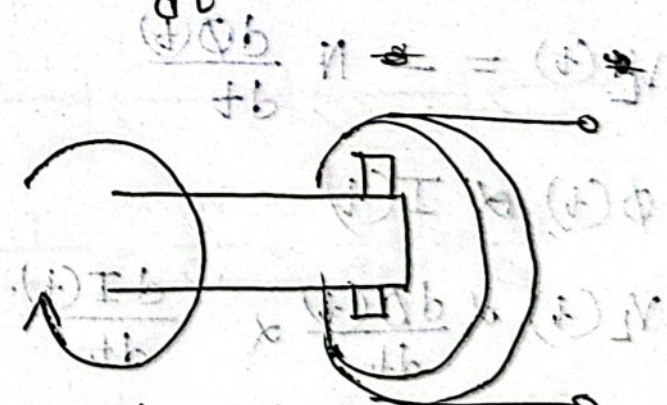
Inductor



Faraday's Law

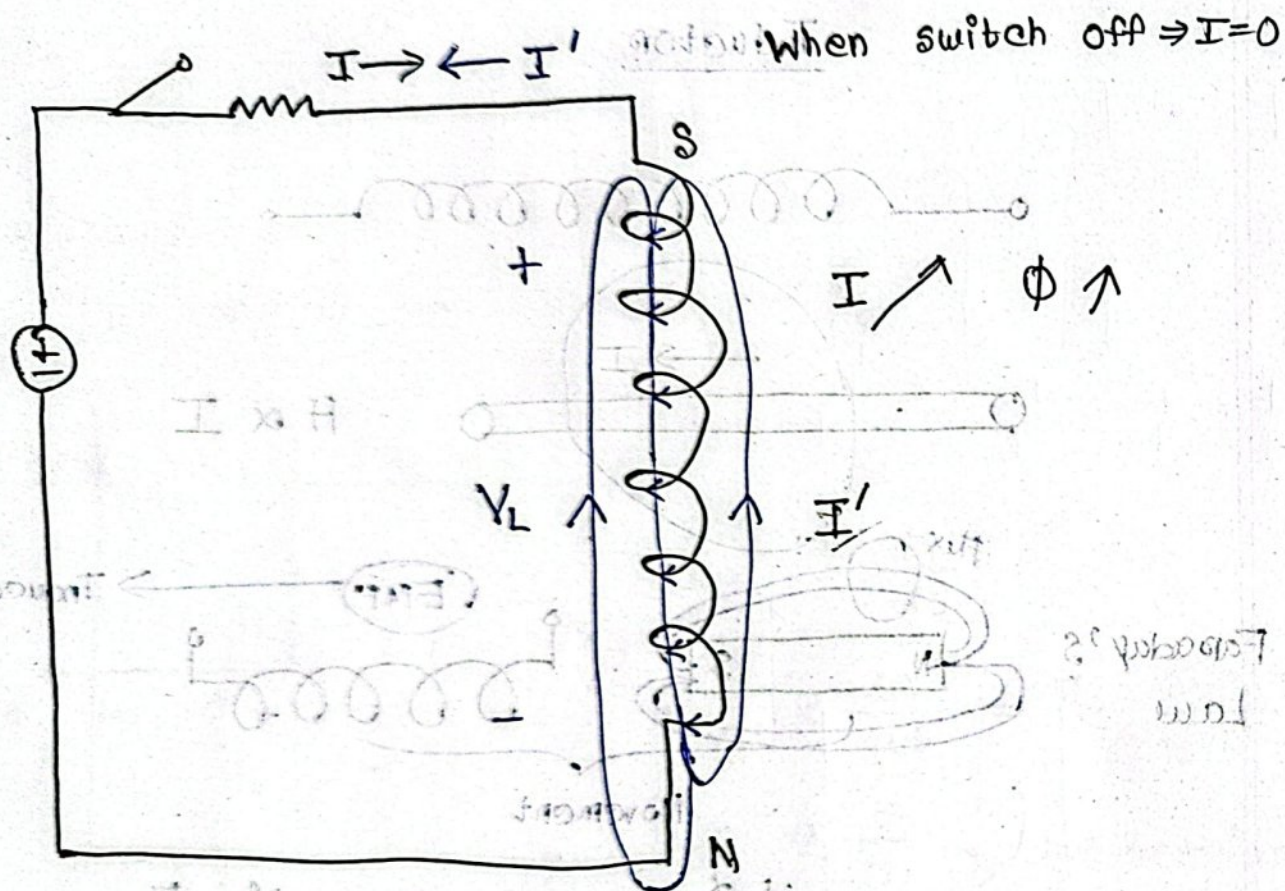
$$EMF = -N \frac{d\Phi}{dt}$$

$\Phi = \text{Magnetic Flux}$



$$EMF = N \frac{d\Phi}{dt}$$

Induced EMF



$$V_L(t) = -N \frac{d\Phi(t)}{dt}$$

$$\Phi(t) \propto I(t)$$

$$V_L(t) \propto \frac{d\Phi(t)}{dt} \propto \frac{dI(t)}{dt}$$

$$\therefore V_L(t) \propto \frac{dI(t)}{dt}$$

$$V_L(t) = L \frac{dI(t)}{dt}$$

$\longrightarrow$ Inductance

$$V_L(t) = N \frac{d\Phi}{dt} = N \frac{d\Phi}{dI} \frac{dI}{dt}$$

$$= L \frac{dI}{dt} \quad \therefore \boxed{L = N \frac{d\Phi}{dI}}$$

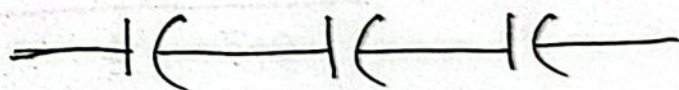
Again,

$$V_L(t) = L \frac{dI(t)}{dt}$$

if $I(t) : dc / \text{constant}$

$$V_L(t) = L \frac{dI}{dt} = 0$$

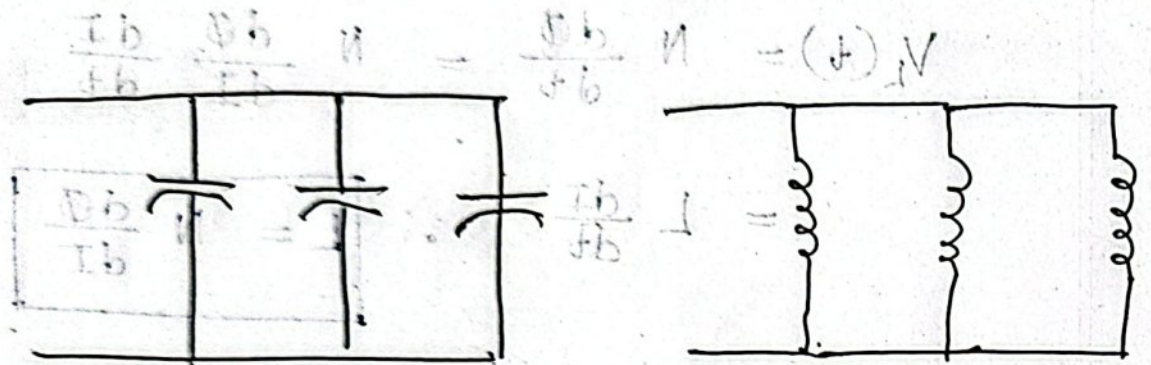
* Inductor to DC is short



$$\Rightarrow \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

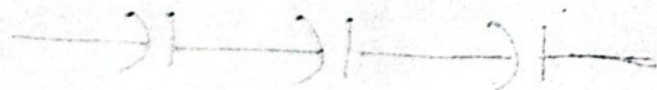
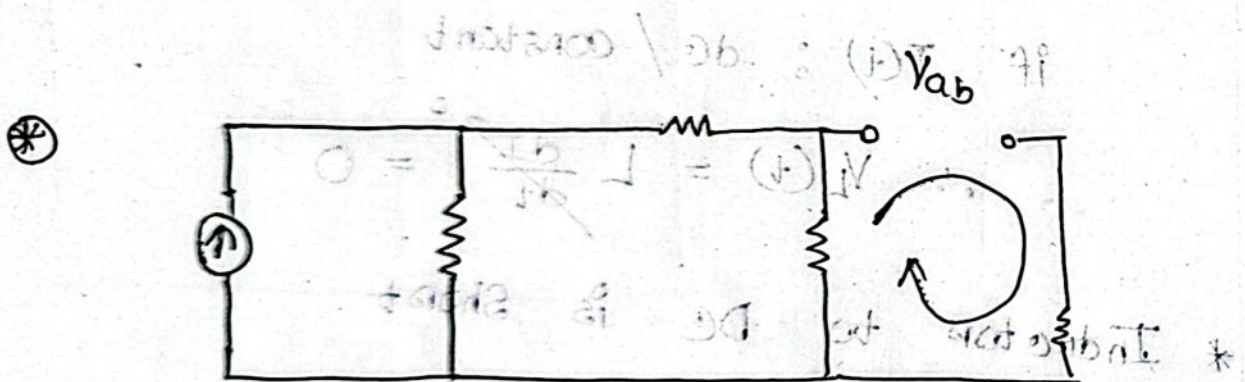


$$L_{eq} = L_1 + L_2 + L_3$$

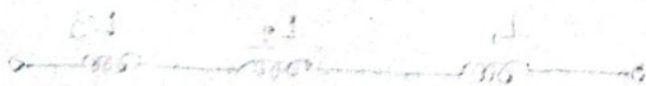


$C_{eq} = C_1 + C_2 + C_3 \frac{I_b}{V_b} \quad I = (I) N$

$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$



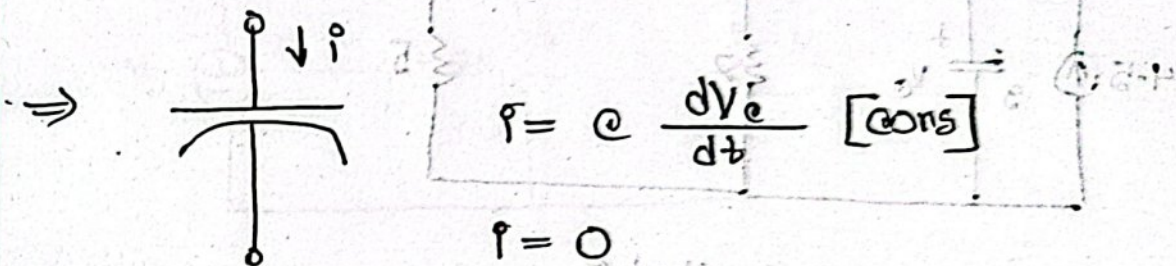
$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$



Thevenin voltage

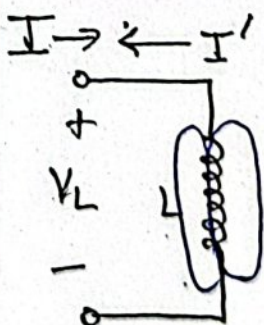
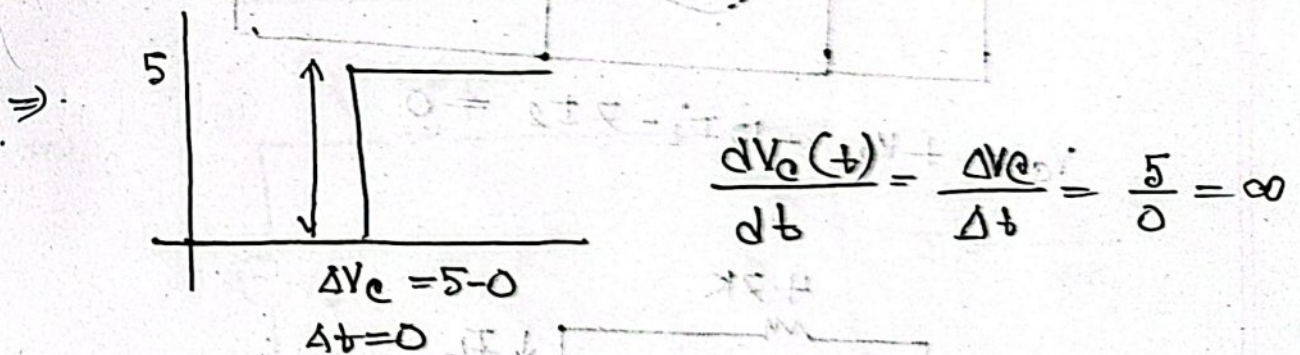
Source Free
RL/RC

L-19
19/08/20



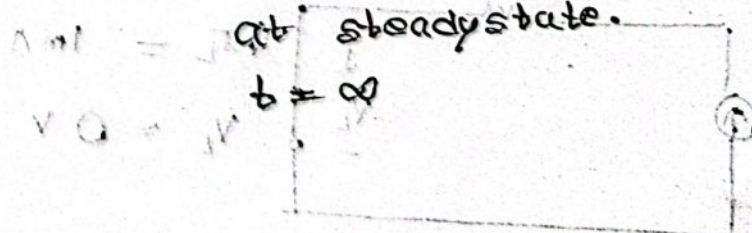
⇒

$i = 0$ * capacitor is open ckt to DC signal at steady state.
 $t = \infty$

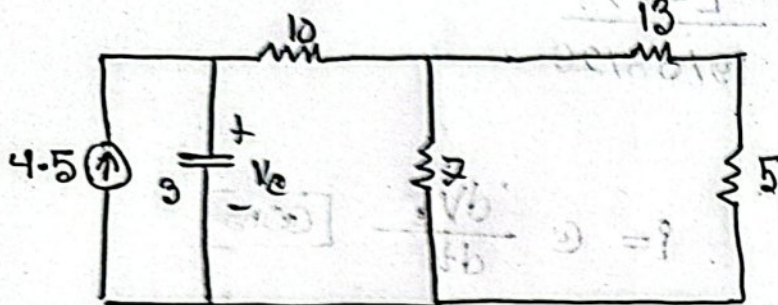


$$v_L(t) = L \frac{di(t)}{dt}$$

* Inductor is short ckt to DC signal at steady state.
 $t = \infty$



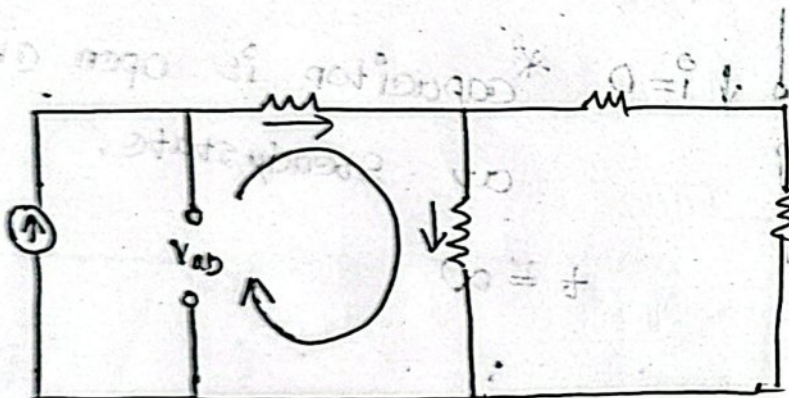
Q1



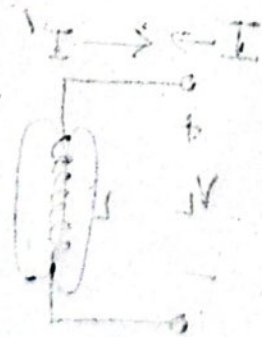
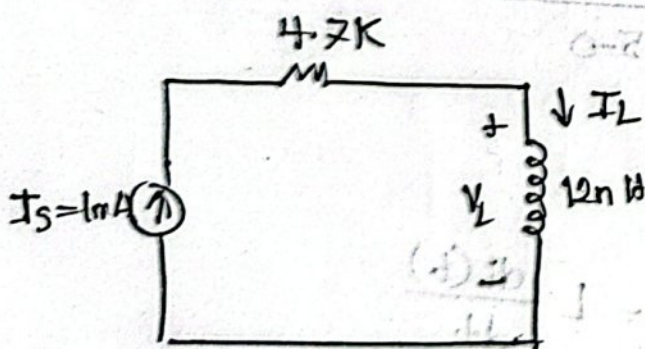
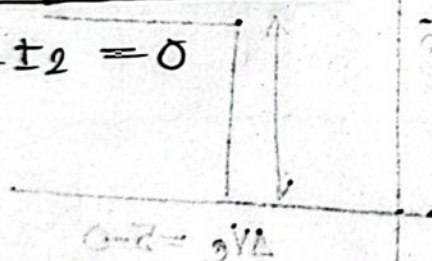
$t = \infty$, steady state



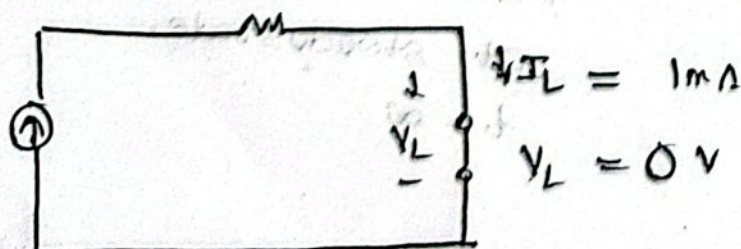
$\Rightarrow$

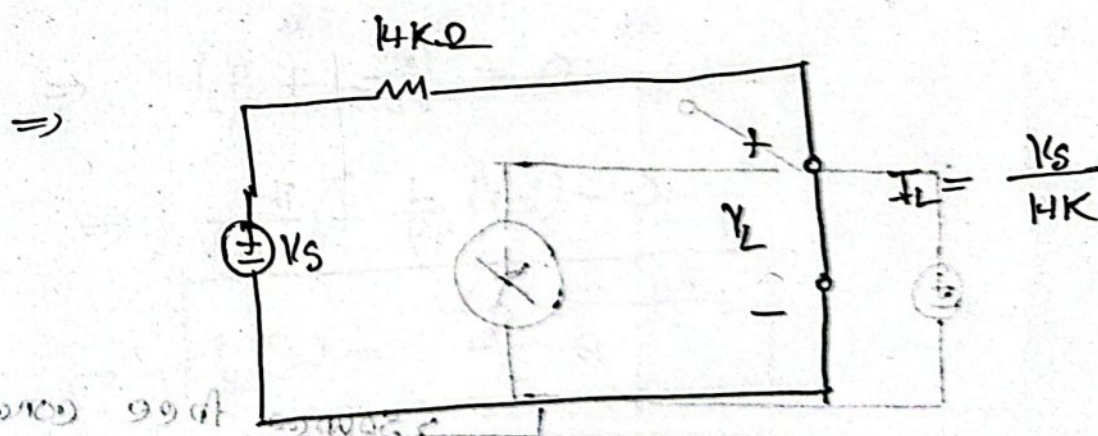
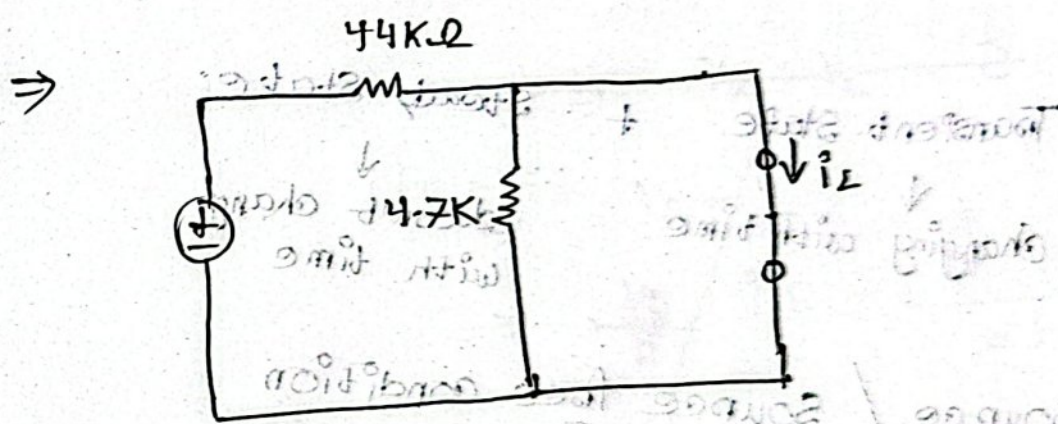
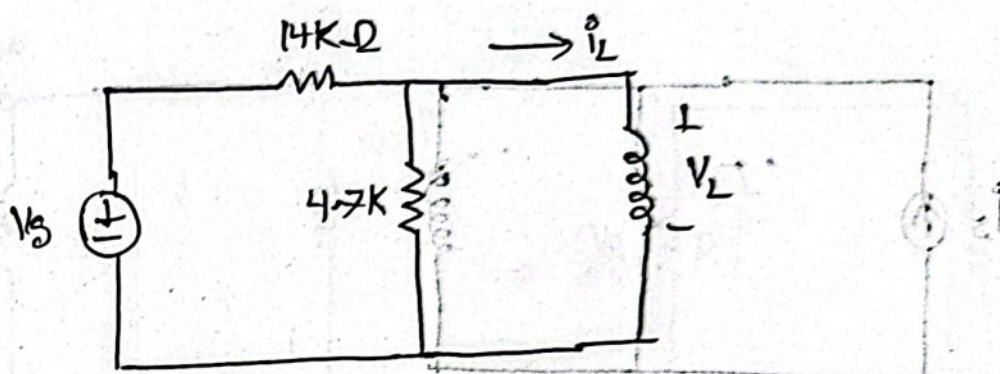


$$V_c = -V_{ab} - 10I_1 - 7I_2 = 0$$

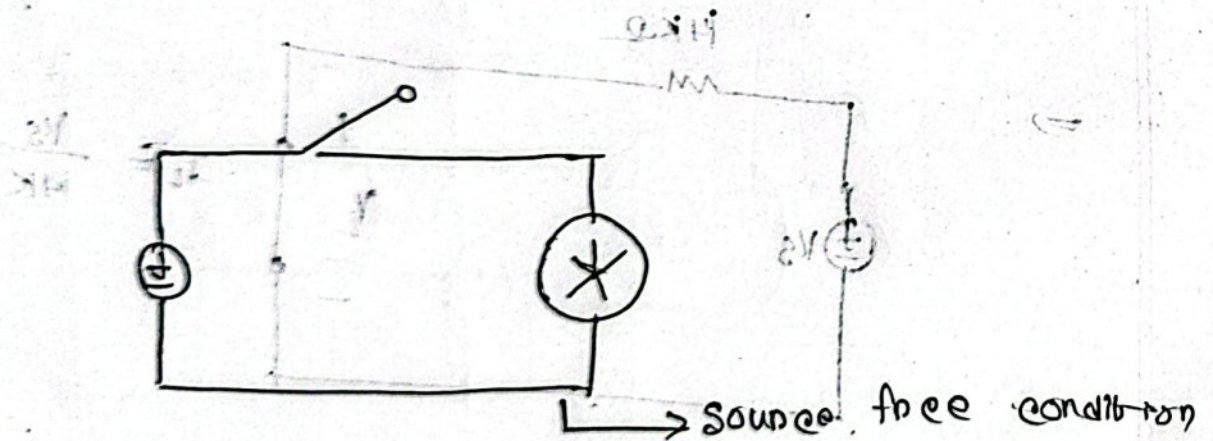
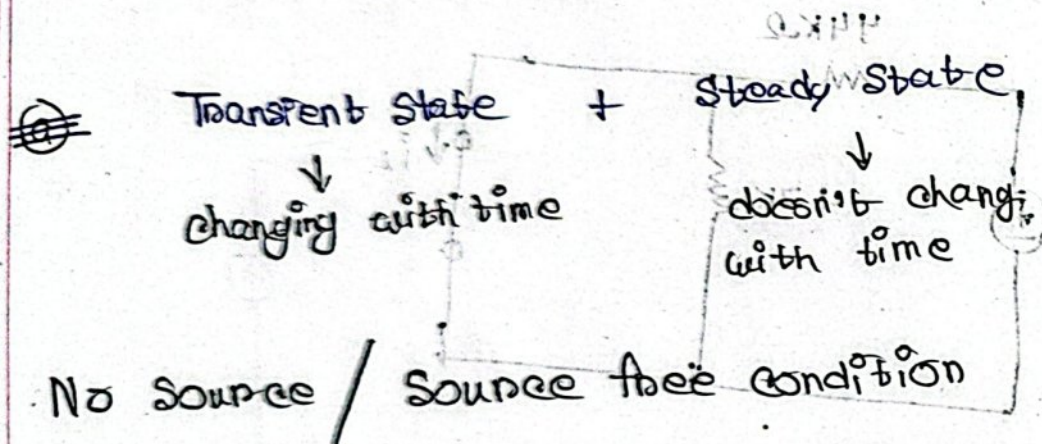
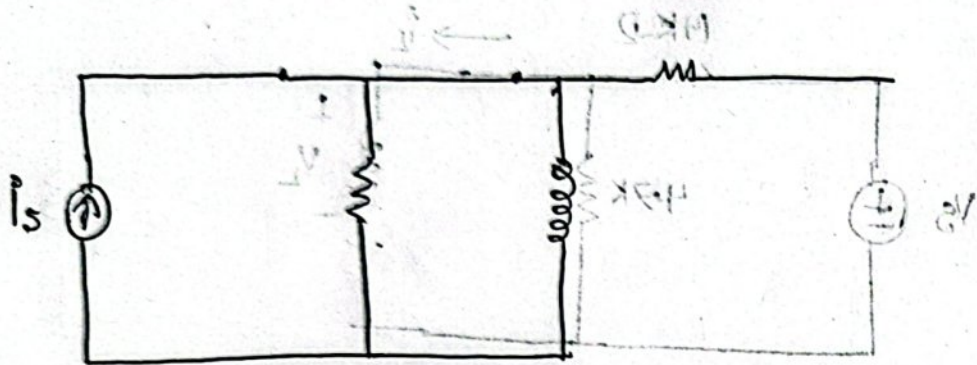


$\Rightarrow$

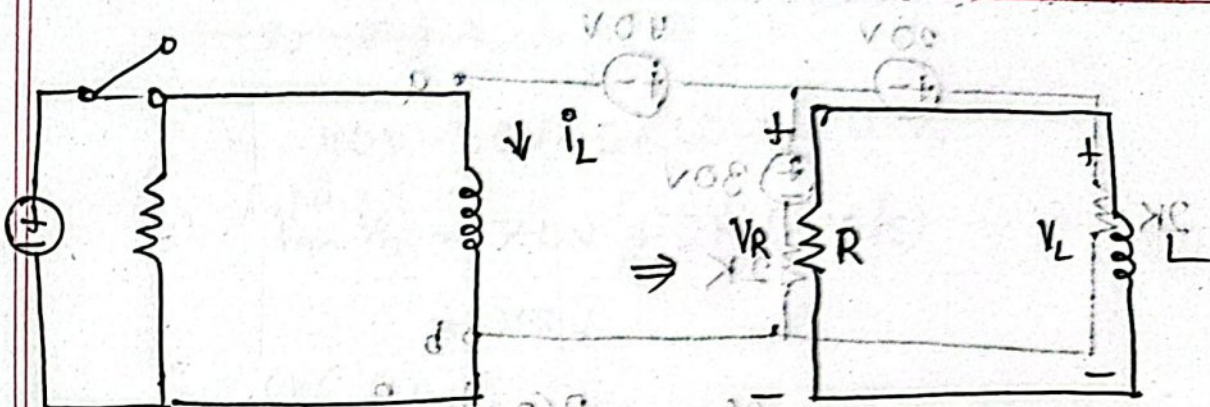




b)



Natural Response



$$\Rightarrow V_R + V_L = 0 \quad (1)$$

$$\Rightarrow V_R = -V_L$$

$$\Rightarrow i_L R = L \frac{di_L(t)}{dt}$$

$$\Rightarrow i_L R + L \frac{di_L}{dt} = 0$$

$$\Rightarrow \frac{di_L}{dt} + \frac{R}{L} i_L(t) = 0$$

$$\Rightarrow i_L(t) = \frac{R}{L} e^{-\frac{R}{L}t}$$

$$\Rightarrow i_L(t) = I_0 e^{-\frac{R}{L}t}$$

$$0 = (0 - I_0) e^{-\frac{R}{L} \cdot 0} = -I_0$$

$$0 = -I_0$$

$$I_0 = 0$$

$$I_0 = 0$$

L20/24-08

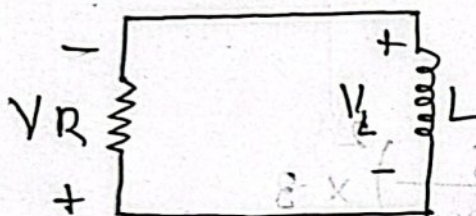
L15/09-08
L17/12-08

Source Free
RL/RC

Lec-20 08-08 (11-07) 2-

24/08/26

Source free RL circuit: (No excitation)



$$V_R + V_L(t) = 0$$

$$i(t)R_L + L \frac{di(t)}{dt} = 0$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = 0$$

$$i(t) = I_0 e^{-\frac{R}{L}t}$$

$$y = Ae^{-\lambda x}$$

$$I_0 e^{-\frac{t}{\tau}}$$

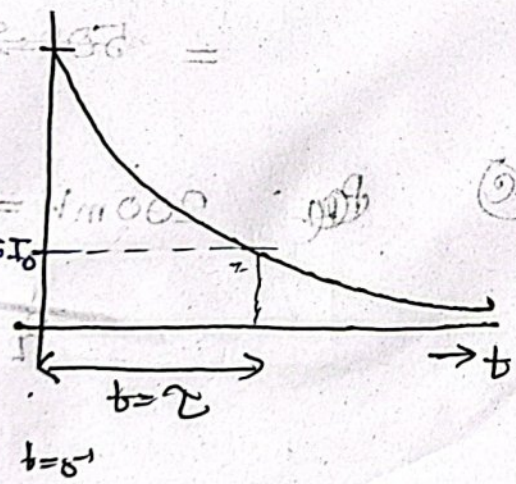
$$\tau = \frac{L}{R} \rightarrow \text{Time constant}$$

@ $t=0$,

$$i(0) = I_0 e^{-\frac{0}{\tau}} = I_0$$

@ $t=\infty$,

$$i(\infty) = 0$$

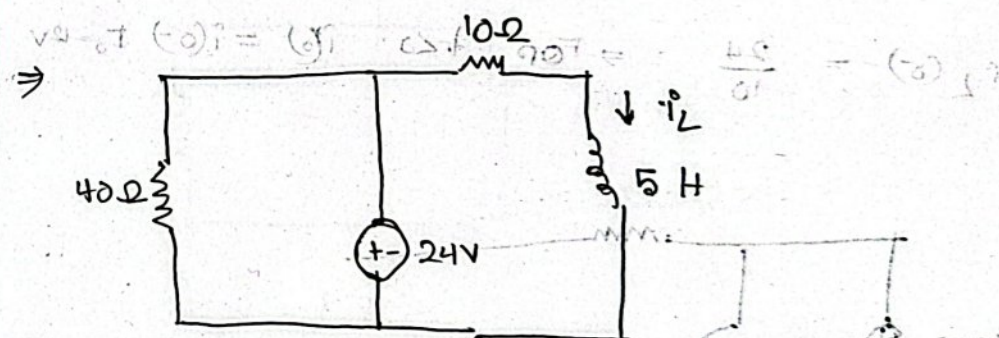
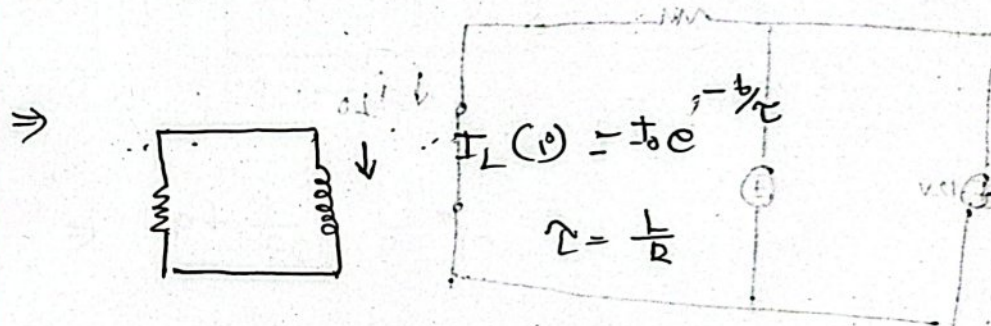


time at which maximum current drops to $e^{-1} \times I_0$

$$i(t) = I_0 e^{-\frac{t}{\tau}} = I_0 e^{-R \times \frac{t}{L}} = I_0 e^{-t/\tau}$$

when $t = \frac{L}{R} = \tau \Rightarrow I_0 e^{-1} \quad (i) \quad I = (i) \quad I$

If $\tau \uparrow \Rightarrow i \uparrow$



Switch is on for long period of time, and at $t=0$, the switch is open. Find $i_L(t)$ for $t \geq 0$

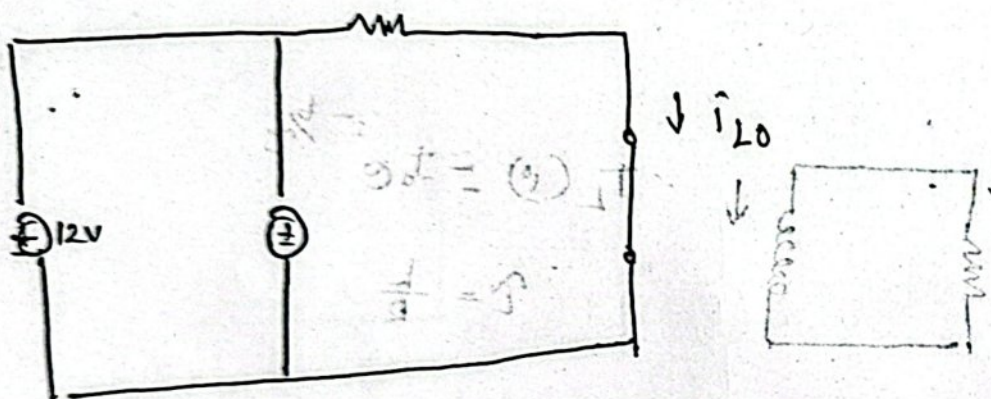
$$i_L(0^-) = \frac{24}{10} = 2.4 \text{ A}$$

$$i_L(0^+) = \frac{24}{10 + 40} = 0.48 \text{ A}$$

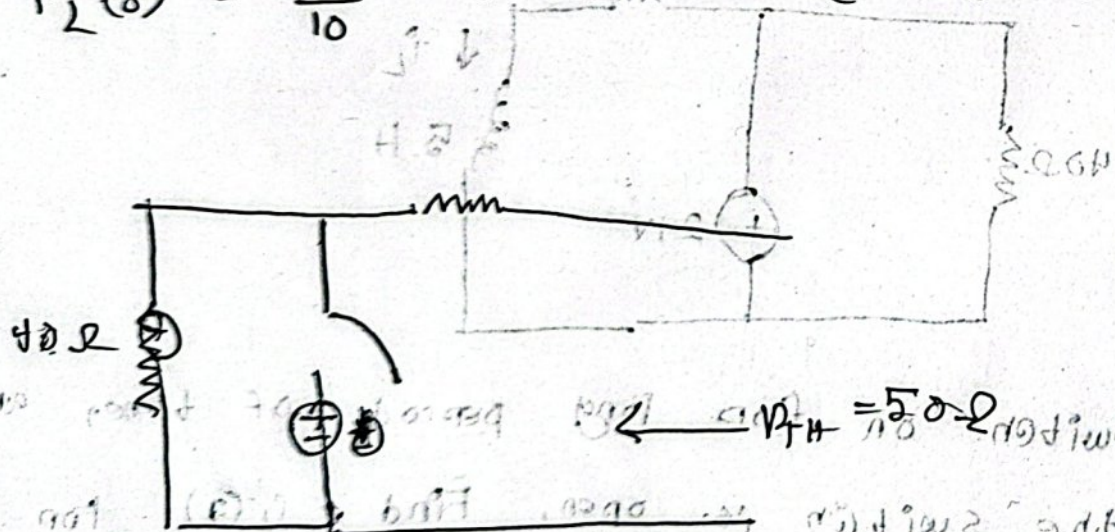
$$P_{TH} \rightarrow \tau_{oc} = \frac{1}{\frac{1}{5} + \frac{1}{10}} = \frac{1}{\frac{2}{10}} = 5 \text{ sec} = (5) \tau$$

$$I'(s) = I(0) \tau_{oc} \quad \Rightarrow \quad \frac{1}{s} = \tau_{oc} \quad \text{when } \tau = \frac{1}{s} = 5 \text{ sec}$$

$$I(s) = \frac{1}{s} \quad \Rightarrow \quad I(t) = 1 \text{ A}$$



$$I_L(0) = \frac{24}{10} = 2.4 \text{ A} \quad \text{For } t < 0 \quad I(0) = I_L(0) \quad I_0 = 12 \text{ V}$$



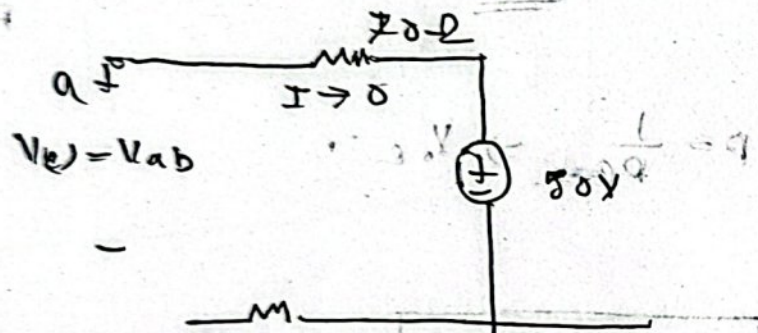
$$\tau = \frac{L}{R} = \frac{1}{\frac{1}{5} + \frac{1}{10}} = 0.1 \text{ sec}$$

$$I(t) = 2.4 e^{-\frac{t}{0.1}} = 2.4 e^{-10t}$$

a. $V_0 \Rightarrow V_C(0) = V_0$ $V_C(0^-) \Rightarrow \text{MAG for } t < 0$

$V_C(0) = +V_A(0^-) = V_{AB}$

$t < 0$



$+V_{AB} \rightarrow 70 \times 50 = 3500$

$\therefore V_{AB} = 70 \Omega$

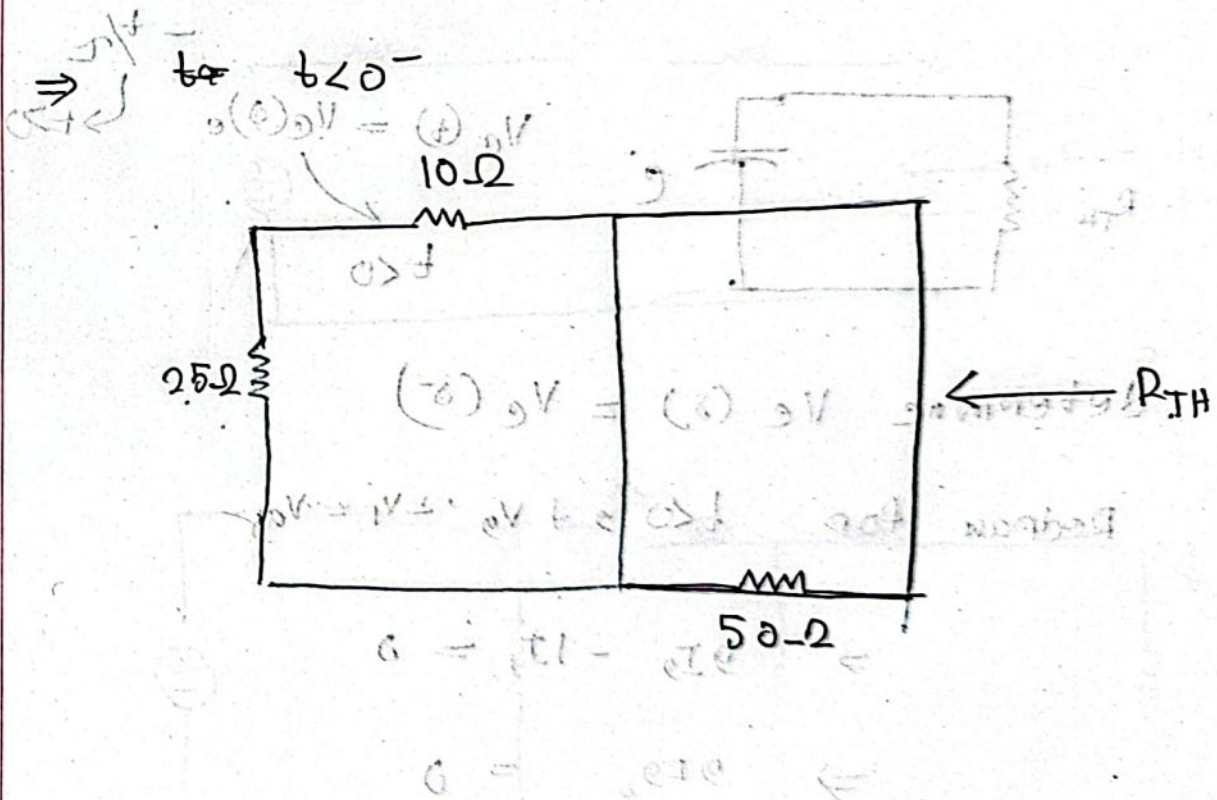
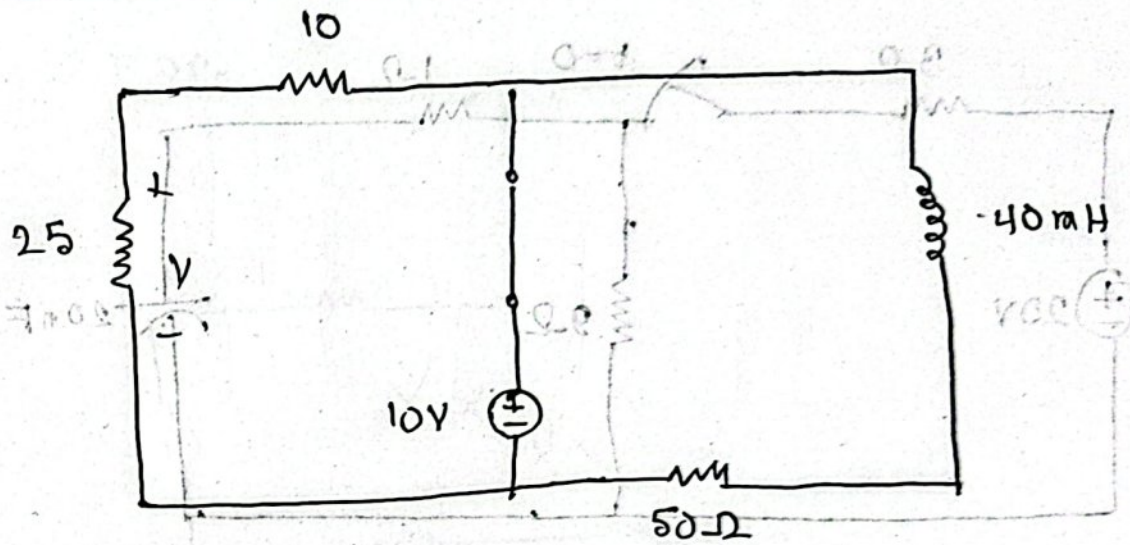
$\tau = R_{TH} C = 80 \times 20 \times 10^{-6} \text{ sec}$

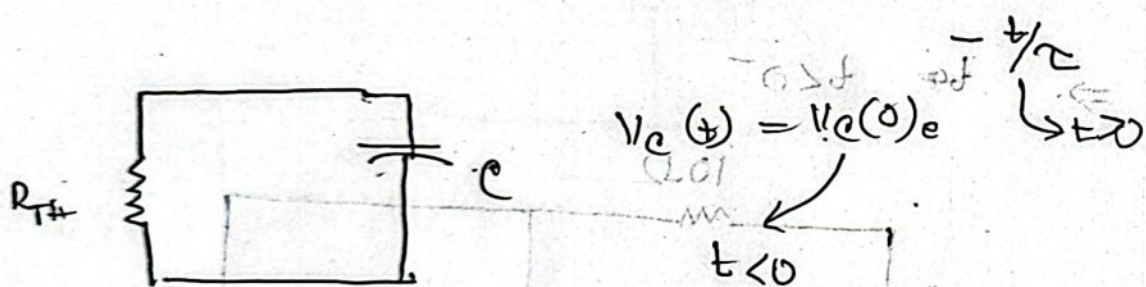
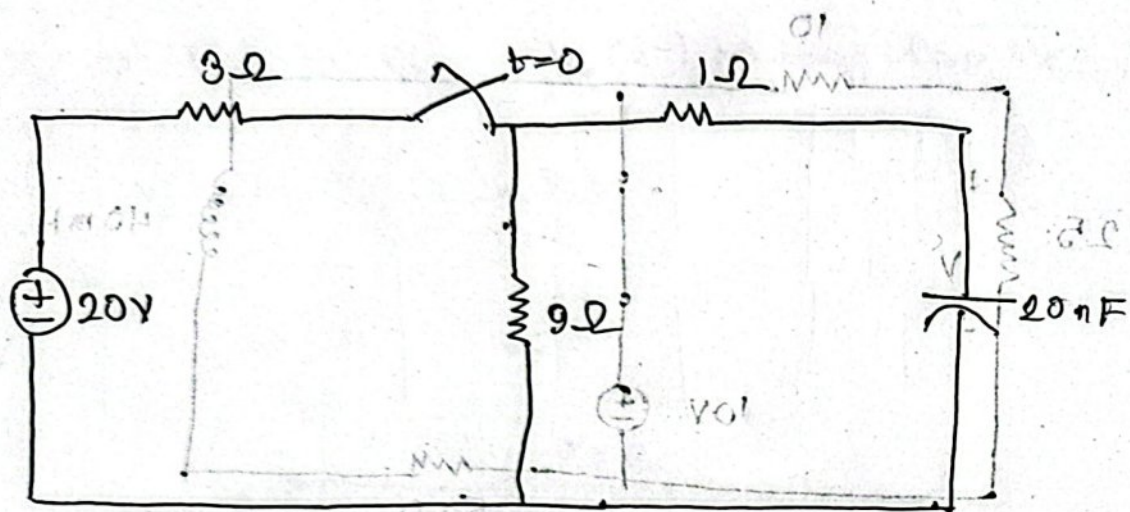
$V_C(t) = 50 e^{-\frac{t}{160 \times 10^{-6}}}$

$i_L(t) = i_L(0) e^{-\frac{t}{\tau}}$

$t < 0$

Next wednesday: 5th Quiz





1. Determine $V_c(t) = V_c(t^-)$

Redraw for $t < 0$, $+V_0$, $-V_1 = V_{01}$

$$\Rightarrow 9I_0 - 1I_1 = 0$$

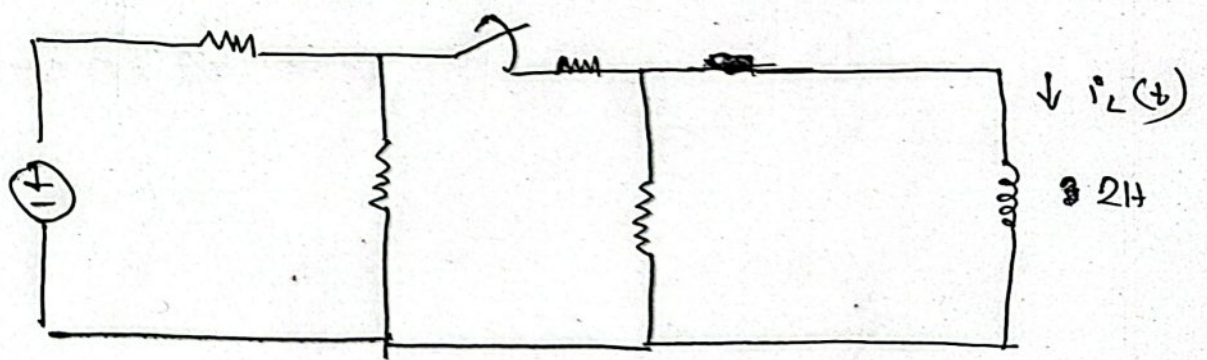
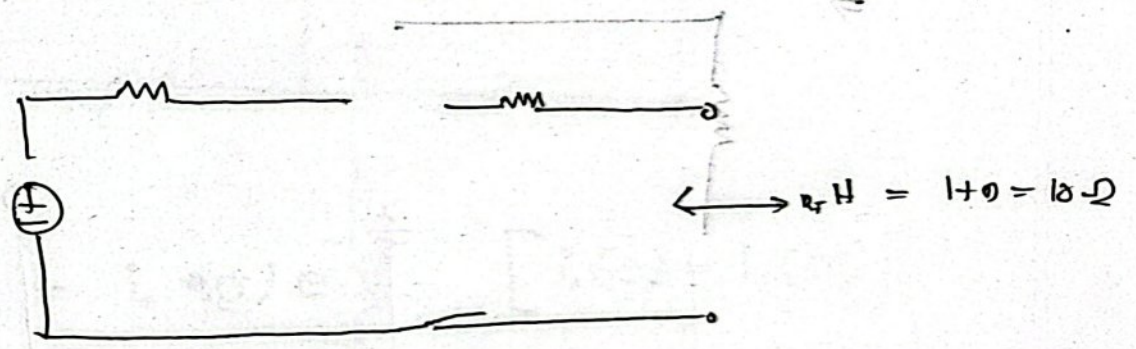
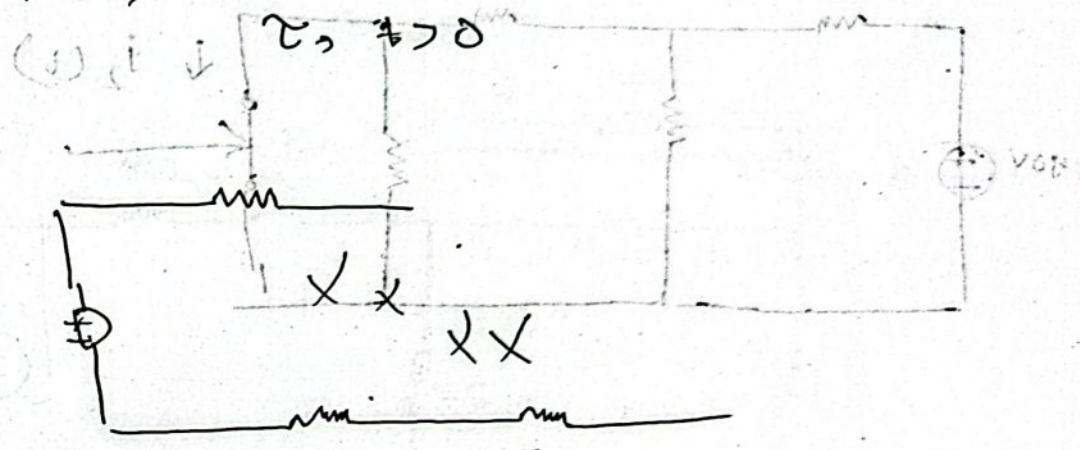
$$\Rightarrow 9I_0 = 0$$

$$\Rightarrow +20 - 3I_1 - 9I_0 = 0$$

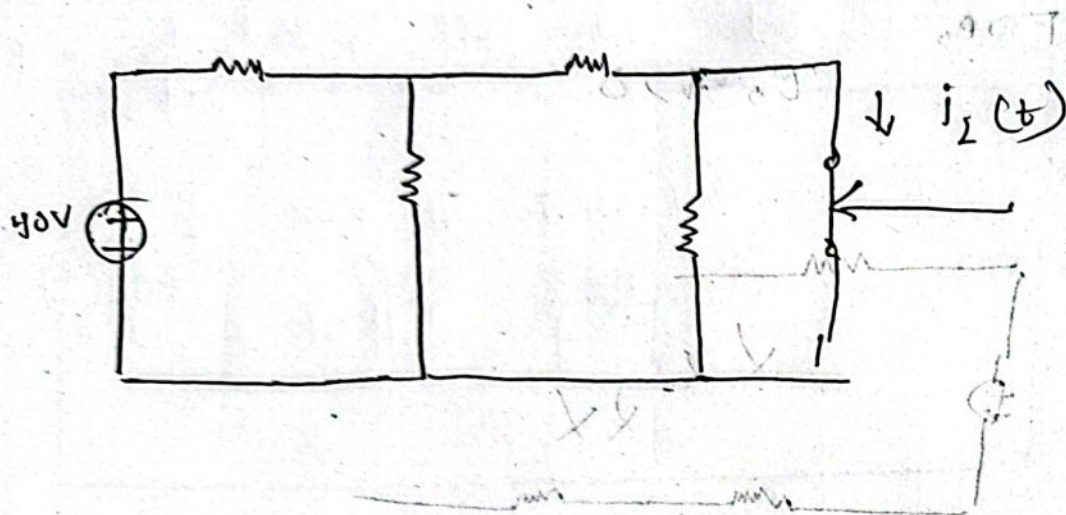
$$\Rightarrow V_{01} = 9 \times \frac{20}{12}$$

Find the value of i_2

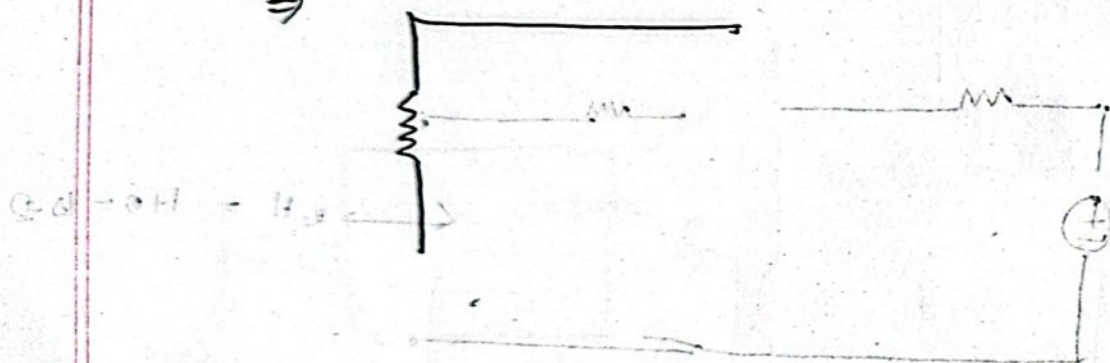
For,



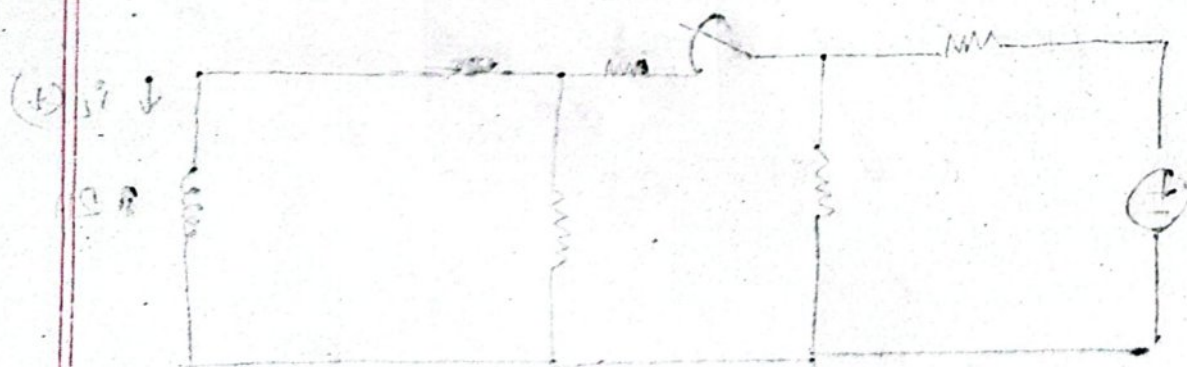
$t < 0$ draw the ckt:



$\Rightarrow$



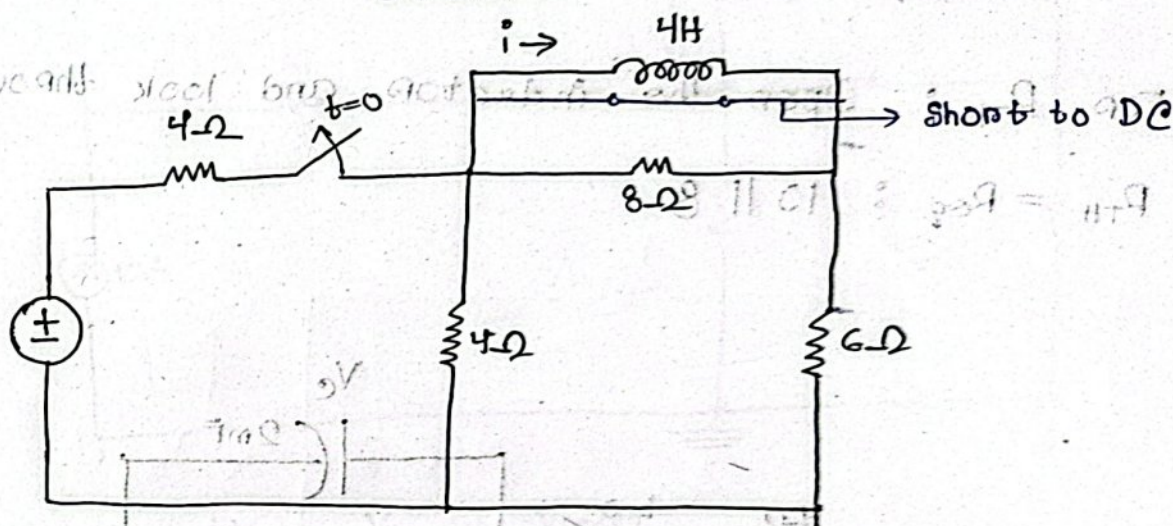
$\Rightarrow$ ~~last time~~:



L21/31-08

Forced Response

Lec-21
31/08/26

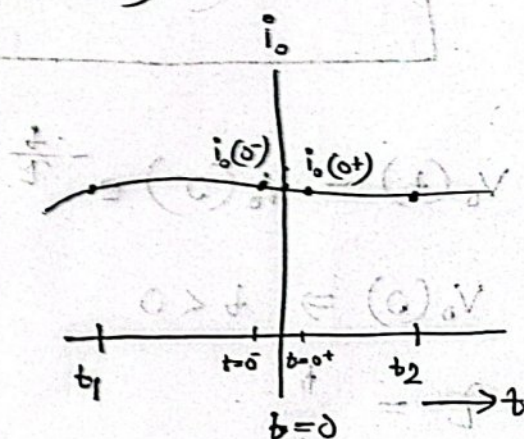


$t \geq 0$

$$i_o(t) = i_o(0) e^{-\frac{t}{\tau}} \quad [i_o(0) = i_o(0^-)]$$

$$v_o(t) = v_o(0) e^{-\frac{t}{\tau}} \quad (\text{capacitor})$$

$i_o(0) \rightarrow \text{ckt for } t < 0$
 $(0^-) \text{ or } (0^+)$

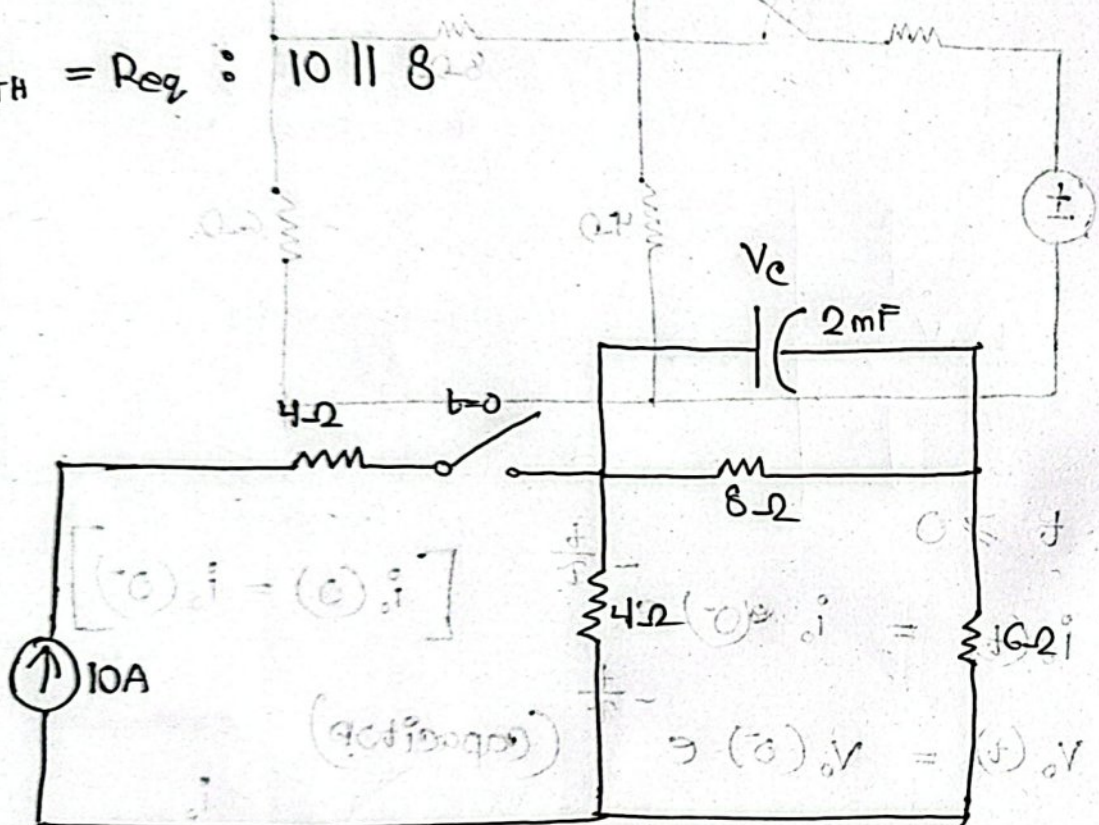


$$i_o(0) = i_o(0^-)$$

$$\tau = \frac{L}{R_{eq}} \text{ for the ckt } t > 0 \text{ [S. Free condition]}$$

For R_{TH} : Open the inductor and look through

$$R_{TH} = R_{eq} : 10 \parallel 8$$



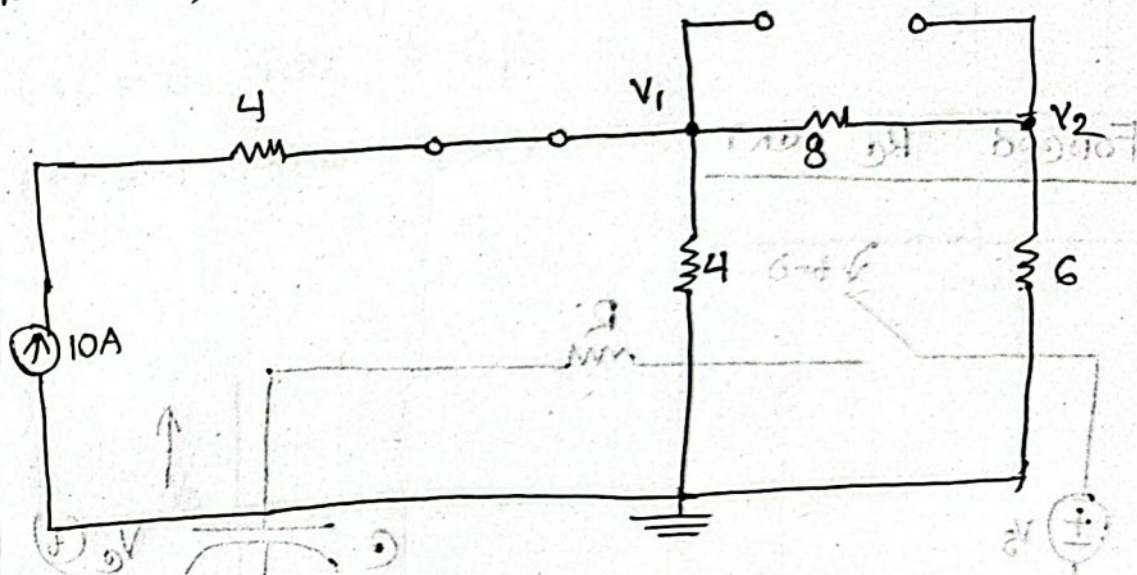
$$V_o(t) = V_o(0^-) e^{-\frac{t}{\tau}}$$

$$V_o(0) \Rightarrow t < 0$$

$$\tau =$$

$$V_o(0) = V_o(0^-)$$

Forced RL and RC Response
For $t > 0$,



$$-10 + \frac{V_1}{4} + \frac{V_1}{4} + \frac{V_1 - V_2}{8} = 0$$

$$\frac{V_2}{6} + \frac{V_2 - V_1}{8} = 0$$

V_{100}

V_{200}

$$\begin{array}{r} 2 \mid 4, 4, 8 \\ 2 \mid 2, 2, 4 \\ \hline 1, 1, 2 \end{array}$$

$$\frac{V_1}{4} + \frac{V_1}{4} + \frac{V_1 - V_2}{8} = 10$$

$$2 \mid 6, 8 \\ \hline 3, 4$$

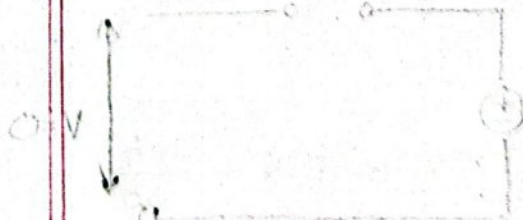
$$0 \geq 0 \Rightarrow 2V_1 + 2V_1 + V_1 - V_2 = 80$$

$$0 < 0 \Rightarrow 1 \times 4 = 5V_1 - V_2 = 80$$

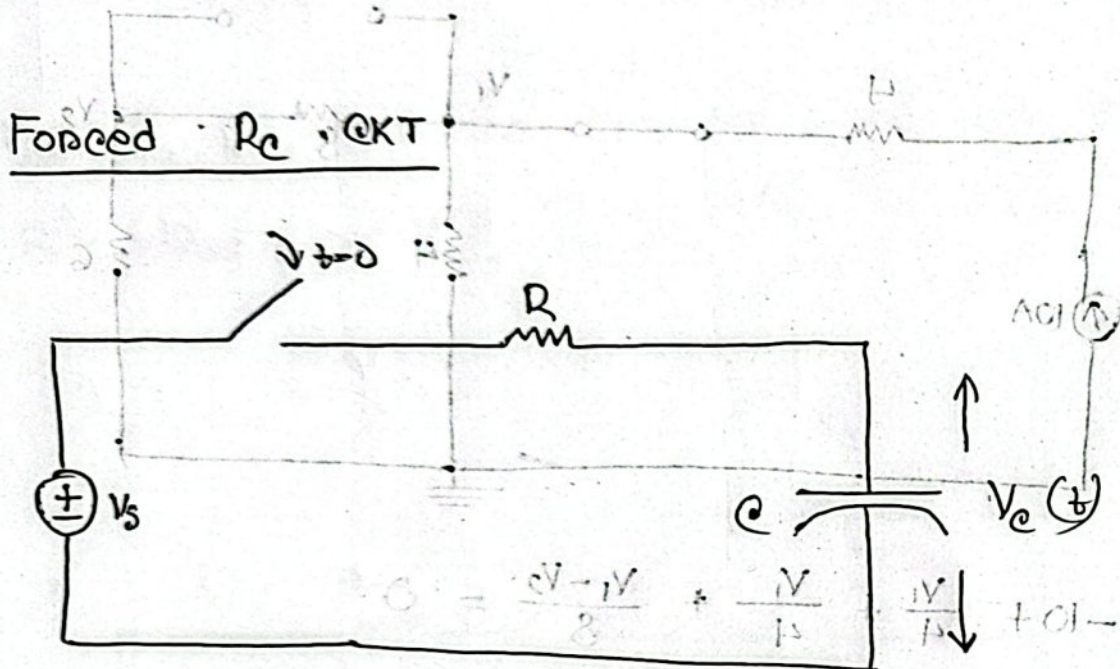
$\Rightarrow$

$0 \geq 0$

$0 < 0$



Forced RL and RC Response



$$0 = \frac{V - V_c}{R} + C \frac{dV_c}{dt}$$

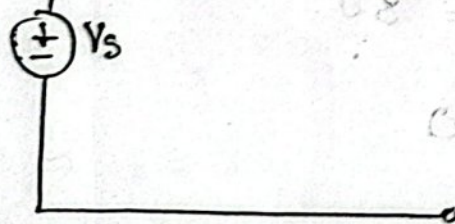
at $t=0^-$

at $t=0^+$

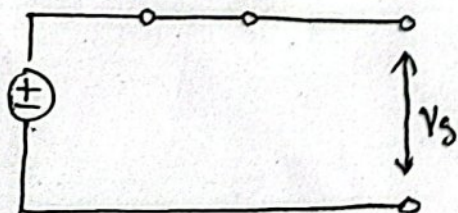
$$0 = \frac{V - V_c}{R} + C \frac{dV_c}{dt}$$

$$V_s u(t) = 0 \text{ for } t \leq 0$$

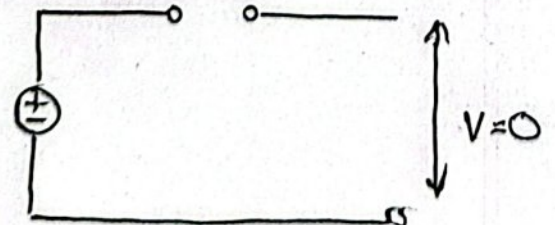
$$V_s u(t) = V_s \times 1 \text{ for } t > 0$$



$t > 0$



or

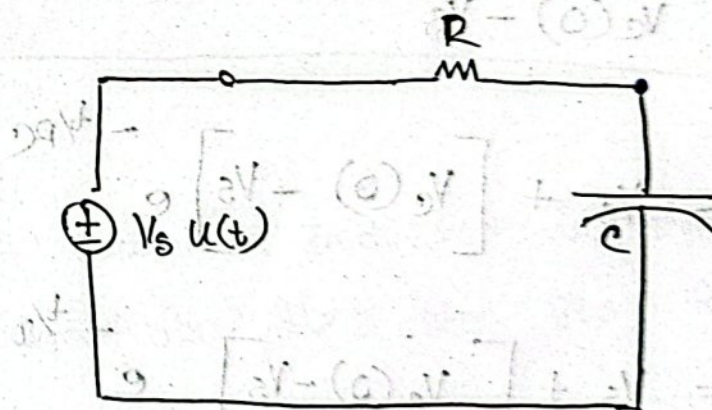
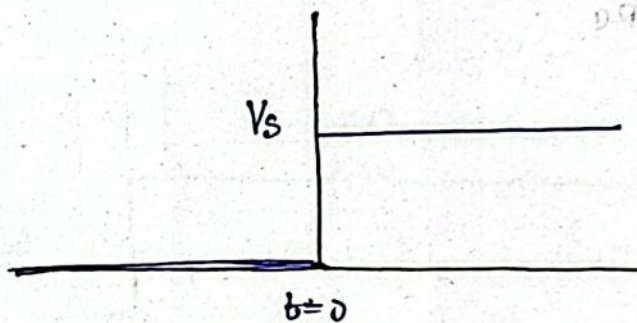
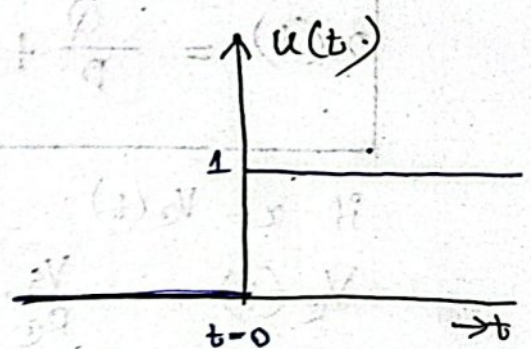


$t \leq 0$

$u(t)$: unit step function

$$u(t) = 0 \text{ for } t \leq 0$$

$$u(t) = 1 \text{ for } t > 0$$



$$\Rightarrow \frac{V_c(t) - V_s u(t)}{R} + C \frac{dV_c(t)}{dt} = 0$$

$$\text{for } t > 0, V_s u(t) = V_s \Rightarrow \frac{V_c(t) - V_s}{R} + C \frac{dV_c(t)}{dt} = 0$$

$$\Rightarrow V_c(t) - V_s + R C \frac{dV_c(t)}{dt} = 0$$

$$\frac{dx}{dt} + P(x) = Q(x)$$

$$\frac{dV_c(t)}{dt} + \frac{V_c(t)}{RC} = \frac{V_s}{RC}$$

$$x(t) = \frac{Q}{p} + A e^{-pt}$$

if $x = V_c(t)$

$$V_c(t) = \frac{V_s}{R_c} \times \frac{R_c}{1} + A e^{-\frac{t}{R_c}}$$

$$V_c(t) = V_s + A e^{-\frac{t}{R_c}}$$

At $t=0$ sec,

$$V_c(0) = V_s + A e^0$$

$$\therefore A = V_c(0) - V_s$$

$$\begin{aligned} \therefore V_c(t) &= V_s + [V_c(0) - V_s] e^{-t/R_c} \\ &= V_s + [V_c(0) - V_s] e^{-t/\tau} \end{aligned}$$

At $t = \infty$ sec,

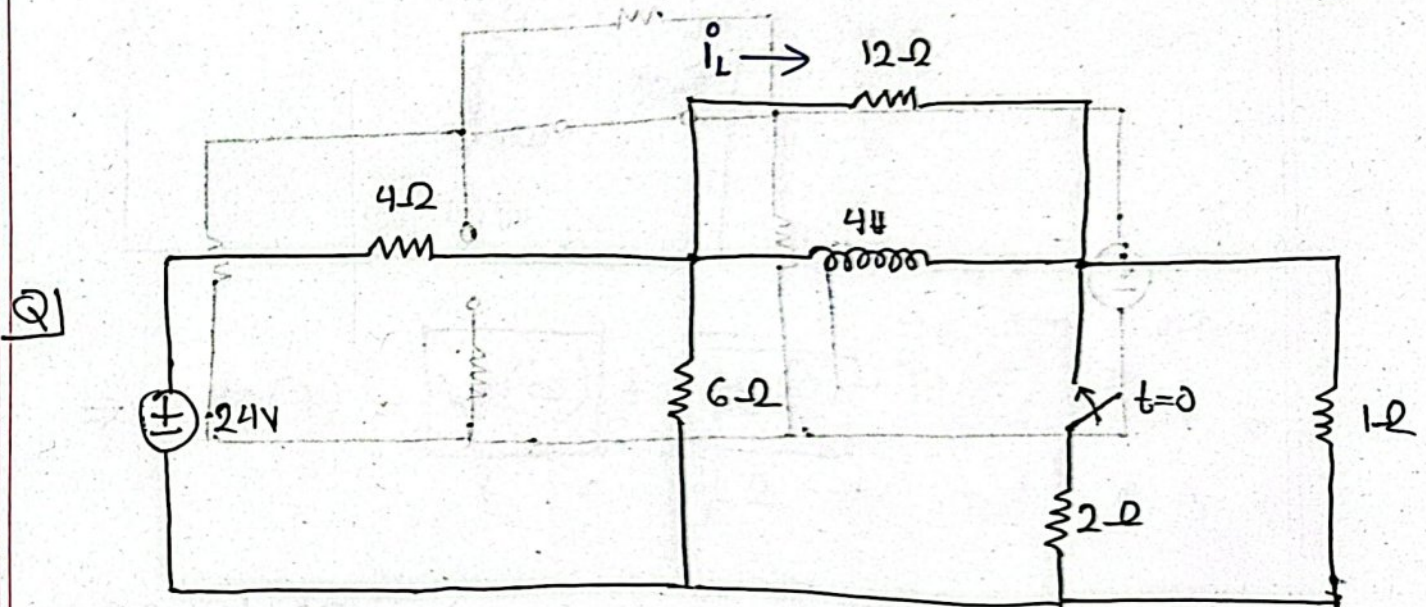
$$V_c(\infty) = V_s + A e^{-\frac{\infty}{R_c}}$$

$$V_c(\infty) = V_s$$

$$\frac{dV}{dt} = \frac{(d)V}{dt} + \frac{(d)V_b}{dt}$$

$$V_C(t) = V_C(\infty) + [V_C(0) - V_C(\infty)] e^{-t/\tau} \quad \tau = R_{eq}C$$

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)] e^{-t/\tau} \quad \tau = \frac{L}{R_{eq}}$$



* If the capacitor/Inductor in the ckt still connected to source for $t > 0$ then its forced condition.

$$* i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)] e^{-t/\tau} \quad [\text{for } t > 0]$$

$\Rightarrow i_L(\infty)$: ckt for $t > 0$

$i_L(0)$: ckt for $t < 0$

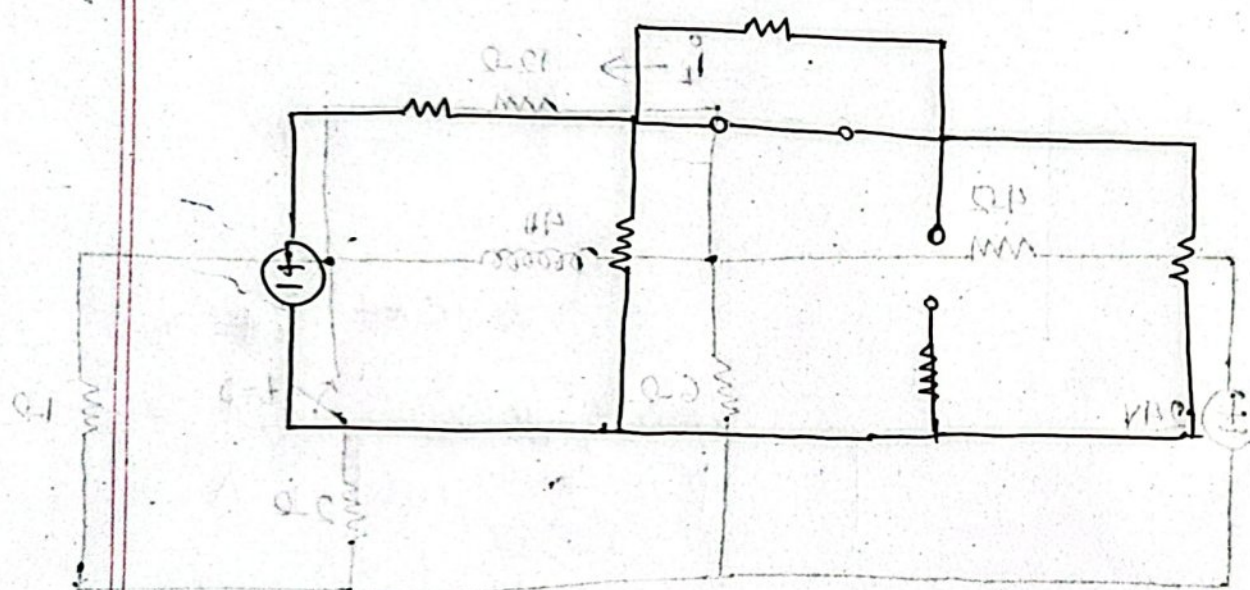
τ : ckt for $t > 0$ [always]

Quiz 5: Source free only RL RC

$$0_{\text{ind}} = t < 0: \quad \left[(0)_{\text{V}} - (0)_{\text{V}} \right] + (\infty)_{\text{V}} = (1)_{\text{V}}$$

$$i_L(\infty) = i_L(0^-)$$

Inductor S.C.



$t > 0:$

Inductor S.C. $i_L(\infty)$ not source of

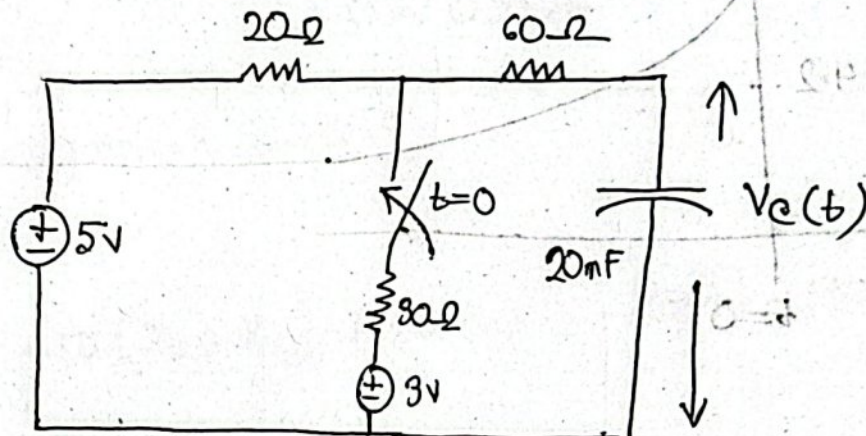
For 2: Open the inductor, Find R_{TH} through

$$R_{TH} : [(4 \parallel 6) + (1 \parallel 2)] \parallel 12$$

[example] $0 < t$ not $t > 0$

lec-22

02/09



$$\Rightarrow V_C(t) = V_C(\infty) + [V_C(0) - V_C(\infty)] e^{-\frac{t}{\tau}}$$

2: $t < 0$

$$V_{ab} = 5V$$

$$V_C(0^-) = V_C(0) = V_{ab} = 5V$$

1: $t > 0$

$$+3 + V_3 - V_C - V_{ab} = 0$$

$R_{th} \Rightarrow$

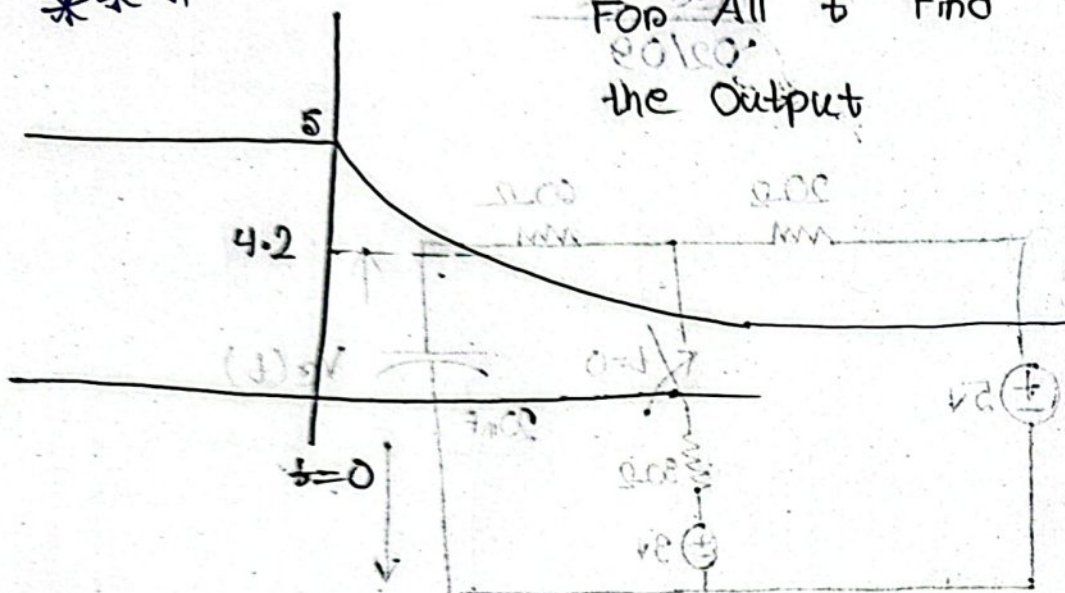
$$\Rightarrow +3 + 30I_2 - 60I_1 - V_{ab} = 0$$

$$\Rightarrow V_{ab} =$$

$$V_C(\infty) = V_{ab} =$$

Final

For All to Find the Output



⇒ Node Analysis

⇒ Mesh Analysis

⇒ Δ -Y [Part of a question]

⇒ Source

⇒ Thevenin / Norton

⇒ -----

$$0 < f < \infty$$

$$V_{th} = 2V$$

$$0 < f < \infty$$

$$C = 10^{-6} F$$

$$C = 10^{-6} F$$

$$= 10^{-6} F$$

$$= 10^{-6} F$$

Lec - 23

Final Lec

07/09/26

Assignment ① Q.4

$$-10(I_b - I_a) + 8 - 30I_b - V_c(b) - 15I_b = 0$$

H.W 10

2 Nodes $\rightarrow$ Voltage $\rightarrow$ KVL Apply

Between 2 nodes যে গঠে যাই থাকুক, সবসময় Voltage

Same. Apply KVL carefully.